CHAPTER 7

Euclidean geometry

Geometry (from the Greek "geo" = earth and "metria" = measure) arose as the field of knowledge dealing with spatial relationships. Analytical geometry deals with space and shape using algebra and a coordinate system. Euclidean geometry deals with space and shape using a system of logical deductions.

Euclidean geometry was first used in surveying and is still used extensively for surveying today. Euclidean geometry is also used in architecture to design new buildings. Other uses of Euclidean geometry are in art and to determine the best packing arrangement for various types of objects.

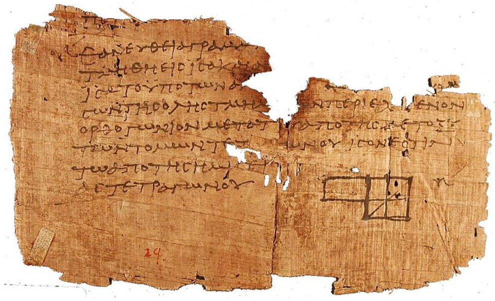


Figure 7.1: A small piece of the original version of Euclid's elements. Euclid is considered to be the father of modern geometry. Euclid's elements was used for many years as the standard text for geometry.

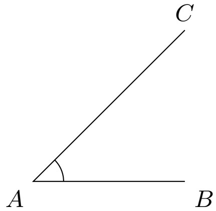
VISIT:
This video highlights some of the basic concepts used in geometry.
See video: 2G5V at www.everythingmaths.co.za

DID YOU KNOW?
In Euclidean geometry we use two fundamental types of measurement: angles and distances.

7.1 Introduction EMA5M

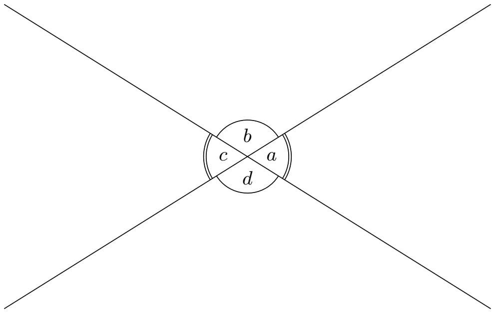
Angles EMA5N

An angle is formed when two straight lines meet at a point, also known as a vertex. Angles are labelled with a caret on a letter, for example, $\hat{B}$. Angles can also be labelled according to the line segments that make up the angle, for example $C\hat{B}A$ or $A\hat{B}C$. The $\angle$ symbol is a short method of writing angle in geometry and is often used in phrases such as "sum of $\angle$s in $\triangle$". Angles are measured in degrees which is denoted by °, a small circle raised above the text, similar to an exponent.



Properties and notation EMA5P

In the diagram below two straight lines intersect at a point, forming the four angles $\hat{a}$, $\hat{b}$, $\hat{c}$ and $\hat{d}$.



The following table summarises the different types of angles, with examples from the figure above.

Term	Property	Examples
Acute angle	$0° <$ angle $< 90°$	$\hat{a}$; $\hat{c}$
Right angle	Angle $= 90°$	
Obtuse angle	$90° <$ angle $< 180°$	$\hat{b}$; $\hat{d}$
Straight angle	Angle $= 180°$	$\hat{a} + \hat{b}$; $\hat{b} + \hat{c}$
Reflex angle	$180° <$ angle $< 360°$	$\hat{a} + \hat{b} + \hat{c}$
Adjacent angles	Angles that share a vertex and a common side.	$\hat{a}$ and $\hat{d}$; $\hat{c}$ and $\hat{d}$
Vertically opposite angles	Angles opposite each other when two lines intersect. They share a vertex and are equal.	$\hat{a} = \hat{c}$; $\hat{b} = \hat{d}$
Supplementary angles	Two angles that add up to $180°$	$\hat{a} + \hat{b} = 180°$; $\hat{b} + \hat{c} = 180°$
Complementary angles	Two angles that add up to $90°$	
Revolution	The sum of all angles around a point.	$\hat{a} + \hat{b} + \hat{c} + \hat{d} = 360°$

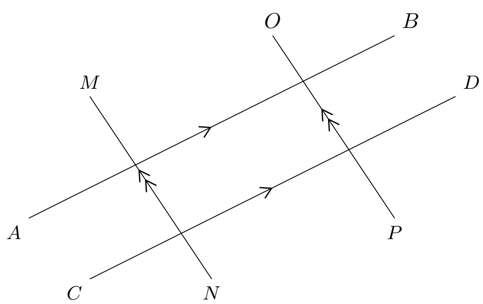
Note that adjacent angles on a straight line are supplementary.

VISIT:
The following video provides a summary of the terms used to refer to angles.
See video: 2G5W at www.everythingmaths.co.za

Parallel lines and transversal lines EMA5Q

Two lines intersect if they cross each other at a point. For example, at a traffic intersection two or more streets intersect; the middle of the intersection is the common point between the streets.

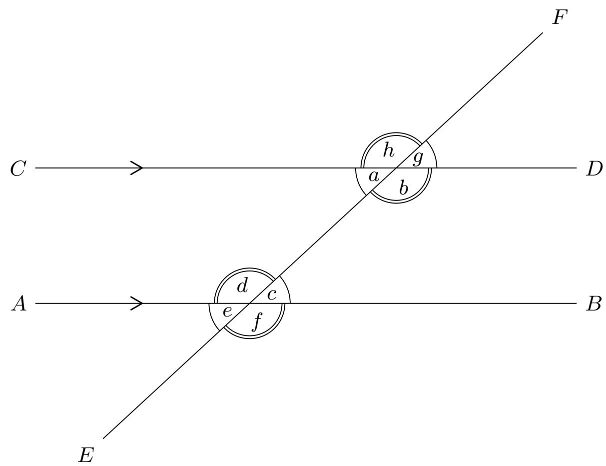
Parallel lines are always the same distance apart and they are denoted by arrow symbols as shown below.



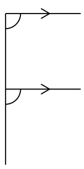
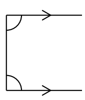
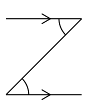
In writing we use two vertical lines to indicate that two lines are parallel:

$$AB \parallel CD \text{ and } MN \parallel OP$$

A transversal line intersects two or more parallel lines. In the diagram below, $AB \parallel CD$ and EF is a transversal line.



The properties of the angles formed by these intersecting lines are summarised in the following table:

Name of angle	Definition	Examples	Notes
Interior angles	Angles that lie in between the parallel lines.	$\hat{a}$, $\hat{b}$, $\hat{c}$ and $\hat{d}$ are interior angles.	Interior means inside.
Exterior angles	Angles that lie outside the parallel lines.	$\hat{e}$, $\hat{f}$, $\hat{g}$ and $\hat{h}$ are exterior angles.	Exterior means outside.
Corresponding angles	Angles on the same side of the lines and the same side of the transversal. If the lines are parallel, the corresponding angles will be equal.	$\hat{a}$ and $\hat{e}$, $\hat{b}$ and $\hat{f}$, $\hat{c}$ and $\hat{g}$, $\hat{d}$ and $\hat{h}$ are pairs of corresponding angles. $\hat{a} = \hat{e}$, $\hat{b} = \hat{f}$, $\hat{c} = \hat{g}$ and $\hat{d} = \hat{h}$.	 F shape
Co-interior angles	Angles that lie in between the lines and on the same side of the transversal. If the lines are parallel, the angles are supplementary.	$\hat{a}$ and $\hat{d}$, $\hat{b}$ and $\hat{c}$ are pairs of co-interior angles. $\hat{a} + \hat{d} = 180°$, $\hat{b} + \hat{c} = 180°$.	 C shape
Alternate interior angles	Equal interior angles that lie inside the lines and on opposite sides of the transversal. If the lines are parallel, the interior angles will be equal.	$\hat{a}$ and $\hat{c}$, $\hat{b}$ and $\hat{d}$ are pairs of alternate interior angles. $\hat{a} = \hat{c}$, $\hat{b} = \hat{d}$	 Z shape

VISIT:
This video provides a short summary of some of the angles formed by intersecting lines.
See video: 2G5X at www.everythingmaths.co.za

If two lines are intersected by a transversal such that:

- corresponding angles are equal; or
- alternate interior angles are equal; or
- co-interior angles are supplementary

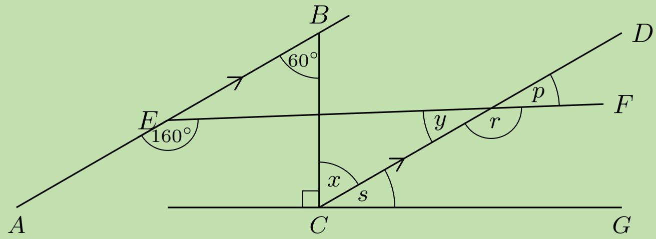
then the two lines are parallel.

NOTE:
When we refer to lines we can either write EF to mean the line through points E and F or $\overline{EF}$ to mean the line segment from point E to point F.

Worked example 1: Finding angles

QUESTION

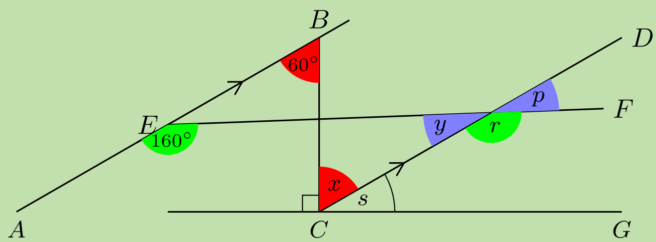
Find all the unknown angles. Is $EF \parallel CG$? Explain your answer.



SOLUTION

Step 1: Use the properties of parallel lines to find all equal angles on the diagram

Redraw the diagram and mark all the equal angles.



Step 2: Determine the unknown angles

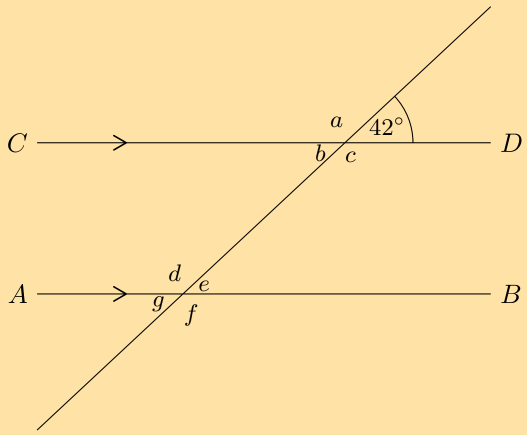
$$
\begin{aligned}
AB &\parallel CD && \text{(given)} \\
\therefore \hat{x} &= 60^\circ && \text{(alt } \angle\text{s; } AB \parallel CD\text{)} \\
\hat{y} + 160^\circ &= 180^\circ && \text{(co-int } \angle\text{s; } AB \parallel CD\text{)} \\
\therefore \hat{y} &= 20^\circ && \\
\hat{p} &= \hat{y} && \text{(vert opp } \angle\text{s } =\text{)} \\
\therefore \hat{p} &= 20^\circ && \\
\hat{r} &= 160^\circ && \text{(corresp } \angle\text{s; } AB \parallel CD\text{)} \\
\hat{s} + \hat{x} + 90^\circ &= 180^\circ && \text{(}\angle\text{s on a str line)} \\
\hat{s} + 60^\circ &= 90^\circ && \\
\therefore \hat{s} &;= 30^\circ &&
\end{aligned}
$$

Step 3: Determine whether $EF \parallel CG$

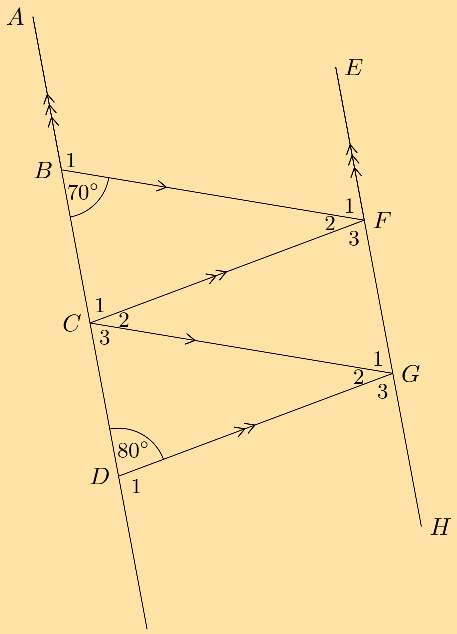
If $EF \parallel CG$ then $\hat{p}$ will be equal to corresponding angle $\hat{s}$, but $\hat{p} = 20^\circ$ and $\hat{s} = 30^\circ$. Therefore EF is not parallel to CG.

Exercise 7 – 1:

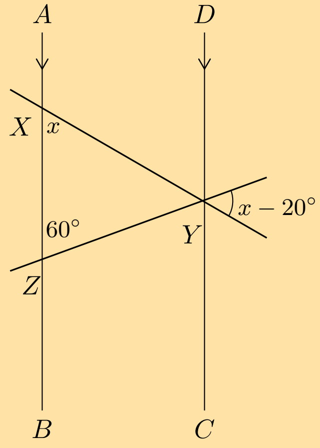
1. Use adjacent, corresponding, co-interior and alternate angles to fill in all the angles labelled with letters in the diagram:



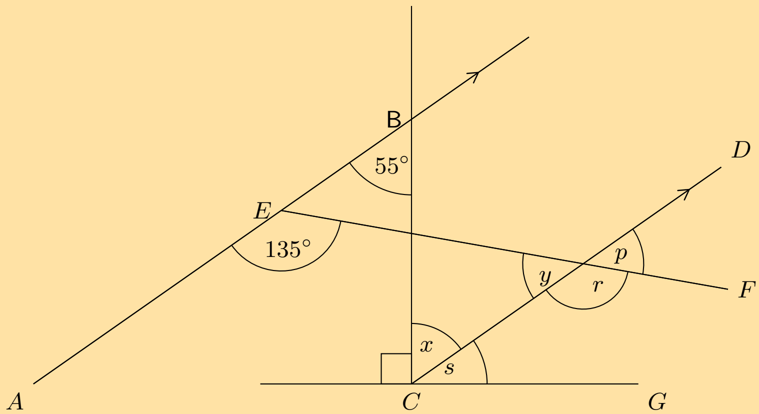
2. Find all the unknown angles in the figure:



3. Find the value of x in the figure:

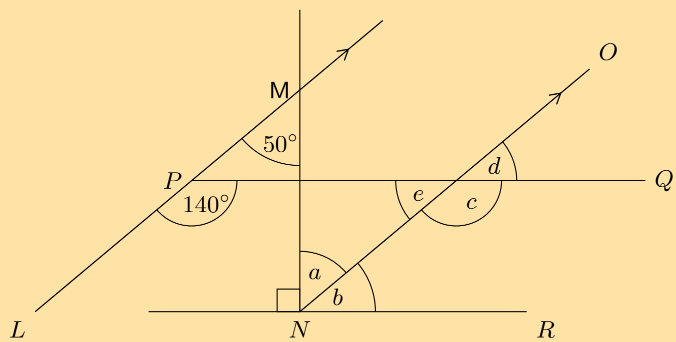


4. Given the figure below:



   a) Find each of the unknown angles marked in the figure below. Find a reason that leads to the answer in a single step.
   b) Based on the results for the angles above, is $EF \parallel CG$?

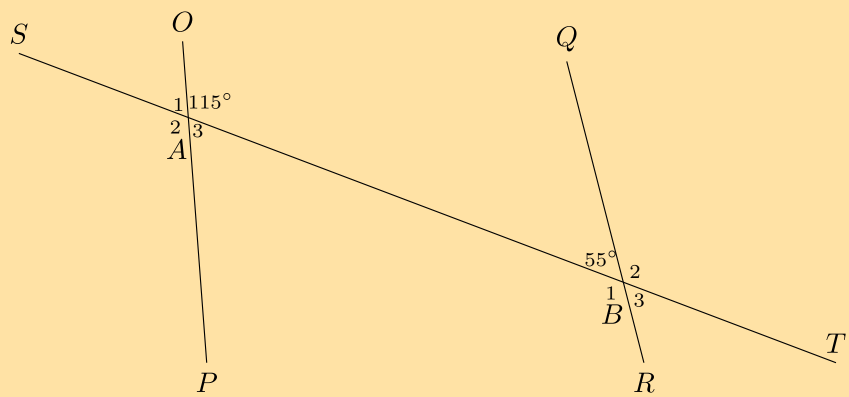
5. Given the figure below:



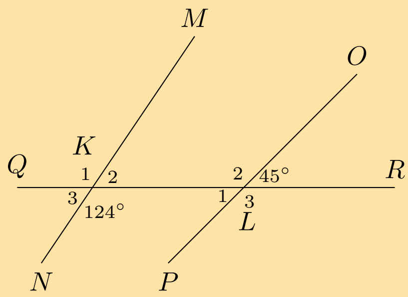
   a) Find each of the unknown angles marked in the figure below. Find a reason that leads to the answer in a single step.
   b) Based on the results for the angles above, is $PQ \parallel NR$?

6. Determine whether the pairs of lines in the following figures are parallel:

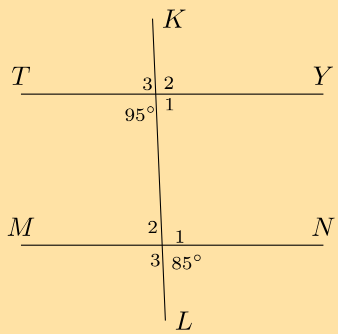
   a)



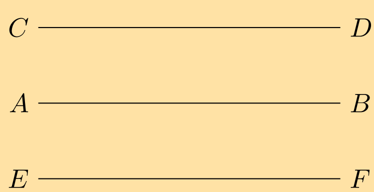
b)



c)



7. If AB is parallel to CD and AB is parallel to EF, explain why CD must be parallel to EF.



For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2G5Y 2. 2G5Z 3. 2G62 4. 2G63 5. 2G64 6a. 2G65 6b. 2G66 6c. 2G67 7. 2G68

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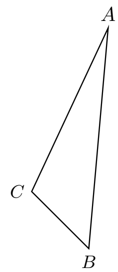
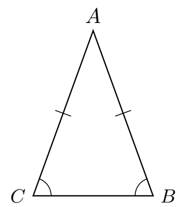
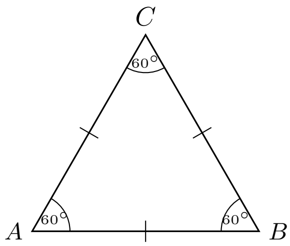
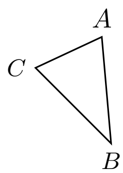
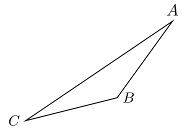
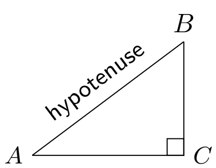
7.2 Triangles EMA5R

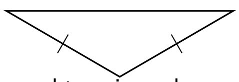
Classification of triangles EMA5S

A triangle is a three-sided polygon. Triangles can be classified according to sides: equilateral, isosceles and scalene. Triangles can also be classified according to angles: acute-angled, obtuse-angled and right-angled.

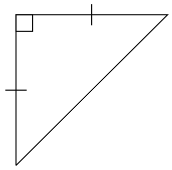
We use the notation $\triangle ABC$ to refer to a triangle with vertices labelled A, B and C.

Name	Diagram	Properties
Scalene		All sides and angles are different.
Isosceles		Two sides are equal in length. The angles opposite the equal sides are also equal.
Equilateral		All three sides are equal in length and all three angles are equal.
Acute		Each of the three interior angles is less than $90°$.
Obtuse		One interior angle is greater than $90°$.
Right-angled		One interior angle is $90°$.

Different combinations of these properties are also possible. For example, an obtuse isosceles triangle and a right-angled isosceles triangle are shown below:



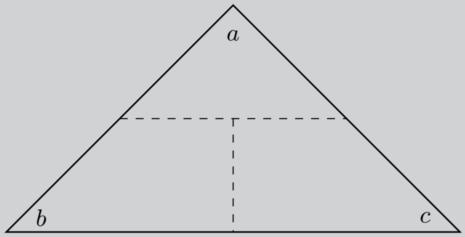
obtuse isosceles



right-angled isosceles

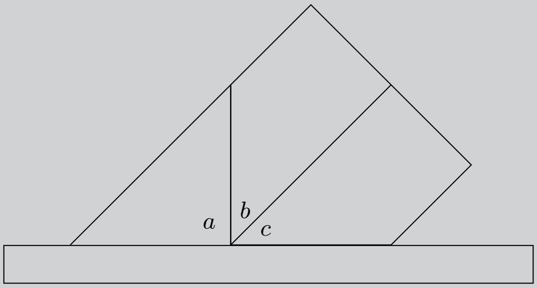
VISIT:
This video shows the different ways to classify triangles.
See video: 2G69 at www.everythingmaths.co.za

Investigation: Interior angles of a triangle

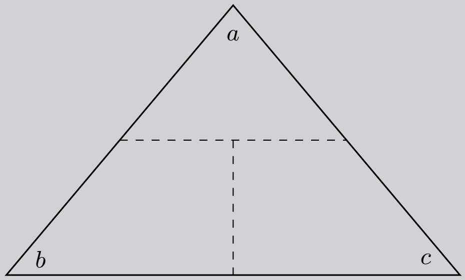


1. On a piece of paper draw a triangle of any size and shape.
2. Cut it out and label the angles $\hat{a}$, $\hat{b}$ and $\hat{c}$ on both sides of the paper.
3. Draw dotted lines as shown and cut along these lines to get three pieces of paper.
4. Place them along your ruler as shown in the figure below.
5. What can we conclude?

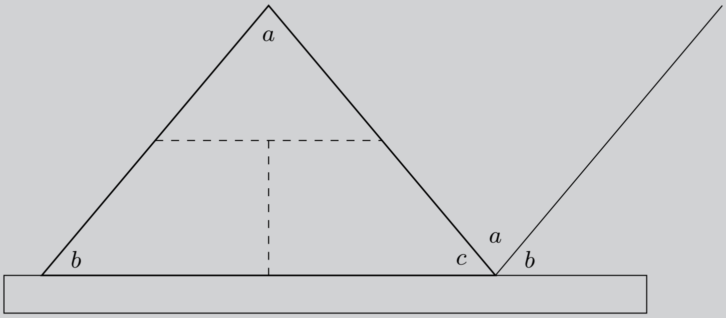
Hint: What is the sum of angles on a straight line?



Investigation: Exterior angles of a triangle



1. On a piece of paper draw a triangle of any size and shape. On another piece of paper, make a copy of the triangle.
2. Cut both out and label the angles of both triangles $\hat{a}$, $\hat{b}$ and $\hat{c}$ on both sides of the paper.
3. Draw dotted lines on **one** triangle as shown and cut along the lines.
4. Place the second triangle and the cut out pieces as shown in the figure below.
5. What can we can conclude?

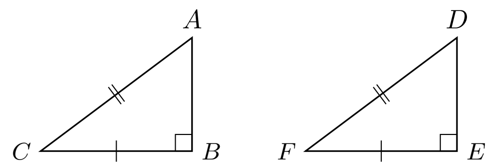
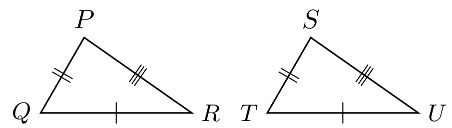
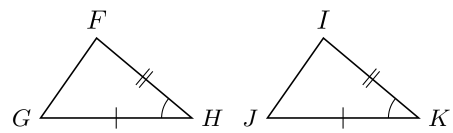
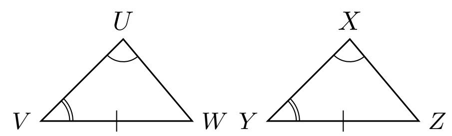


VISIT:
We can use the fact that the angles in a triangle add up to $180°$ to work out the sum of the exterior angles in a pentagon. This video shows you how.
See video: 2G6B at www.everythingmaths.co.za

Congruency EMA5T

Two triangles are congruent if one fits exactly over the other. This means that the triangles have equal corresponding angles and sides. To determine whether two triangles are congruent, it is not necessary to check every side and every angle. We indicate congruency using $\equiv$.

The following table describes the requirements for congruency:

Rule	Description	Diagram
RHS or 90°HS (90°, hypotenuse, side)	If the hypotenuse and one side of a right-angled triangle are equal to the hypotenuse and the corresponding side of another right-angled triangle, then the two triangles are congruent.	 $\triangle ABC \equiv \triangle DEF$
SSS (side, side, side)	If three sides of a triangle are equal in length to the corresponding sides of another triangle, then the two triangles are congruent.	 $\triangle PQR \equiv \triangle STU$
SAS or S∠S (side, angle, side)	If two sides and the **included** angle of a triangle are equal to the corresponding two sides and **included** angle of another triangle, then the two triangles are congruent.	 $\triangle FGH \equiv \triangle IJK$
AAS or ∠∠S (angle, angle, side)	If one side and two angles of a triangle are equal to the corresponding one side and two angles of another triangle, then the two triangles are congruent.	 $\triangle UVW \equiv \triangle XYZ$

The order of letters when labelling congruent triangles is very important.

$$\triangle ABC \equiv \triangle DEF$$

This notation indicates the following properties of the two triangles: $\hat{A} = \hat{D}$, $\hat{B} = \hat{E}$, $\hat{C} = \hat{F}$, $AB = DE$, $AC = DF$ and $BC = EF$.

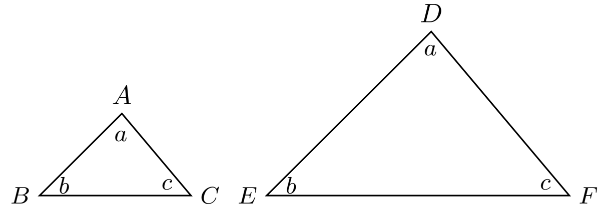
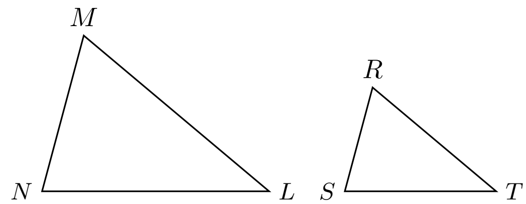
NOTE:
You might see $\cong$ used to show that two triangles are congruent. This is the internationally recognised symbol for congruency.

VISIT:
This video shows some practice examples of finding congruent triangles.
See video: 2G6C at www.everythingmaths.co.za

Similarity EMA5V

Two triangles are similar if one triangle is a scaled version of the other. This means that their corresponding angles are equal in measure and the ratio of their corresponding sides are in proportion. The two triangles have the same shape, but different scales. Congruent triangles are similar triangles, but not all similar triangles are congruent. We use $|||$ to indicate that two triangles are similar.

The following table describes the requirements for similarity:

Rule	Description	Diagram
AAA (angle, angle, angle)	If all three pairs of corresponding angles of two triangles are equal, then the triangles are similar.	 $\hat{A} = \hat{D}$, $\hat{B} = \hat{E}$, $\hat{C} = \hat{F}$ $\therefore \triangle ABC \ ||| \ \triangle DEF$
SSS (side, side, side)	If all three pairs of corresponding sides of two triangles are in proportion, then the triangles are similar.	 $\frac{MN}{RS} = \frac{ML}{RT} = \frac{NL}{ST}$ $\therefore \triangle MNL \ ||| \ \triangle RST$

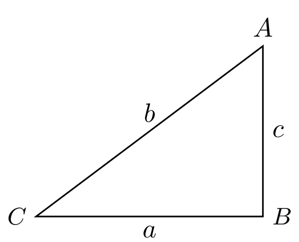
The order of letters for similar triangles is very important. Always label similar triangles in corresponding order. For example,

$$\triangle MNL \ ||| \ \triangle RST \text{ is correct; but}$$
$$\triangle MNL \ ||| \ \triangle RTS \text{ is incorrect.}$$

NOTE:
You might see $\sim$ used to show that two triangles are similar. This is the internationally recognised symbol for similarity.

VISIT:
The following video explains similar triangles.
See video: 2G6D at www.everythingmaths.co.za

The theorem of Pythagoras EMA5W



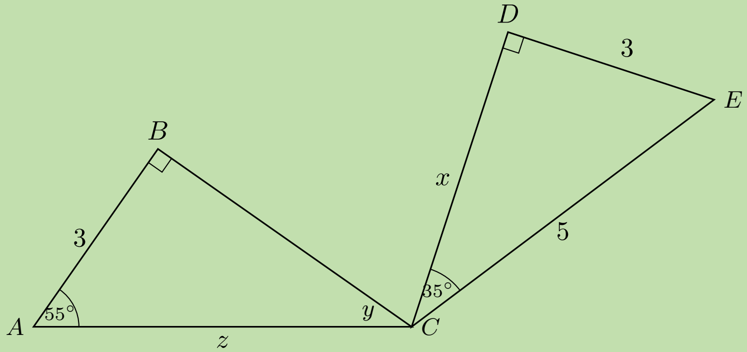
If $\triangle ABC$ is right-angled with $\hat{B} = 90°$, then $b^2 = a^2 + c^2$.
Converse: If $b^2 = a^2 + c^2$, then $\triangle ABC$ is right-angled with $\hat{B} = 90°$.

VISIT:
The following video explains the theorem of Pythagoras and shows some examples of working with the theorem of Pythagoras.
See video: 2G6F at www.everythingmaths.co.za

Worked example 2: Triangles

QUESTION

Determine if the two triangles are congruent. Use the result to find x, $\hat{y}$ and z.



SOLUTION

Step 1: Examine the information given for both triangles

Step 2: Determine whether $\triangle CDE \equiv \triangle CBA$

In $\triangle CDE$:

$$\begin{aligned} \hat{D} + \hat{C} + \hat{E} &= 180° && \text{(sum of } \angle\text{s in } \triangle) \\ 90° + 35° + \hat{E} &= 180° \\ \therefore \hat{E} &= 55° \end{aligned}$$

In $\triangle CDE$ and $\triangle CBA$:

$$\begin{aligned} D\hat{E}C &= B\hat{A}C = 55° && \text{(proved)} \\ C\hat{D}E &= C\hat{B}A = 90° && \text{(given)} \\ DE &= BA = 3 && \text{(given)} \\ \therefore \triangle CDE &\equiv \triangle CBA && \text{(AAS)} \end{aligned}$$

Step 3: Determine the unknown angles and sides

In $\triangle CDE$:

$$\begin{aligned} CE^2 &= DE^2 + CD^2 && \text{(Pythagoras)} \\ 5^2 &= 3^2 + x^2 \\ x^2 &= 16 \\ \therefore x &= 4 \end{aligned}$$

In $\triangle CBA$:

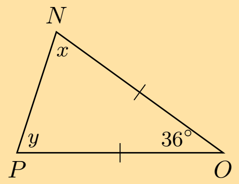
$$\begin{aligned} \hat{B} + \hat{A} + \hat{y} &= 180° \qquad \text{(sum of } \angle\text{s in } \triangle) \\ 90° + 55° + \hat{y} &= 180° \\ \therefore \hat{y} &= 35° \end{aligned}$$

$$\begin{aligned} \triangle CDE &\equiv \triangle CBA \qquad \text{(proved)} \\ \therefore CE &= CA \\ \therefore z &= 5 \end{aligned}$$

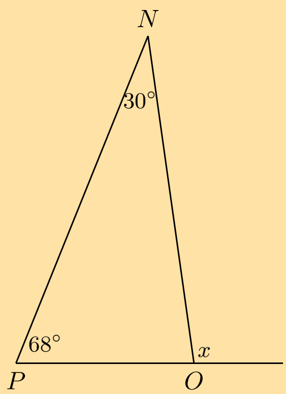
Exercise 7 – 2:

1. Calculate the unknown variables in each of the following figures.

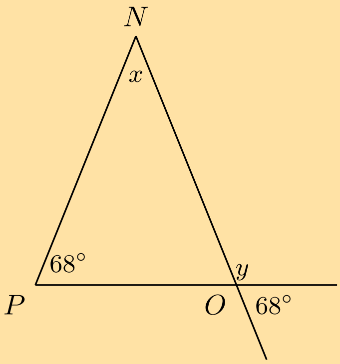
   a)



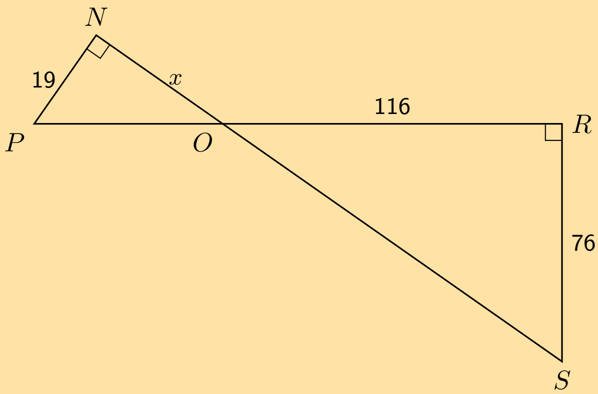
   b)



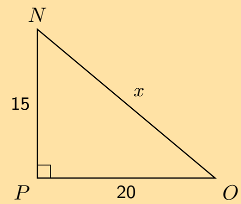
   c)



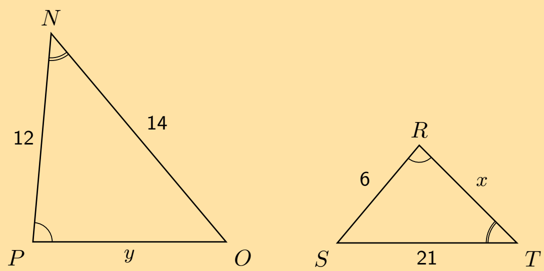
d)



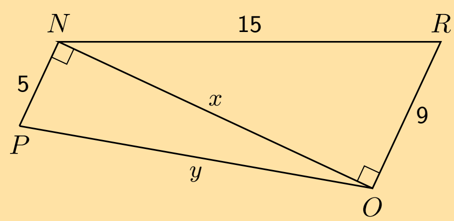
e)



f)



g)



2. Given the following diagrams:

Diagram A

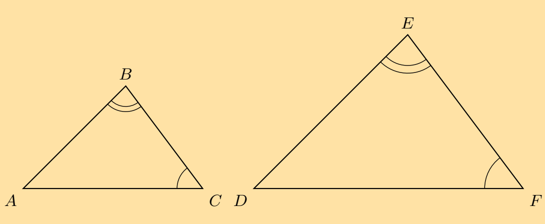
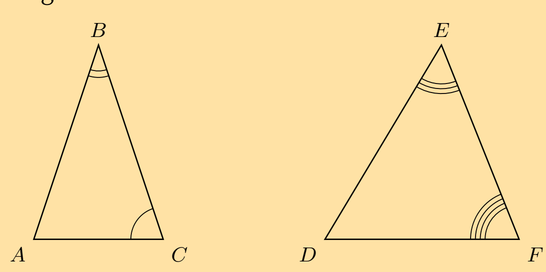


Diagram B



Which diagram correctly gives a pair of similar triangles?

3. Given the following diagrams:

Diagram A

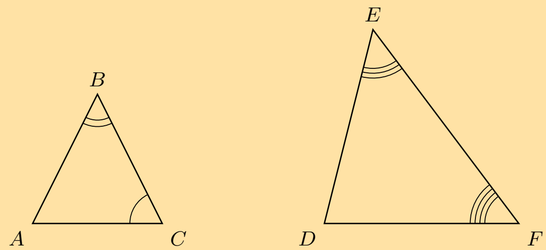
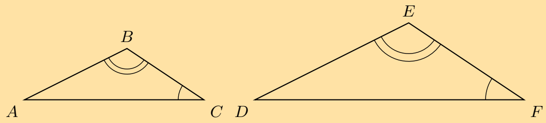
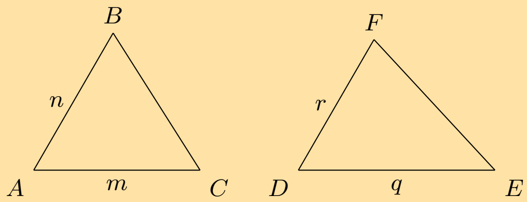


Diagram B



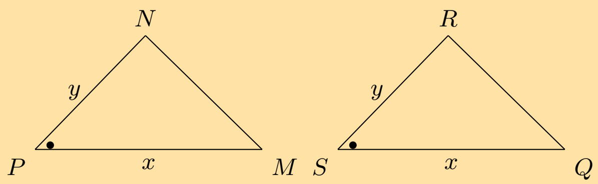
Which diagram correctly gives a pair of similar triangles?

4. Have a look at the following triangles, which are drawn to scale:



Are the two triangles congruent? If so state the reason and use the correct notation to state that they are congruent.

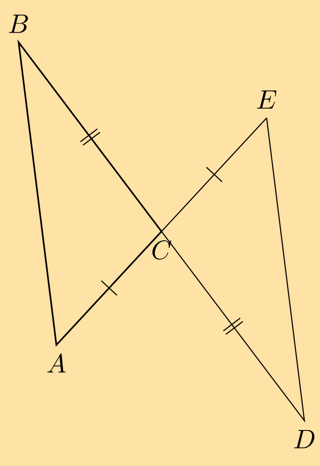
5. Have a look at the following triangles, which are drawn to scale:



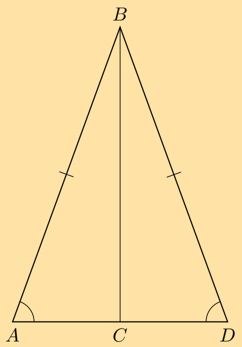
Are the two triangles congruent? If so state the reason and use the correct notation to state that they are congruent.

6. State whether the following pairs of triangles are congruent or not. Give reasons for your answers. If there is not enough information to make a decision, explain why.

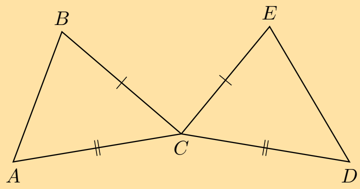
a)



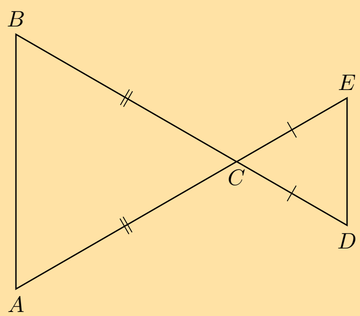
b)



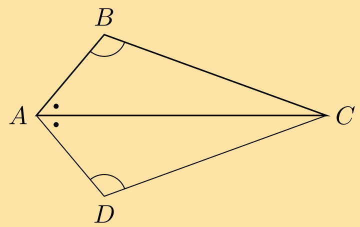
c)



d)



e)



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1a. 2G6G 1b. 2G6H 1c. 2G6J 1d. 2G6K 1e. 2G6M 1f. 2G6N 1g. 2G6P 2. 2G6Q
3. 2G6R 4. 2G6S 5. 2G6T 6a. 2G6V 6b. 2G6W 6c. 2G6X 6d. 2G6Y 6e. 2G6Z

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7.3 Quadrilaterals EMA5X

DEFINITION: *Quadrilateral*

A quadrilateral is a closed shape consisting of four straight line segments.

NOTE:
The interior angles of a quadrilateral add up to 360°.

Parallelogram EMA5Y

DEFINITION: *Parallelogram*

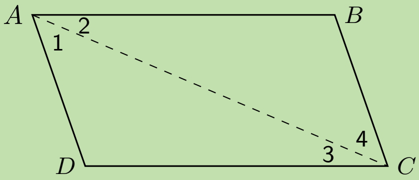
A parallelogram is a quadrilateral with both pairs of opposite sides parallel.

Worked example 3: Properties of a parallelogram

QUESTION

$ABCD$ is a parallelogram with $AB \parallel DC$ and $AD \parallel BC$. Show that:

1. $AB = DC$ and $AD = BC$
2. $\hat{A} = \hat{C}$ and $\hat{B} = \hat{D}$



SOLUTION

Step 1: Connect AC to form $\triangle ABC$ and $\triangle CDA$
Redraw the diagram and draw line AC.

Step 2: Use properties of parallel lines to indicate all equal angles on the diagram
On your diagram mark all the equal angles.

Step 3: Prove $\triangle ABC \equiv \triangle CDA$
In $\triangle ABC$ and $\triangle CDA$:

$$\begin{aligned}
\hat{A}_2 &= \hat{C}_3 && \text{(alt } \angle\text{s; } AB \parallel DC) \\
\hat{C}_4 &= \hat{A}_1 && \text{(alt } \angle\text{s; } BC \parallel AD) \\
AC & && \text{(common side)} \\
\therefore \triangle ABC &\equiv \triangle CDA && \text{(AAS)} \\
\therefore AB = CD & \quad \text{and} \quad BC = DA &&
\end{aligned}$$

$\therefore$ Opposite sides of a parallelogram have equal length.

We have already shown $\hat{A}_2 = \hat{C}_3$ and $\hat{A}_1 = \hat{C}_4$. Therefore,

$$\hat{A} = \hat{A}_1 + \hat{A}_2 = \hat{C}_3 + \hat{C}_4 = \hat{C}$$

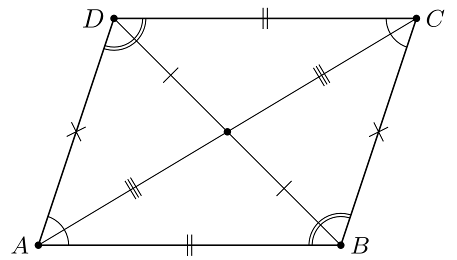
Furthermore,

$$\hat{B} = \hat{D} \quad (\triangle ABC \equiv \triangle CDA)$$

Therefore opposite angles of a parallelogram are equal.

Summary of the properties of a parallelogram:

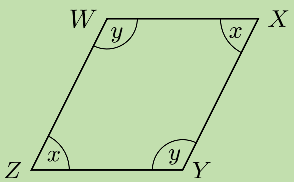
- Both pairs of opposite sides are parallel.
- Both pairs of opposite sides are equal in length.
- Both pairs of opposite angles are equal.
- Both diagonals bisect each other.



Worked example 4: Proving a quadrilateral is a parallelogram

QUESTION

Prove that if both pairs of opposite angles in a quadrilateral are equal, the quadrilateral is a parallelogram.



SOLUTION

Step 1: Find the relationship between $\hat{x}$ and $\hat{y}$

In $WXYZ$:

$$\begin{aligned}
\hat{W} = \hat{Y} &= \hat{y} && \text{(given)}\\
\hat{Z} = \hat{X} &= \hat{x} && \text{(given)}\\
\hat{W} + \hat{X} + \hat{Y} + \hat{Z} &= 360° && \text{(sum of } \angle\text{s in a quad)}\\
\therefore 2\hat{x} + 2\hat{y} &= 360°\\
\therefore \hat{x} + \hat{y} &= 180°\\
\hat{W} + \hat{Z} &= \hat{x} + \hat{y}\\
&= 180°
\end{aligned}$$

But these are co-interior angles between lines WX and ZY. Therefore $WX \parallel ZY$.

Step 2: Find parallel lines

Similarly $\hat{W} + \hat{X} = 180°$. These are co-interior angles between lines XY and WZ. Therefore $XY \parallel WZ$.

Both pairs of opposite sides of the quadrilateral are parallel, therefore $WXYZ$ is a parallelogram.

Investigation: Proving a quadrilateral is a parallelogram

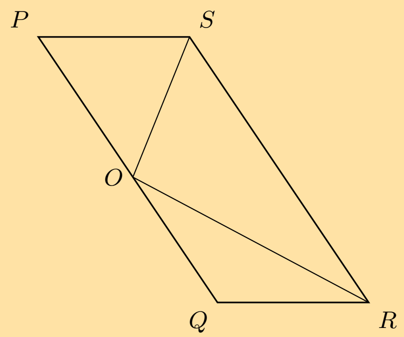
1. Prove that if both pairs of opposite sides of a quadrilateral are equal, then the quadrilateral is a parallelogram.
2. Prove that if the diagonals of a quadrilateral bisect each other, then the quadrilateral is a parallelogram.
3. Prove that if one pair of opposite sides of a quadrilateral are both equal and parallel, then the quadrilateral is a parallelogram.

A quadrilateral is a parallelogram if:

- Both pairs of opposite sides are parallel.
- Both pairs of opposite sides are equal.
- Both pairs of opposite angles are equal.
- The diagonals bisect each other.
- One pair of opposite sides are both equal and parallel.

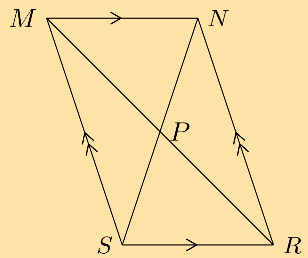
Exercise 7 – 3:

1. $PQRS$ is a parallelogram. $PS = OS$ and $QO = QR$. $S\hat{O}R = 96°$ and $Q\hat{O}R = x$.



   a) Find with reasons, two other angles equal to x.
   b) Write $\hat{P}$ in terms of x.
   c) Calculate the value of x.

2. Prove that the diagonals of parallelogram $MNRS$ bisect one another at P.



   Hint: Use congruency.

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Rectangle EMA5Z

DEFINITION: *Rectangle*

A rectangle is a parallelogram that has all four angles equal to $90°$.

A rectangle has all the properties of a parallelogram:

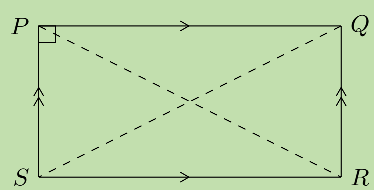
- Both pairs of opposite sides are parallel.
- Both pairs of opposite sides are equal in length.
- Both pairs of opposite angles are equal.
- Both diagonals bisect each other.

It also has the following special property:

Worked example 5: Special property of a rectangle

QUESTION

$PQRS$ is a rectangle. Prove that the diagonals are of equal length.



SOLUTION

Step 1: Connect P to R and Q to S to form $\triangle PSR$ and $\triangle QRS$

Step 2: Use the definition of a rectangle to fill in on the diagram all equal angles and sides

Step 3: Prove $\triangle PSR \equiv \triangle QRS$

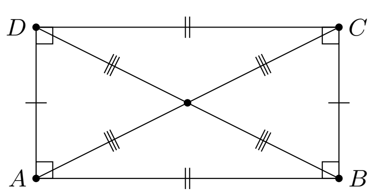
In $\triangle PSR$ and $\triangle QRS$:

$$\begin{aligned} PS &= QR && \text{(opp sides of rectangle)} \\ SR & && amp; \text{(common side)} \\ P\hat{S}R &= Q\hat{R}S = 90° && (\angle\text{s of rectangle}) \\ \therefore \triangle PSR &\equiv \triangle QRS && \text{(RHS)} \\ \text{Therefore } PR &= QS \end{aligned}$$

The diagonals of a rectangle are of equal length.

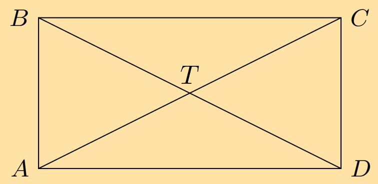
Summary of the properties of a rectangle:

- Both pairs of opposite sides are parallel.
- Both pairs of opposite sides are of equal length.
- Both pairs of opposite angles are equal.
- Both diagonals bisect each other.
- Diagonals are equal in length.
- All interior angles are equal to 90°



Exercise 7 – 4:

1. $ABCD$ is a quadrilateral. Diagonals AC and BD intersect at T. $AC = BD$, $AT = TC$, $DT = TB$.



Prove that:

a) $ABCD$ is a parallelogram
b) $ABCD$ is a rectangle

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Rhombus EMA62

DEFINITION: *Rhombus*

A rhombus is a parallelogram with all four sides of equal length.

A rhombus has all the properties of a parallelogram:

- Both pairs of opposite sides are parallel.
- Both pairs of opposite sides are equal in length.
- Both pairs of opposite angles are equal.
- Both diagonals bisect each other.

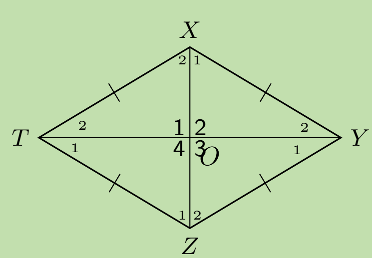
It also has two special properties:

Worked example 6: Special properties of a rhombus

QUESTION

$XYZT$ is a rhombus. Prove that:

1. the diagonals bisect each other perpendicularly;
2. the diagonals bisect the interior angles.



SOLUTION

Step 1: Use the definition of a rhombus to fill in on the diagram all equal angles and sides

Step 2: Prove $\triangle XTO \equiv \triangle ZTO$

$$
\begin{aligned}
XT &= ZT && \text{(sides of rhombus)} \\
TO & amp; && \text{(common side)} \\
XO &= OZ && \text{(diags of rhombus)} \\
\therefore \triangle XTO &\equiv \triangle ZTO && \text{(SSS)} \\
\therefore \hat{O}_1 &= \hat{O}_4 && \\
\text{But } \hat{O}_1 + \hat{O}_4 &= 180° && (\angle\text{s on a str line}) \\
\therefore \hat{O}_1 &= \hat{O}_4 = 90° &&
\end{aligned}
$$

We can further conclude that $\hat{O}_1 = \hat{O}_2 = \hat{O}_3 = \hat{O}_4 = 90°$.

Therefore the diagonals bisect each other perpendicularly.

Step 3: Use properties of congruent triangles to prove diagonals bisect interior angles

$$
\begin{aligned}
\hat{X}_2 &= \hat{Z}_1 && (\triangle XTO \equiv \triangle ZTO) \\
\text{and } \hat{X}_2 &= \hat{Z}_2 && (\text{alt } \angle\text{s; } XT \parallel YZ) \\
\therefore \hat{Z}_1 &= \hat{Z}_2 &&
\end{aligned}
$$

Therefore diagonal XZ bisects $\hat{Z}$. Similarly, we can show that XZ also bisects $\hat{X}$; and that diagonal TY bisects $\hat{T}$ and $\hat{Y}$.

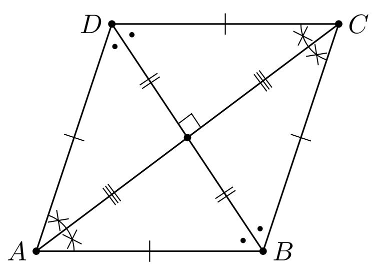
We conclude that the diagonals of a rhombus bisect the interior angles.

To prove a parallelogram is a rhombus, we need to show any one of the following:

- All sides are equal in length.
- Diagonals intersect at right angles.
- Diagonals bisect interior angles.

Summary of the properties of a rhombus:

- Both pairs of opposite sides are parallel.
- Both pairs of opposite sides are equal in length.
- Both pairs of opposite angles are equal.
- Both diagonals bisect each other.
- All sides are equal in length.
- The diagonals bisect each other at 90°
- The diagonals bisect both pairs of opposite angles.



Square

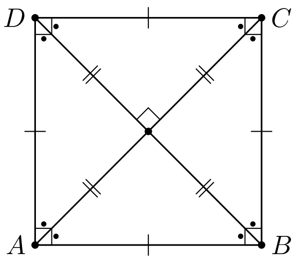
EMA63

DEFINITION: *Square*

A square is a rhombus with all four interior angles equal to $90°$
OR
A square is a rectangle with all four sides equal in length.

A square has all the properties of a rhombus:

- Both pairs of opposite sides are parallel.
- Both pairs of opposite sides are equal in length.
- Both pairs of opposite angles are equal.
- Both diagonals bisect each other.
- All sides are equal in length.
- The diagonals bisect each other at $90°$
- The diagonals bisect both pairs of opposite angles.



It also has the following special properties:

- All interior angles equal $90°$.
- Diagonals are equal in length.
- Diagonals bisect both pairs of interior opposite angles (i.e. all are $45°$).

To prove a parallelogram is a square, we need to show either one of the following:

- It is a rhombus (all four sides of equal length) with interior angles equal to $90°$.
- It is a rectangle (interior angles equal to $90°$).

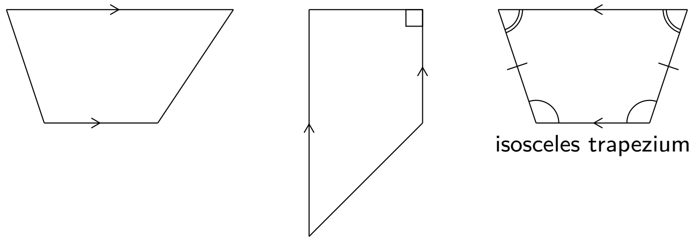
Trapezium EMA64

DEFINITION: *Trapezium*

A trapezium is a quadrilateral with one pair of opposite sides parallel.

NOTE:
A trapezium is sometimes called a trapezoid.

Some examples of trapeziums are given below:



Kite EMA65

DEFINITION: *Kite*

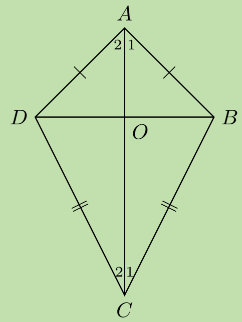
A kite is a quadrilateral with two pairs of adjacent sides equal.

Worked example 7: Properties of a kite

QUESTION

$ABCD$ is a kite with $AD = AB$ and $CD = CB$. Prove that:

1. $A\hat{D}C = A\hat{B}C$
2. Diagonal AC bisects $\hat{A}$ and $\hat{C}$



SOLUTION

Step 1: Prove $\triangle ADC \equiv \triangle ABC$

In $\triangle ADC$ and $\triangle ABC$:

$$\begin{aligned} AD &= AB && \text{(given)} \\ CD &= CB && \text{(given)} \\ AC & && \text{(common side)} \\ \therefore \triangle ADC &\equiv \triangle ABC && \text{(SSS)} \\ \therefore A\hat{D}C &= A\hat{B}C \end{aligned}$$

Therefore one pair of opposite angles are equal in kite $ABCD$.

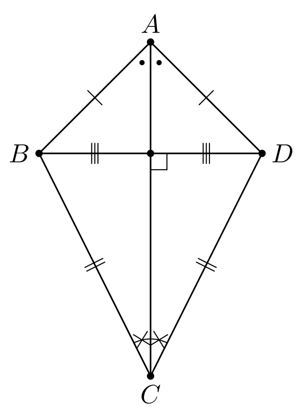
Step 2: Use properties of congruent triangles to prove AC bisects $\hat{A}$ and $\hat{C}$

$$\begin{aligned} \hat{A}_1 &= \hat{A}_2 \quad (\triangle ADC \equiv \triangle ABC) \\ \text{and } \hat{C}_1 &= \hat{C}_2 \quad (\triangle ADC \equiv \triangle ABC) \end{aligned}$$

Therefore diagonal AC bisects $\hat{A}$ and $\hat{C}$.

We conclude that the diagonal between the equal sides of a kite bisects the two interior angles and is an axis of symmetry.

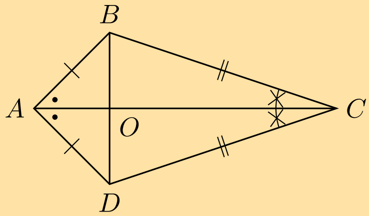
Summary of the properties of a kite:



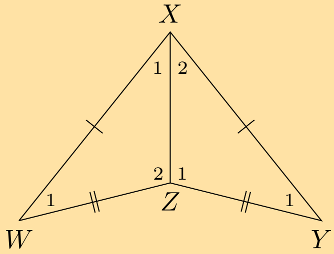
- Diagonal between equal sides bisects the other diagonal.
- One pair of opposite angles are equal (the angles between unequal sides).
- Diagonal between equal sides bisects the interior angles and is an axis of symmetry.
- Diagonals intersect at $90°$

Exercise 7 – 5:

1. Use the sketch of quadrilateral $ABCD$ to prove the diagonals of a kite are perpendicular to each other.



2. Explain why quadrilateral $WXYZ$ is a kite. Write down all the properties of quadrilateral $WXYZ$.



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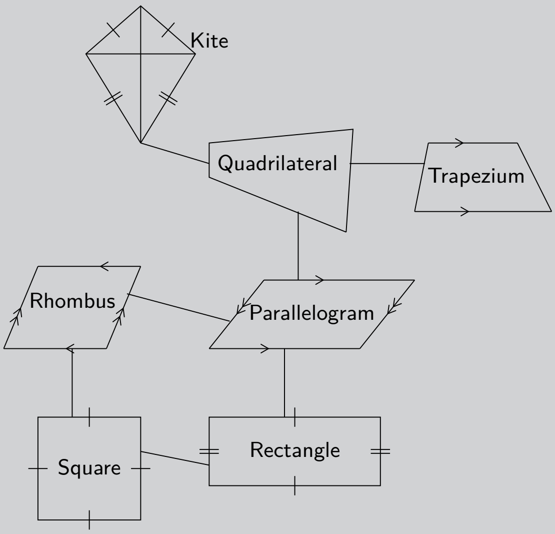
VISIT:
This video provides a summary of the different types of quadrilaterals and their properties.
See video: 2G77 at www.everythingmaths.co.za

Investigation: Relationships between the different quadrilaterals

Heather has drawn the following diagram to illustrate her understanding of the relationships between the different quadrilaterals. The following diagram summarises the different types of special quadrilaterals.



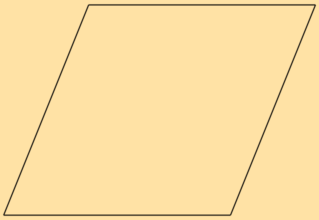
1. Explain her possible reasoning for structuring the diagram as shown.
2. Design your own diagram to show the relationships between the different quadrilaterals and write a short explanation of your design.

Exercise 7 – 6:

1. The following shape is drawn **to scale**:

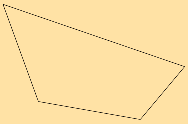
Give the most specific name for the shape.

2. The following shape is drawn **to scale**:

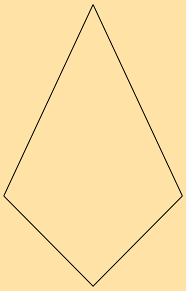


Give the most specific name for the shape.

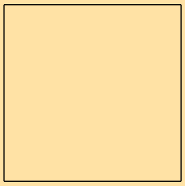
3. Based on the shape that you see list the all the names of the shape. The figure is drawn to scale.



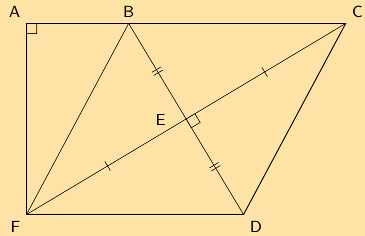
4. Based on the shape that you see list the all the names of the shape. The figure is drawn to scale.



5. Based on the shape that you see list the all the names of the shape. The figure is drawn to scale.



6. Find the area of $ACDF$ if $AB = 8,\ BF = 17,\ FE = EC,\ BE = ED,\ \hat{A} = 90°,\ C\hat{E}D = 90°$.



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1. 2G78 2. 2G79 3. 2G7B 4. 2G7C 5. 2G7D 6. 2G7F

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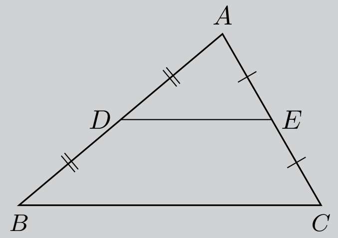
7.4 The mid-point theorem EMA66

Investigation: Proving the mid-point theorem

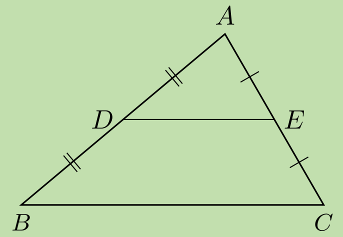


1. Draw a large scalene triangle on a sheet of paper.
2. Name the vertices A, B and C. Find the mid-points (D and E) of two sides and connect them.
3. Cut out $\triangle ABC$ and cut along line DE.
4. Place $\triangle ADE$ on quadrilateral $BDEC$ with vertex E on vertex C. Write down your observations.
5. Shift $\triangle ADE$ to place vertex D on vertex B. Write down your observations.
6. What do you notice about the lengths DE and BC?
7. Make a conjecture regarding the line joining the mid-point of two sides of a triangle.

Worked example 8: Mid-point theorem

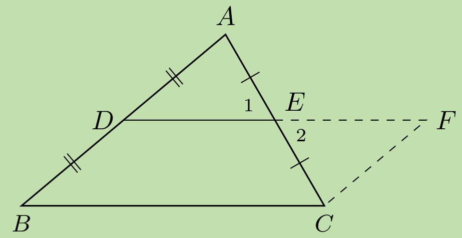
QUESTION

Prove that the line joining the mid-points of two sides of a triangle is parallel to the third side and equal to half the length of the third side.



SOLUTION

Step 1: Extend DE to F so that $DE = EF$ and join FC



Step 2: Prove $BCFD$ is a parallelogram

In $\triangle EAD$ and $\triangle ECF$:

$$\begin{aligned} \hat{E}_1 &= \hat{E}_2 && \text{(vert opp } \angle\text{s} = \text{)} \\ AE &= CE && \text{(given)} \\ DE &= EF && \text{(by construction)} \\ \therefore \triangle EAD &\equiv \triangle ECF && \text{(SAS)} \\ \therefore A\hat{D}E &= C\hat{F}E \end{aligned}$$

But these are alternate interior angles, therefore $BD \parallel FC$

$$\begin{aligned} BD &= DA && \text{(given)} \\ DA &= FC && (\triangle EAD \equiv \triangle ECF) \\ \therefore BD &= FC \\ \therefore BCFD &\text{ is a parallelogram} && \text{(one pair opp. sides } = \text{ and } \parallel \text{)} \end{aligned}$$

Therefore $DE \parallel BC$.

We conclude that the line joining the two mid-points of two sides of a triangle is parallel to the third side.

Step 3: Use properties of parallelogram $BCFD$ to prove that $DE = \frac{1}{2}BC$

$$\begin{aligned} DF &= BC && \text{(opp sides } \parallel \text{m)} \\ \text{and } DF &= 2\,(DE) && \text{(by construction)} \\ \therefore 2DE &= BC \\ \therefore DE &= \tfrac{1}{2}BC \end{aligned}$$

We conclude that the line joining the mid-point of two sides of a triangle is equal to half the length of the third side.

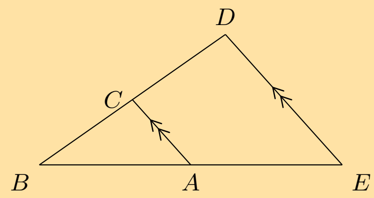
Converse

The converse of this theorem states: If a line is drawn through the mid-point of a side of a triangle parallel to the second side, it will bisect the third side.

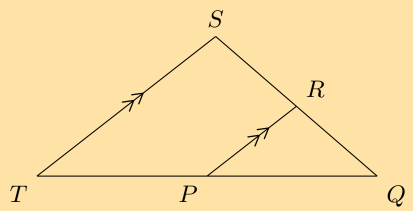
VISIT:
You can use GeoGebra to show that the converse of the mid-point theorem is true.

Exercise 7 – 7:

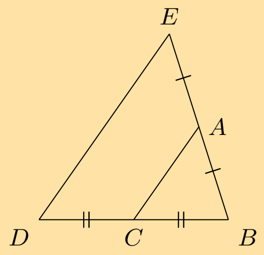
1. Points C and A are the mid-points on lines BD and BE. Study $\triangle EDB$ carefully. Identify the third side of this triangle, using the information as shown, together with what you know about the mid-point theorem. Name the third side by its endpoints, e.g., FG.



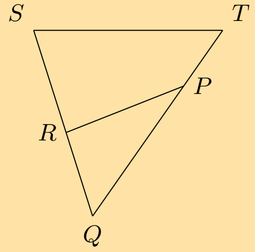
2. Points R and P are the mid-points on lines QS and QT. Study $\triangle TSQ$ carefully. Identify the third side of this triangle, using the information as shown, together with what you know about the mid-point theorem. Name the third side by its endpoints, e.g., FG.



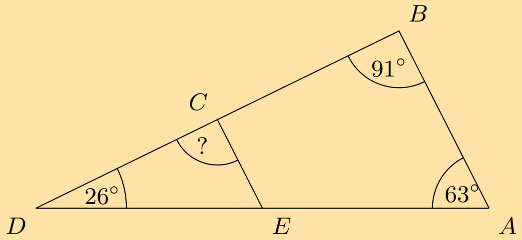
3. Points C and A are given on the lines BD and BE. Study the triangle carefully, then identify and name the parallel lines.



4. Points R and P are given on the lines QS and QT. Study the triangle carefully, then identify and name the parallel lines.



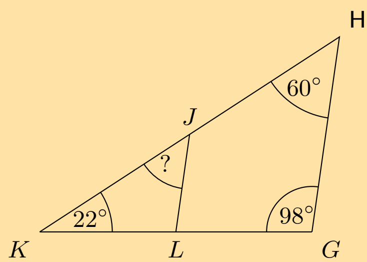
5. The figure below shows a large triangle with vertices A, B and D, and a smaller triangle with vertices at C, D and E. Point C is the mid-point of BD and point E is the mid-point of AD.



   a) Three angles are given: $\hat{A} = 63°$, $\hat{B} = 91°$ and $\hat{D} = 26°$; determine the value of $D\hat{C}E$.

   b) The two triangles in this question are similar triangles. Complete the following statement correctly by giving the three vertices in the correct order (there is only one correct answer).
   $\triangle DEC \; ||| \; \triangle ?$

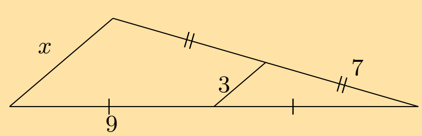
6. The figure below shows a large triangle with vertices G, H and K, and a smaller triangle with vertices at J, K and L. Point J is the mid-point of HK and point L is the mid-point of GK.



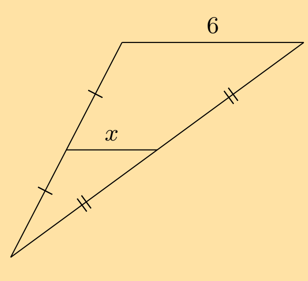
   a) Three angles are given: $\hat{G} = 98°$, $\hat{H} = 60°$, and $\hat{K} = 22°$; determine the value of $K\hat{J}L$.

   b) The two triangles in this question are similar triangles. Complete the following statement correctly by giving the three vertices in the correct order (there is only one correct answer).
   $\triangle HKG \; ||| \; \triangle ?$

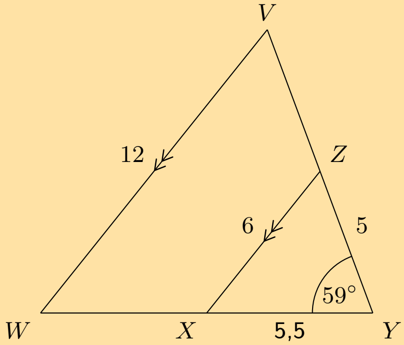
7. Consider the triangle in the diagram below. There is a line crossing through a large triangle. Notice that some lines in the figure are marked as equal to each other. One side of the triangle has a given length of 3. Determine the value of x.



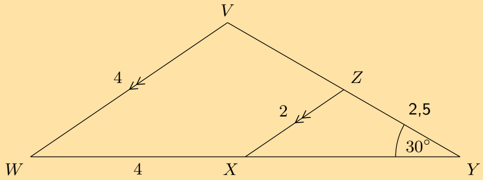
8. Consider the triangle in the diagram below. There is a line crossing through a large triangle. Notice that some lines in the figure are marked as equal to each other. One side of the triangle has a given length of 6. Determine the value of x.



9. In the figure below, $VW \parallel ZX$, as labelled. Furthermore, the following lengths and angles are given: $VW = 12$; $ZX = 6$; $XY = 5{,}5$; $YZ = 5$ and $\hat{V} = 59°$. The figure is drawn to scale. Determine the length of WY.

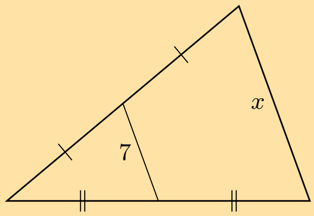


10. In the figure below, $VW \parallel ZX$, as labelled. Furthermore, the following lengths and angles are given: $VW = 4$; $ZX = 2$; $WX = 4$; $YZ = 3{,}5$ and $\hat{Y} = 30°$. The figure is drawn to scale. Determine the length of XY.

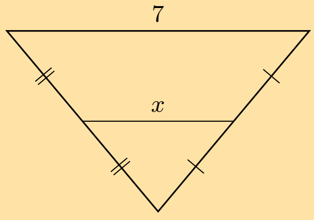


11. Find x and y in the following:

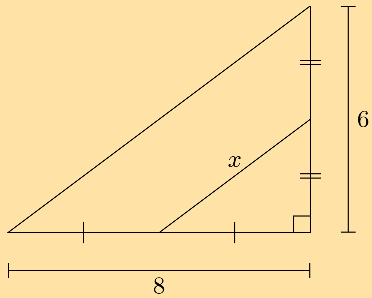
a)



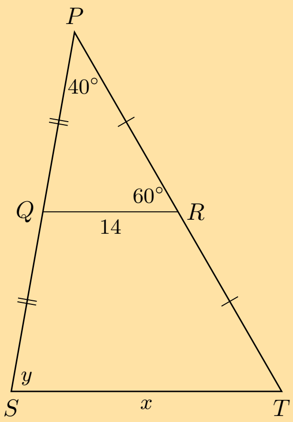
b)



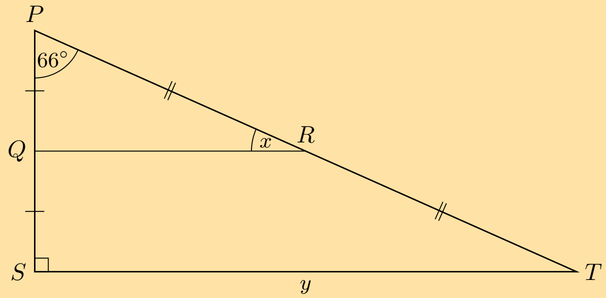
c)



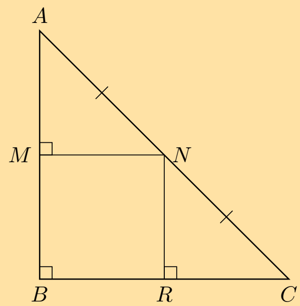
d)



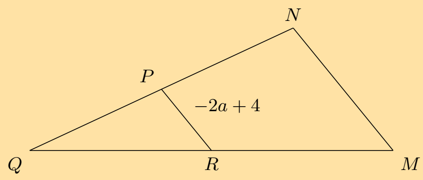
e) In the following diagram $PQ = 2{,}5$ and $RT = 6{,}5$.



12. Show that M is the mid-point of AB and that $MN = RC$.



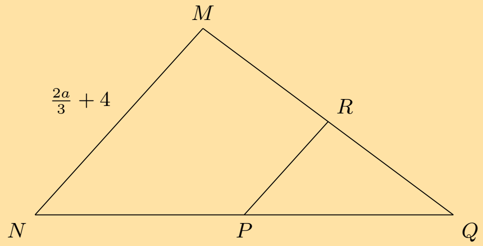
13. In the diagram below, P is the mid-point of NQ and R is the mid-point of MQ. The segment inside of the large triangle is labelled with a length of $-2a + 4$.



a) Calculate the value of MN in terms of a.

b) You are now told that MN has a length of 18. What is the value of a? Give your answer as a fraction.

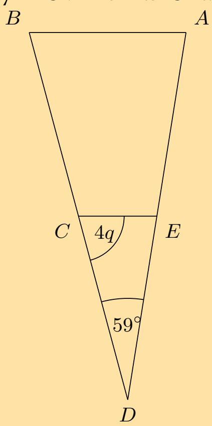
14. In the diagram below, P is the mid-point of NQ and R is the mid-point of MQ. One side of the triangle has a given length of $\frac{2a}{3} + 4$.



a) Find the value of PR in terms of a.

b) You are now told that PR has a length of 8. What is the value of a?

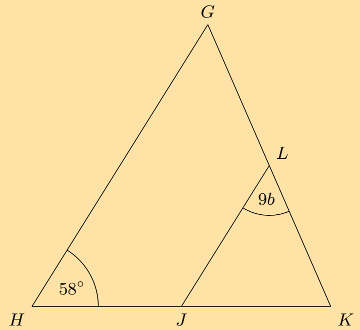
15. The figure below shows $\triangle ABD$ crossed by EC. Points C and E bisect their respective sides of the triangle.



a) The angles $\hat{D} = 59°$ and $E\hat{C}D = 4q$ are given; determine the value of $\hat{A}$ in terms of q.

b) You are now told that $E\hat{C}D$ has a measure of $72°$. Calculate for the value of q.

16. The figure below shows $\triangle GHK$ crossed by LJ. Points J and L bisect their respective sides of the triangle.



a) Given the angles $\hat{H} = 58°$ and $K\hat{L}J = 9b$, determine the value of $\hat{K}$ in terms of b.

b) You are now told that $\hat{K}$ has a measure of $74°$. Solve for the value of b. Give your answer as a fraction.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2G7G	2. 2G7H	3. 2G7J	4. 2G7K	5. 2G7M	6. 2G7N	7. 2G7P	8. 2G7Q
9. 2G7R	10. 2G7S	11a. 2G7T	11b. 2G7V	11c. 2G7W	11d. 2G7X	11e. 2G7Y	12. 2G7Z
13. 2G82	14. 2G83	15. 2G84	16. 2G85				

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7.5 Chapter summary

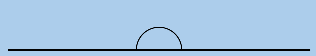
See presentation: 2G86 at www.everythingmaths.co.za

- A quadrilateral is a closed shape consisting of four straight line segments.
- A parallelogram is a quadrilateral with both pairs of opposite sides parallel.
  - Both pairs of opposite sides are equal in length.
  - Both pairs of opposite angles are equal.
  - Both diagonals bisect each other.
- A rectangle is a parallelogram that has all four angles equal to $90°$
  - Both pairs of opposite sides are parallel.
  - Both pairs of opposite sides are equal in length.
  - The diagonals bisect each other.
  - The diagonals are equal in length.
  - All interior angles are equal to $90°$.
- A rhombus is a parallelogram that has all four sides equal in length.
  - Both pairs of opposite sides are parallel.
  - All sides are equal in length.
  - Both pairs of opposite angles are equal.
  - The diagonals bisect each other at $90°$.
  - The diagonals of a rhombus bisect both pairs of opposite angles.
- A square is a rhombus that has all four interior angles equal to $90°$.
  - Both pairs of opposite sides are parallel.
  - The diagonals bisect each other at $90°$.
  - All interior angles are equal to $90°$.
  - The diagonals are equal in length.
  - The diagonals bisect both pairs of interior opposite angles (i.e. all are $45°$
- A trapezium is a quadrilateral with one pair of opposite sides parallel.
- A kite is a quadrilateral with two pairs of adjacent sides equal.
  - One pair of opposite angles are equal (the angles are between unequal sides).
  - The diagonal between equal sides bisects the other diagonal.
  - The diagonal between equal sides bisects the interior angles.
  - The diagonals intersect at $90°$.
- The mid-point theorem states that the line joining the mid-points of two sides of a triangle is parallel to the third side and equal to half the length of the third side.

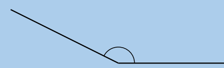
End of chapter Exercise 7 – 8:

1. Identify the types of angles shown below:

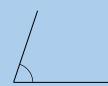
   a)



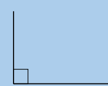
b)



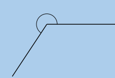
c)



d)



e)



f) An angle of 91°

g) An angle of 180°

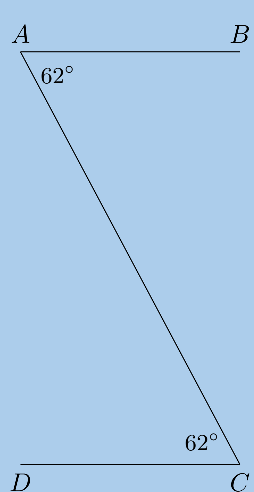
h) An angle of 210°

2. Assess whether the following statements are true or false. If the statement is false, explain why:

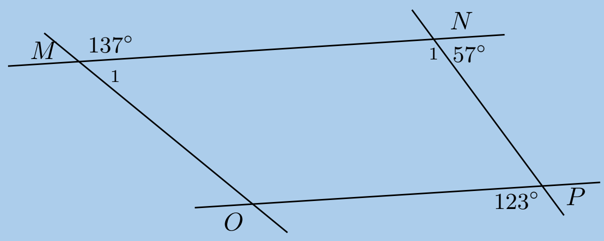
   a) A trapezium is a quadrilateral with two pairs of opposite sides that are parallel.
   b) Both diagonals of a parallelogram bisect each other.
   c) A rectangle is a parallelogram that has all interior angles equal to 90°.
   d) Two adjacent sides of a rhombus have different lengths.
   e) The diagonals of a kite intersect at right angles.
   f) All squares are parallelograms.
   g) A rhombus is a kite with a pair of equal, opposite sides.
   h) The diagonals of a parallelogram are axes of symmetry.
   i) The diagonals of a rhombus are equal in length.
   j) Both diagonals of a kite bisect the interior angles.

3. Find all pairs of parallel lines in the following figures, giving reasons in each case.

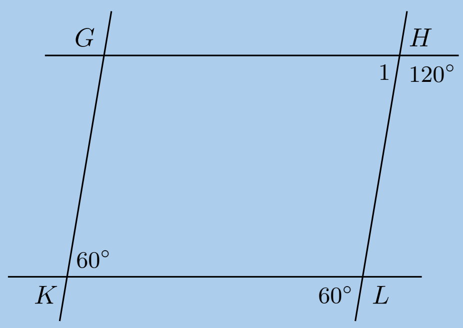
   a)



b)

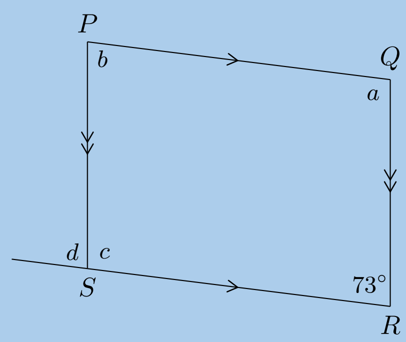


c)

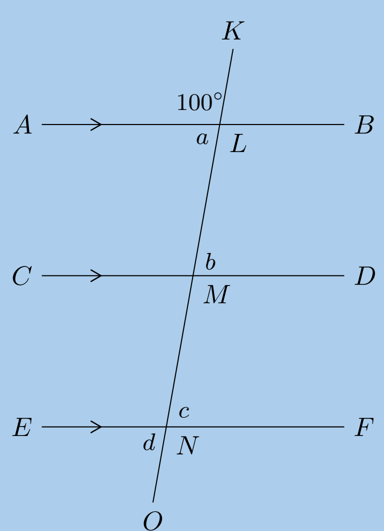


4. Find angles a, b, c and d in each case, giving reasons:

a)



b)



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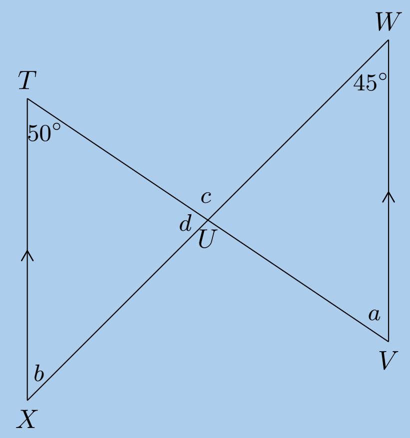
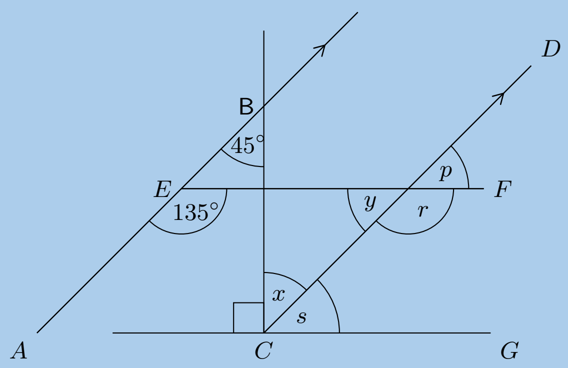


5. Given the figure below.



a) Find each of the unknown angles marked in the figure below. Find a reason that leads to the answer in a single step.
b) Based on the results for the angles above, is $EF \parallel CG$?

6. Given the following diagrams:

Diagram A

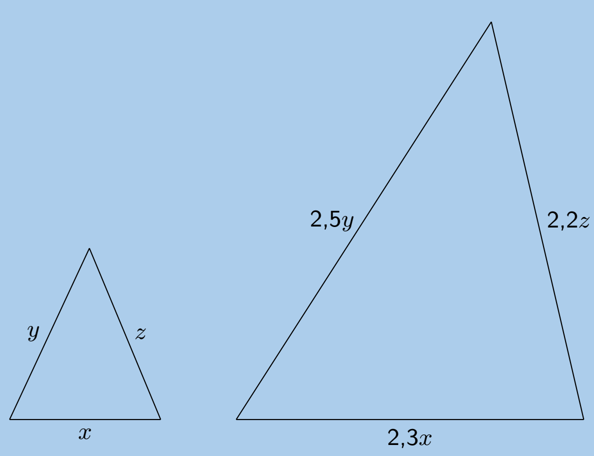
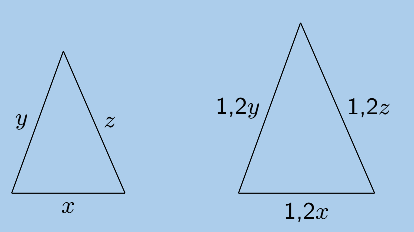


Diagram B



Which diagram correctly gives a pair of similar triangles?

7. Given the following diagrams:

Diagram A

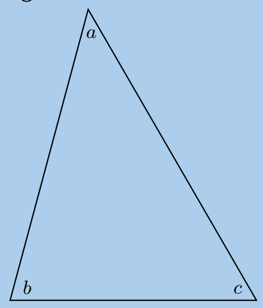
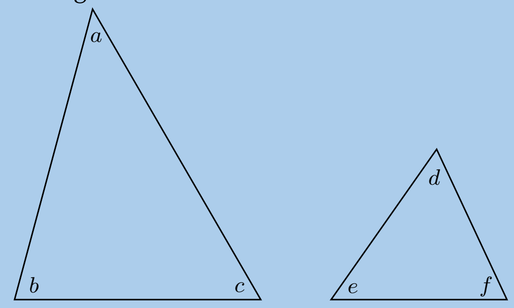
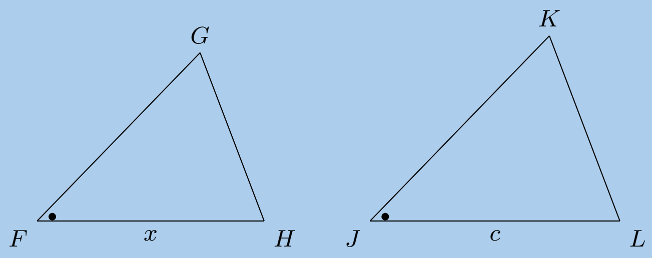


Diagram B



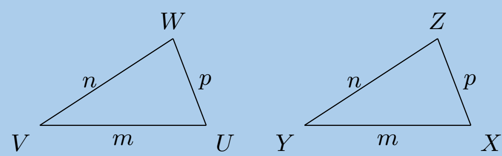
Which diagram correctly gives a pair of similar triangles?

8. Have a look at the following triangles, which are drawn to scale:



Are the triangles congruent? If so state the reason and use correct notation to state that they are congruent.

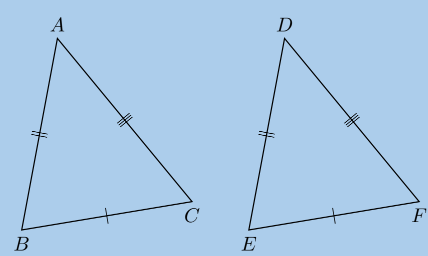
9. Have a look at the following triangles, which are drawn to scale:



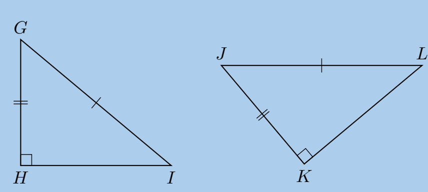
Are the triangles congruent? If so state the reason and use correct notation to state that they are congruent.

10. Say which of the following pairs of triangles are congruent with reasons.

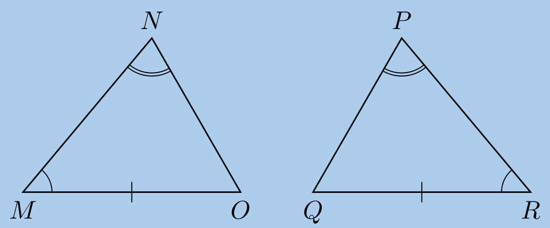
a)



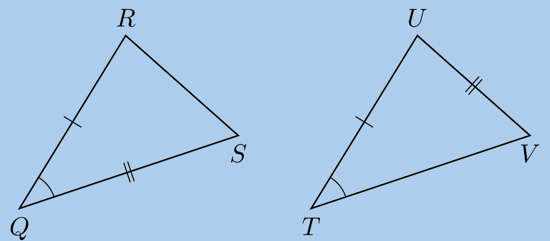
b)



c)

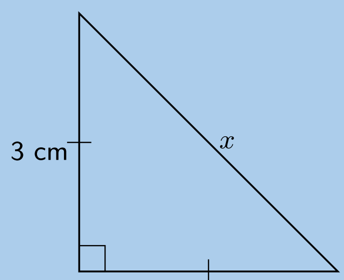


d)

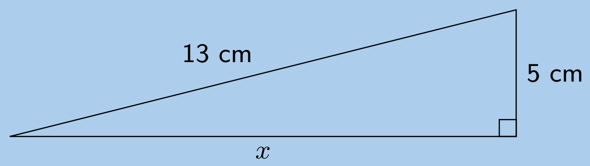


11. Using the theorem of Pythagoras, calculate the length x:

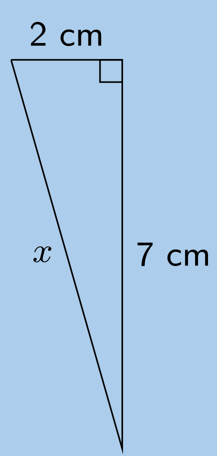
a)



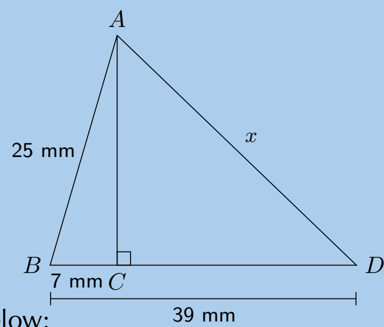
b)



c)

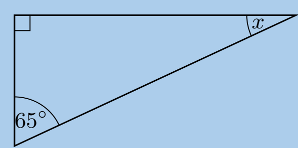


d)

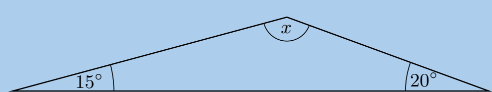


12. Calculate x and y in the diagrams below:

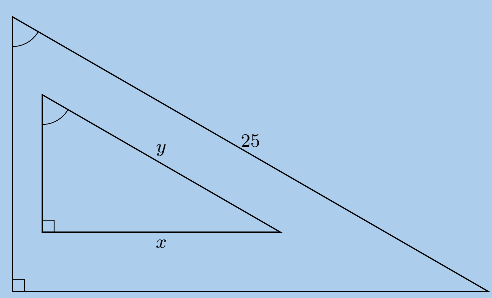
a)



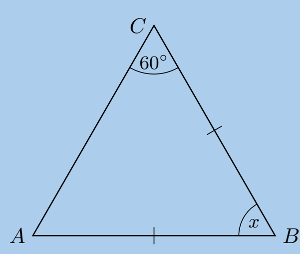
b)



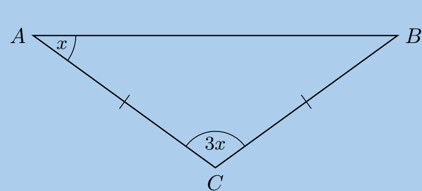
c)



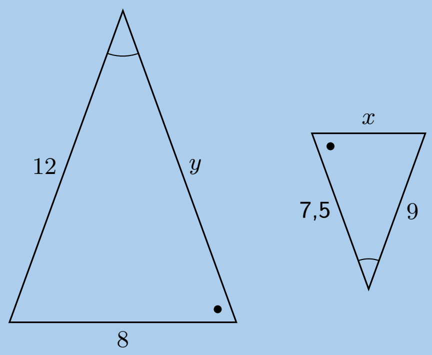
d)



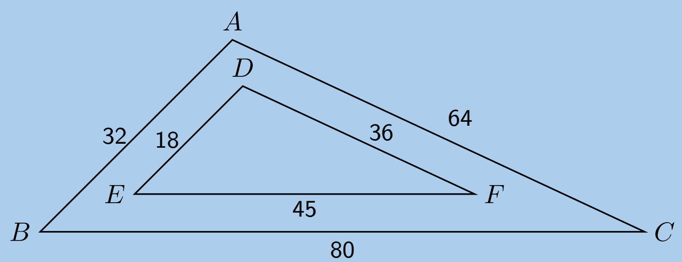
e)



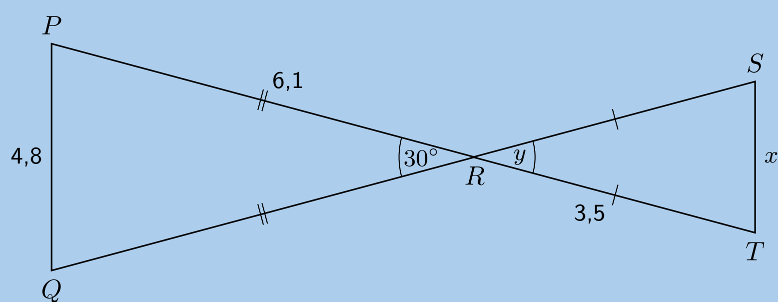
f)



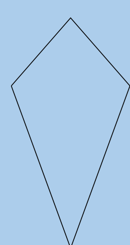
13. Consider the diagram below. Is $\triangle ABC \ ||| \ \triangle DEF$? Give reasons for your answer.



14. Explain why $\triangle PQR$ is similar to $\triangle TSR$ and calculate the values of x and y.

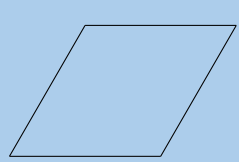


15. The following shape is drawn to scale:

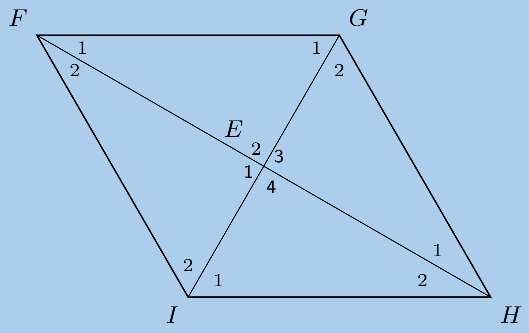


Give the most specific name for the shape.

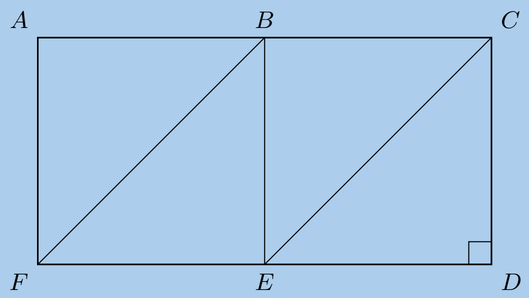
16. Based on the shape that you see list the all the names of the shape. The figure is drawn to scale.



17. $FGHI$ is a rhombus. $\hat{F}_1 = 3x + 20°$; $\hat{G}_1 = x + 10°$. Determine the value of x.



18. In the diagram below, $AB = BC = CD = DE = EF = FA = BE$.



Name:

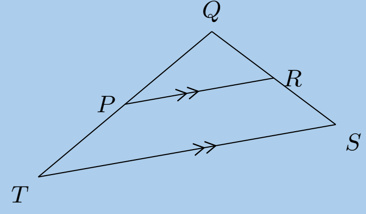
a) 3 rectangles

b) 4 parallelograms

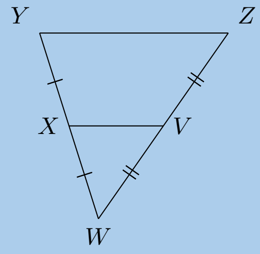
c) 2 trapeziums

d) 2 rhombi

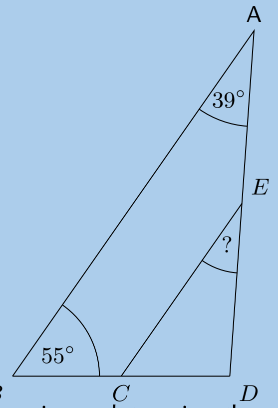
19. Points R and P are the mid-points on lines QS and QT. Study $\triangle TSQ$ carefully. Identify the third side of this triangle, using the information as shown, together with what you know about the mid-point theorem. (Name the third side by its endpoints, e.g., FG.)



20. Points X and V are given on the segments WY and WZ. Study the triangle carefully, then identify and name the parallel line segments.



21. The figure below shows a large triangle with vertices A, B and D, and a smaller triangle with vertices at C, D and E. Point C is the mid-point of BD and point E is the mid-point of AD.

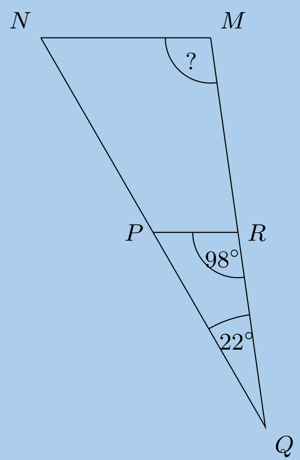


a) The angles $\hat{A} = 39°$ and $\hat{B} = 55°$ are given; determine the value of $D\hat{E}C$.

b) The two triangles in this question are similar triangles. Complete the following statement correctly by giving the three vertices in the correct order (there is only one correct answer).

$\triangle DEC \ ||| \ \triangle ?$

22. The figure below shows a large triangle with vertices M, N and Q, and a smaller triangle with vertices at P, Q and R. Point P is the mid-point of NQ and point R is the mid-point of MQ.

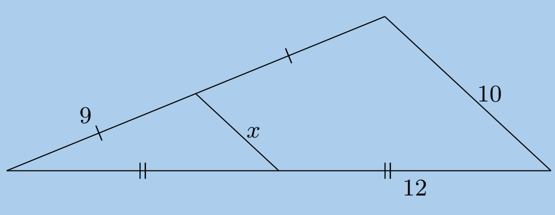


a) With the two angles given, $\hat{Q} = 22°$ and $Q\hat{R}P = 98°$, determine the value of $\hat{M}$.

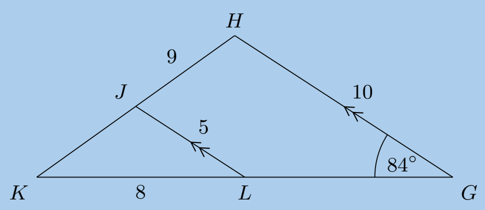
b) The two triangles in this question are similar triangles. Complete the following statement correctly by giving the three vertices in the correct order (there is only one correct answer).

$\triangle QMN \ ||| \ \triangle ?$

23. Consider the triangle in the diagram below. There is a line segment crossing through a large triangle. Notice that some segments in the figure are marked as equal to each other. One side of the triangle has a given length of 10. Determine the value of x.

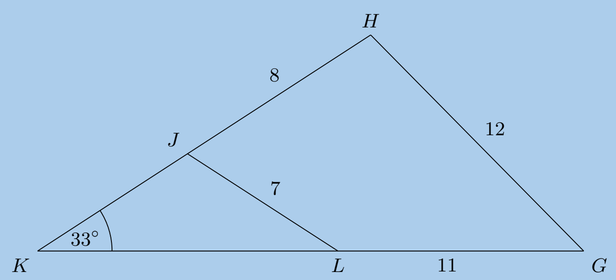


24. In the figure below, $GH \parallel LJ$, as labelled. Furthermore, the following lengths and angles are given: $GH = 10$; $LJ = 5$; $HJ = 9$; $KL = 8$ and $\hat{G} = 84°$. The figure is drawn to scale.



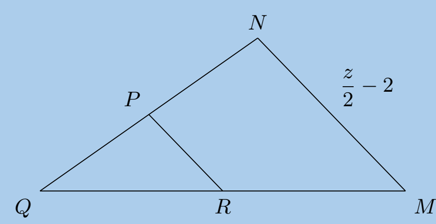
Calculate the length of JK.

25. The figure below shows triangle GHK with the smaller triangle JKL sitting inside of it. Furthermore, the following lengths and angles are given: $GH = 12; LJ = 7; HJ = 8; LG = 11$; $\hat{K} = 33°$. The figure is drawn to scale.



Find the length of KL.

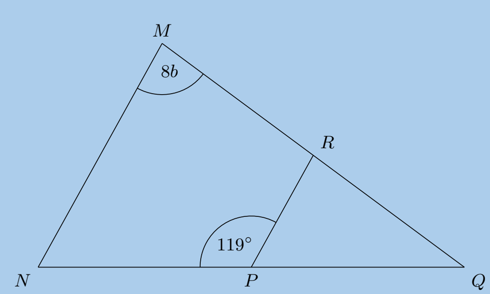
26. In the diagram below, P is the mid-point of NQ and R is the mid-point of MQ. One side of the triangle has a given length of $\frac{z}{2} - 2$.



a) Determine the value of PR in terms of z.

b) You are now told that PR has a length of 2. What is the value of z?

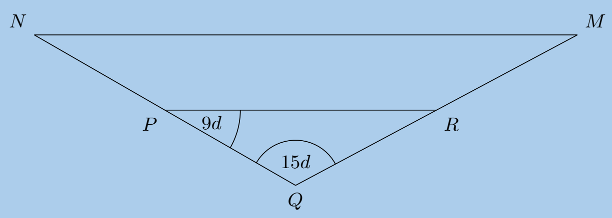
27. The figure below shows $\triangle MNQ$ crossed by RP. Points P and R bisect their respective sides of the triangle.



a) With the two angles given, $\hat{M} = 8b$ and $\angle NPR = 119°$, determine the value of $\hat{Q}$ in terms of b.

b) You are now told that $\hat{M}$ has a measure of 76°. Determine for the value of b. Give your answer as an exact fractional value.

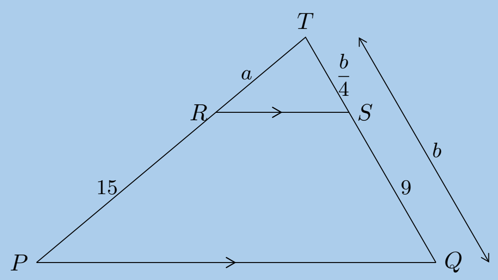
28. The figure below shows $\triangle MNQ$ crossed by RP. Points P and R bisect their respective sides of the triangle.



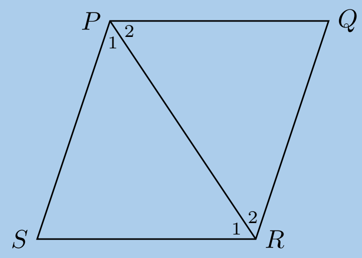
a) The angles $\hat{Q} = 15d$ and $R\hat{P}Q = 9d$ are given in the large triangle; determine the value of $\hat{M}$ in terms of d.

b) You are now told that $R\hat{P}Q$ has a measure of $60°$. Solve for the value of d. Give your answer as an exact fractional value.

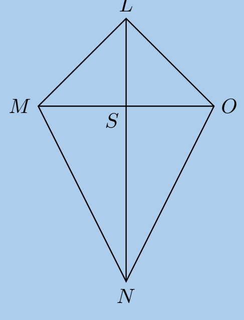
29. Calculate a and b:



30. $\triangle PQR$ and $\triangle PSR$ are equilateral triangles. Prove that $PQRS$ is a rhombus.



31. $LMNO$ is a quadrilateral with $LM = LO$ and diagonals that intersect at S such that $MS = SO$. Prove that:

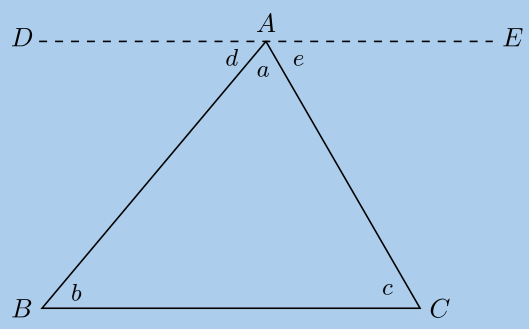


a) $M\hat{L}S = S\hat{L}O$

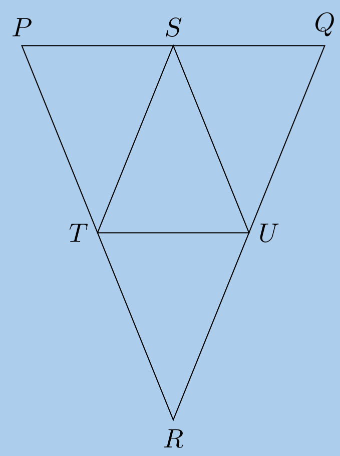
b) $\triangle LON \equiv \triangle LMN$

c) $MO \perp LN$

32. Using the figure below, show that the sum of the three angles in a triangle is 180°. Line DE is parallel to BC.

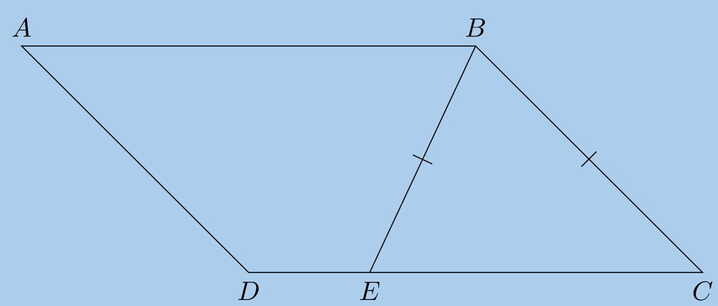


33. PQR is an isosceles triangle with $PR = QR$. S is the mid-point of PQ, T is the mid-point of PR and U is the mid-point of RQ.

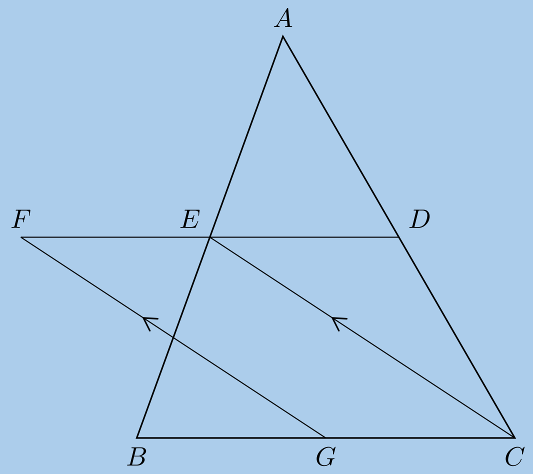


a) Prove $\triangle STU$ is also isosceles.
b) What type of quadrilateral is $STRU$? Motivate your answer.
c) If $R\hat{T}U = 68°$ calculate, with reasons, the size of $T\hat{S}U$.

34. $ABCD$ is a parallelogram. $BE = BC$. Prove that $A\hat{B}E = B\hat{C}D$.



35. In the diagram below, D, E and G are the mid-points of AC, AB and BC respectively. $EC \parallel FG$.



a) Prove that $FECG$ is a parallelogram.

b) Prove that $FE = ED$.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1a. 2G87	1b. 2G88	1c. 2G89	1d. 2G8B	1e. 2G8C	1f. 2G8D
1g. 2G8F	1h. 2G8G	2a. 2G8H	2b. 2G8J	2c. 2G8K	2d. 2G8M
2e. 2G8N	2f. 2G8P	2g. 2G8Q	2h. 2G8R	2i. 2G8S	2j. 2G8T
3a. 2G8V	3b. 2G8W	3c. 2G8X	4a. 2G8Y	4b. 2G8Z	4c. 2G92
5. 2G93	6. 2G94	7. 2G95	8. 2G96	9. 2G97	10a. 2G98
10b. 2G99	10c. 2G9B	10d. 2G9C	11a. 2G9D	11b. 2G9F	11c. 2G9G
11d. 2G9H	12a. 2G9J	12b. 2G9K	12c. 2G9M	12d. 2G9N	12e. 2G9P
12f. 2G9Q	13. 2G9R	14. 2G9S	15. 2G9T	16. 2G9V	17. 2G9W
18. 2G9X	19. 2G9Y	20. 2G9Z	21. 2GB2	22. 2GB3	23. 2GB4
24. 2GB5	25. 2GB6	26. 2GB7	27. 2GB8	28. 2GB9	29. 2GBB
30. 2GBC	31. 2GBD	32. 2GBF	33. 2GBG	34. 2GBH	35. 2GBJ

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CHAPTER 8

Analytical geometry

Analytical geometry is the study of geometric properties, relationships and measurement of points, lines and angles in the Cartesian plane. Geometrical shapes are defined using a coordinate system and algebraic principles. Some consider the introduction of analytical geometry, also called coordinate or Cartesian geometry, to be the beginning of modern mathematics.

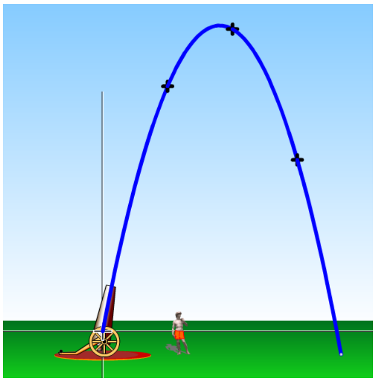
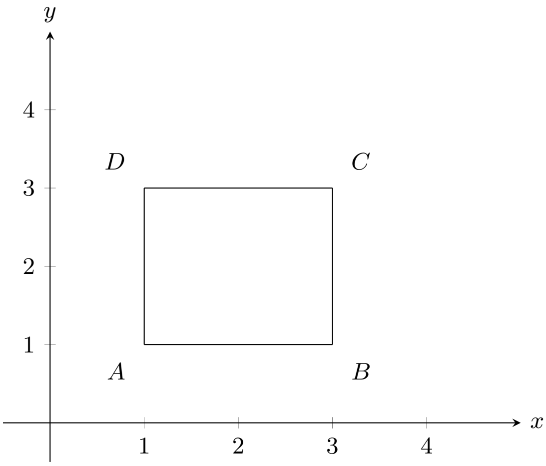


Figure 8.1: The motion of a projectile can be plotted on the Cartesian plane. Animators use this information to help them create animations.

8.1 Drawing figures on the Cartesian plane EMA68

If we are given the coordinates of the vertices of a figure, we can draw the figure on the Cartesian plane. For example, quadrilateral $ABCD$ with coordinates $A\,(1;1)$, $B\,(3;1)$, $C\,(3;3)$ and $D\,(1;3)$.

NOTE:
You might also see coordinates written as $A(1,1)$.

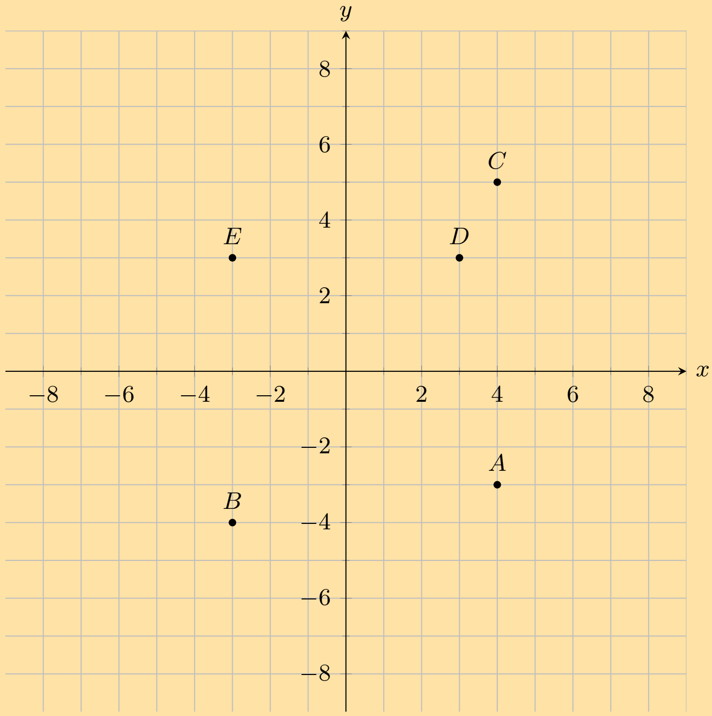


The order of the letters for naming a figure is important. It indicates the order in which points must be joined: A to B, B to C, C to D and D back to A. So the above quadrilateral can be referred to as quadrilateral $ABCD$ or $CBAD$ or $BADC$. However it is conventional to write the letters in alphabetical order and so we only refer to the quadrilateral as $ABCD$.

VISIT:
You can use an online tool to help you when plotting points on the Cartesian plane. Click here to try this one on mathsisfun.com.

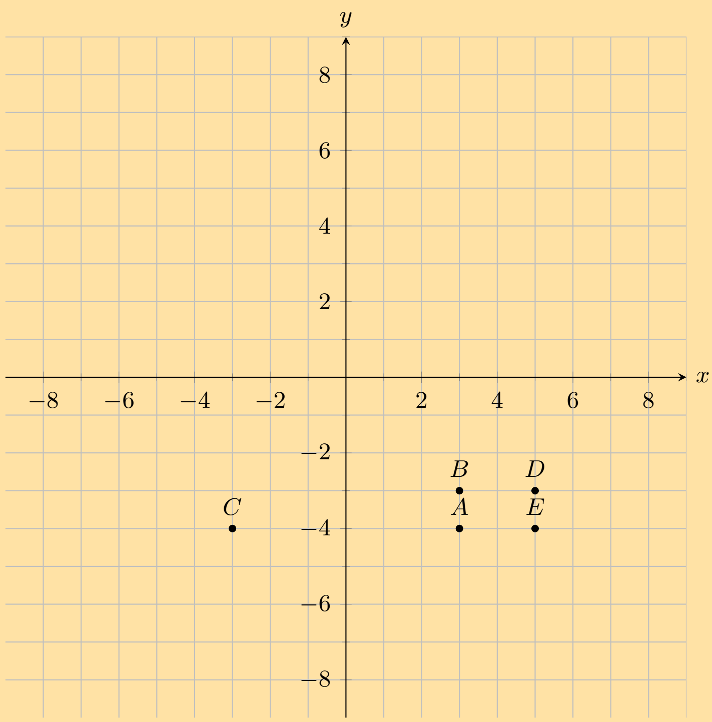
Exercise 8 – 1:

1. You are given the following diagram, with various points shown:



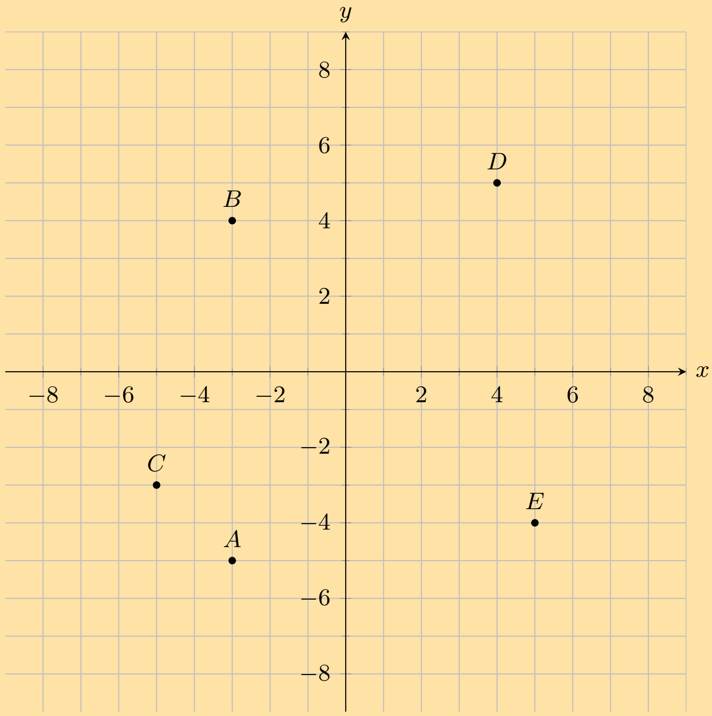
Find the coordinates of point D.

2. You are given the following diagram, with various points shown:



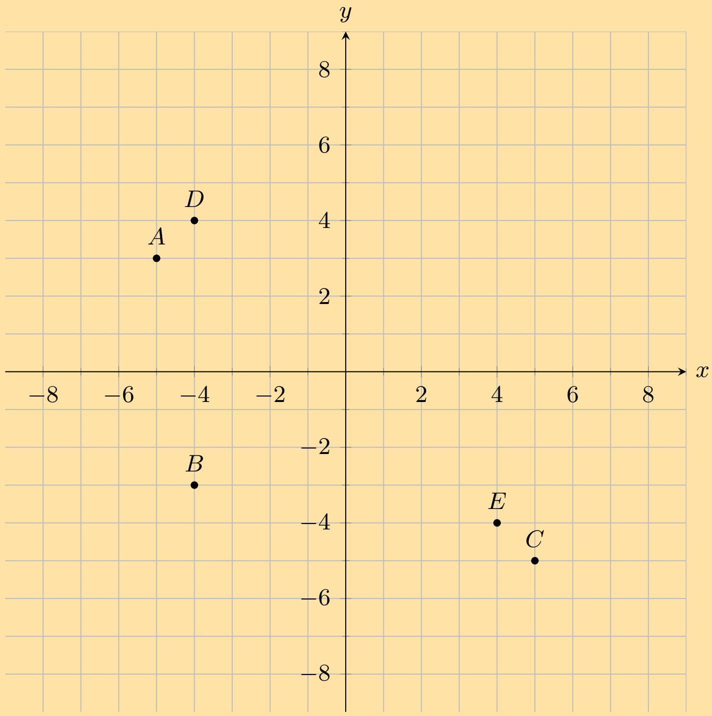
Find the coordinates of all the labelled points.

3. You are given the following diagram, with various points shown:



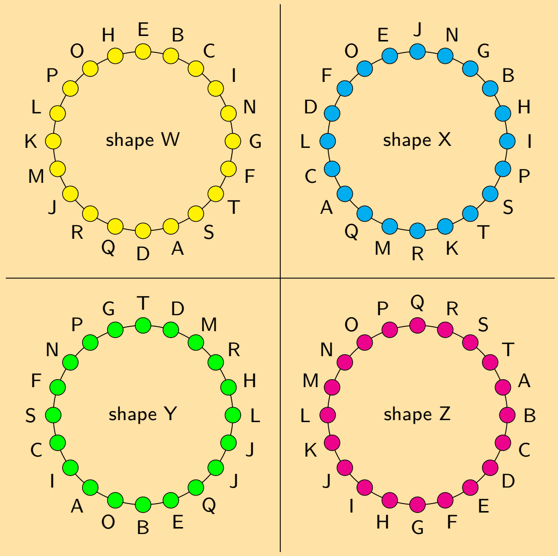
Which point lies at the coordinates $(5; -4)$?

4. You are given the following diagram, with various points shown:



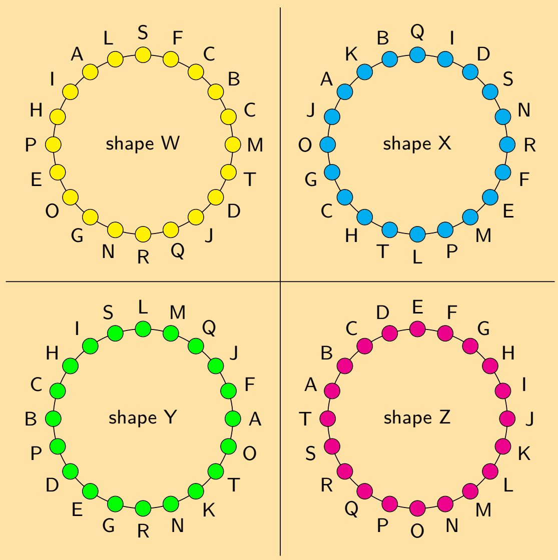
Which point lies at the coordinates $(-4; -3)$?

5. You are given the following diagram, with 4 shapes drawn.
All the shapes are identical, but each shape uses a different naming convention:



Which shape uses the correct naming convention?

6. You are given the following diagram, with 4 shapes drawn.
All the shapes are identical, but each shape uses a different naming convention:
Which shape uses the correct naming convention?



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1. 2GBK 2. 2GBM 3. 2GBN 4. 2GBP 5. 2GBQ 6. 2GBR

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8.2 Distance between two points

EMA69

A point is a simple geometric object having location as its only property.

DEFINITION: *Point*

A point is an ordered pair of numbers written as $(x; y)$.

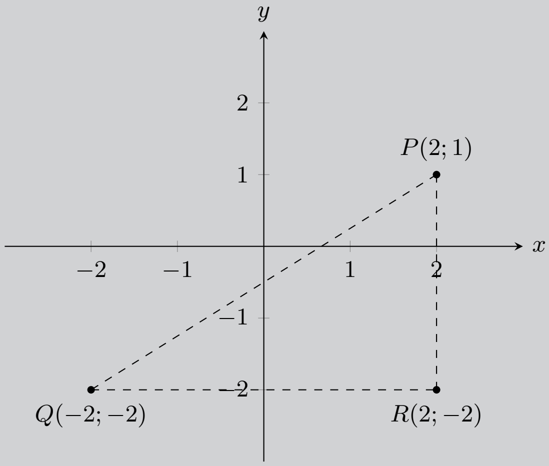
DEFINITION: *Distance*

Distance is a measure of the length between two points.

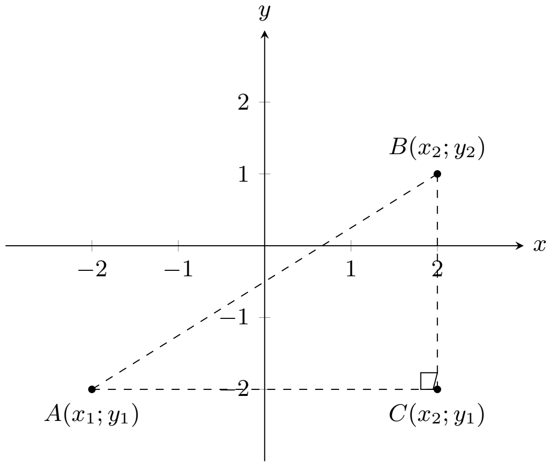
Investigation: Distance between two points

Points $P\,(2; 1)$, $Q\,(-2; -2)$ and $R\,(2; -2)$ are given.

- Can we assume that $\hat{R} = 90°$? If so, why?
- Apply the theorem of Pythagoras in $\triangle PQR$ to find the length of PQ.



To derive a general formula for the distance between two points $A\,(x_1; y_1)$ and $B\,(x_2; y_2)$ we use the theorem of Pythagoras.



$$AB^2 = AC^2 + BC^2$$
$$\therefore AB = \sqrt{AC^2 + BC^2}$$

And:

$$AC = x_2 - x_1$$
$$BC = y_2 - y_1$$

Therefore:

$$AB = \sqrt{AC^2 + BC^2}$$
$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Therefore to calculate the distance between any two points, $(x_1; y_1)$ and $(x_2; y_2)$, we use:

distance $(d) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

Note that $(x_1 - x_2)^2 = (x_2 - x_1)^2$.

VISIT:
The following video gives two examples of working with the distance formula and shows how to determine the distance formula.
See video: 2GBS at www.everythingmaths.co.za

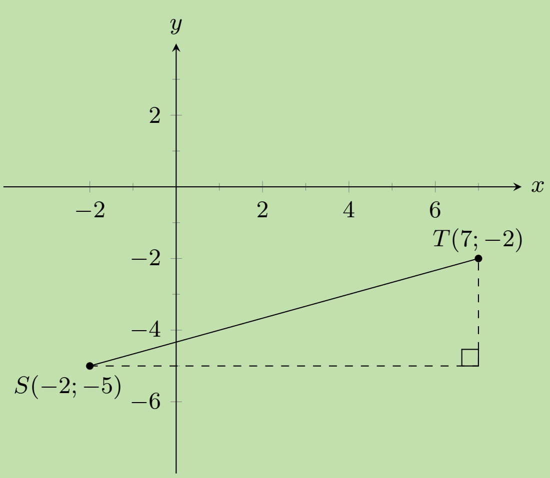
Worked example 1: Using the distance formula

QUESTION

Find the distance between $S(-2; -5)$ and $Q(7; -2)$.

SOLUTION

Step 1: Draw a sketch



Step 2: Assign values to $(x_1; y_1)$ and $(x_2; y_2)$

Let the coordinates of S be $(x_1; y_1)$ and the coordinates of T be $(x_2; y_2)$.

$$x_1 = -2 \quad y_1 = -5 \quad x_2 = 7 \quad y_2 = -2$$

Step 3: Write down the distance formula

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Step 4: Substitute values

$$\begin{aligned} d_{ST} &= \sqrt{(-2-7)^2 + (-5-(-2))^2} \\ &= \sqrt{(-9)^2 + (-3)^2} \\ &= \sqrt{81+9} \\ &= \sqrt{90} \\ &= 9{,}5 \end{aligned}$$

Step 5: Write the final answer

The distance between S and T is 9,5 units.

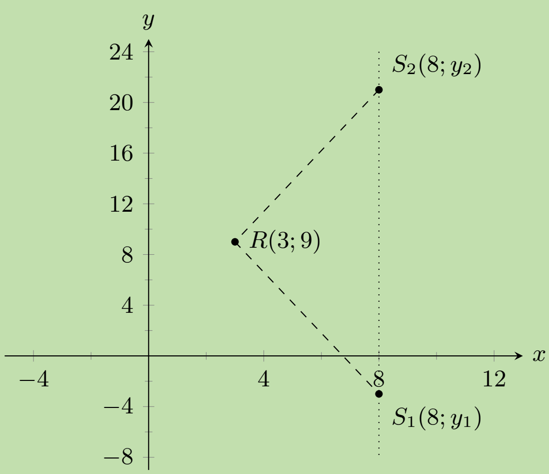
Worked example 2: Using the distance formula

QUESTION

Given $RS = 13$, $R\,(3; 9)$ and $S\,(8; y)$. Find y.

SOLUTION

Step 1: Draw a sketch



Note that we expect two possible values for y. This is because the distance formula includes the term $(y_1 - y_2)^2$ which results in a quadratic equation when we substitute in the y coordinates.

Step 2: Assign values to $(x_1; y_1)$ and $(x_2; y_2)$

Let the coordinates of R be $(x_1; y_1)$ and the coordinates of S be $(x_2; y_2)$.

$$x_1 = 3 \quad y_1 = 9 \quad x_2 = 8 \quad y_2 = y$$

Step 3: Write down the distance formula

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Step 4: Substitute values and solve for y

$$\begin{aligned}
13 &= \sqrt{(3-8)^2 + (9-y)^2} \\
13^2 &= (-5)^2 + \left(81 - 18y + y^2\right) \\
0 &= y^2 - 18y - 63 \\
&= (y+3)(y-21) \\
\therefore y &= -3 \text{ or } y = 21
\end{aligned}$$

Step 5: Check both values for y

Check $y = -3$:

$$\begin{aligned}
d &= \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \\
&= \sqrt{(3-8)^2 + (9+3)^2} \\
&= \sqrt{25 + 144} \\
&= \sqrt{169} \\
&= 13
\end{aligned}$$

Check $y = 21$:

$$\begin{aligned}
d &= \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \\
&= \sqrt{(3-8)^2 + (9-21)^2} \\
&= \sqrt{25 + 144} \\
&= \sqrt{169} \\
&= 13
\end{aligned}$$

Step 6: Write the final answer

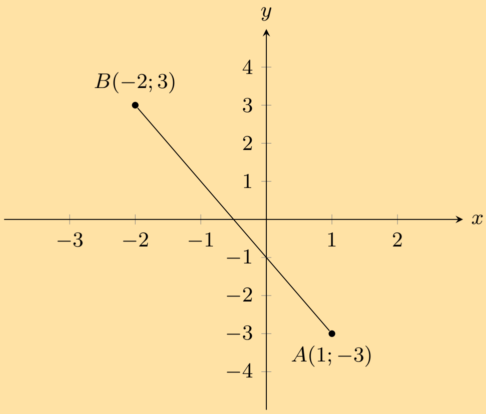
S is $(8; -3)$ or $(8; 21)$.

Therefore $y = -3$ or $y = 21$.

NOTE:
Drawing a sketch helps with your calculation and makes it easier to check if your answer is correct.

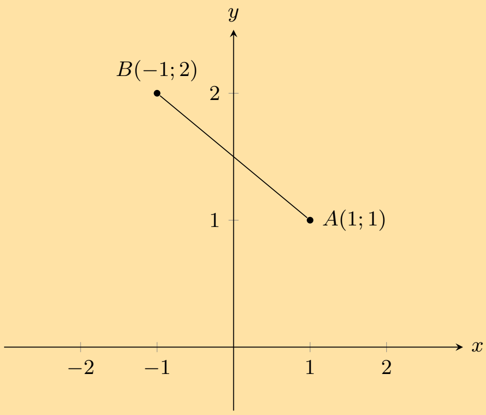
Exercise 8 – 2:

1. You are given the following diagram:



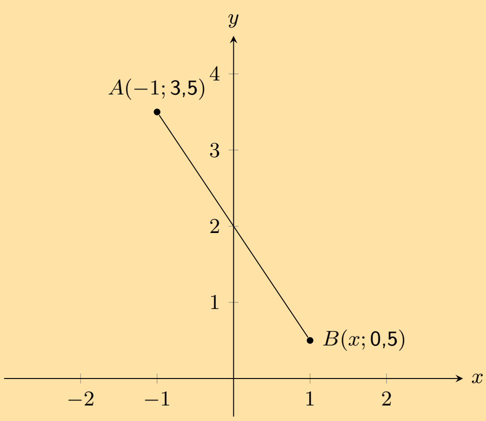
   Calculate the length of line AB, correct to 2 decimal places.

2. You are given the following diagram:



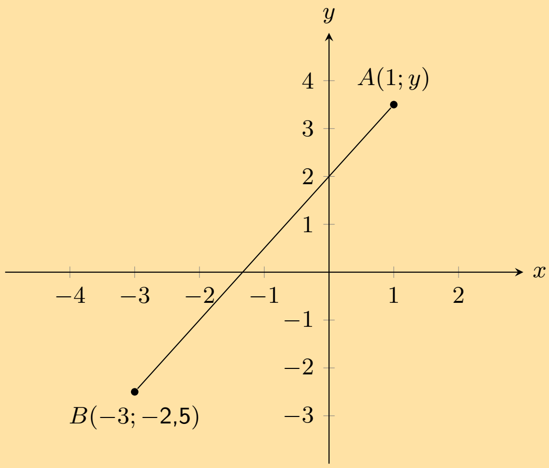
   Calculate the length of line AB, correct to 2 decimal places.

3. The following picture shows two points on the Cartesian plane, A and B.



   The distance between the points is 3,6056. Calculate the missing coordinate of point B.

4. The following picture shows two points on the Cartesian plane, A and B.



The line AB has a length of 7,2111. Calculate the missing coordinate of point B. Round your answer to one decimal place.

5. Find the length of AB for each of the following. Leave your answer in surd form.

a) $A(2;7)$ and $B(-3;5)$ b) $A(-3;5)$ and $B(-9;1)$ c) $A(x;y)$ and $B(x+4;y-1)$

6. The length of $CD = 5$. Find the missing coordinate if:

a) $C(6;-2)$ and $D(x;2)$. b) $C(4;y)$ and $D(1;-1)$.

7. If the distance between $C(0;-3)$ and $F(8;p)$ is 10 units, find the possible values of p.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GBT 2. 2GBV 3. 2GBW 4. 2GBX 5a. 2GBY 5b. 2GBZ
5c. 2GC2 6a. 2GC3 6b. 2GC4 7. 2GC5

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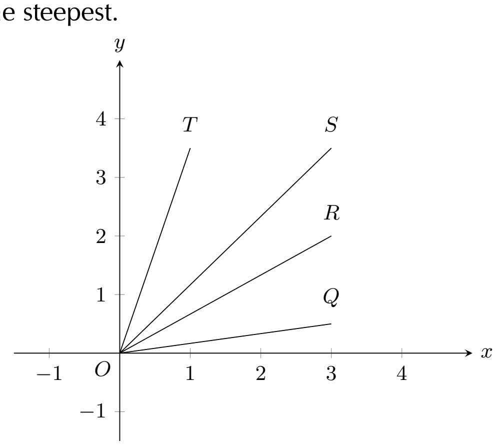
8.3 Gradient of a line EMA6B

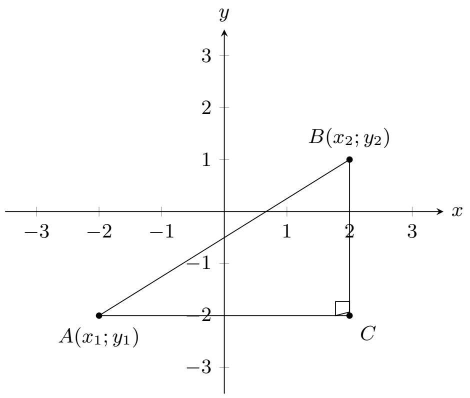
DEFINITION: *Gradient*

The gradient of a line is determined by the ratio of vertical change to horizontal change.

Gradient (m) describes the slope or steepness of the line joining two points. In the figure below, line OQ is the least steep and line OT is the steepest.



To derive the formula for gradient, we consider any right-angled triangle formed from $A(x_1; y_1)$ and $B(x_2; y_2)$ with hypotenuse AB as shown in the diagram alongside. The gradient is determined by the ratio of the length of the vertical side of the triangle to the length of the horizontal side of the triangle. The length of the vertical side of the triangle is the difference in y-values of points A and B. The length of the horizontal side of the triangle is the difference in x-values of points A and B.



Therefore, gradient is determined using the following formula:

Gradient $(m) = \dfrac{y_2 - y_1}{x_2 - x_1}$ or $\dfrac{y_1 - y_2}{x_1 - x_2}$

IMPORTANT!

Remember to be consistent: $m \neq \dfrac{y_1 - y_2}{x_2 - x_1}$

VISIT:
To learn more about determining the gradient you can watch the following video.
See video: 2GC6 at www.everythingmaths.co.za

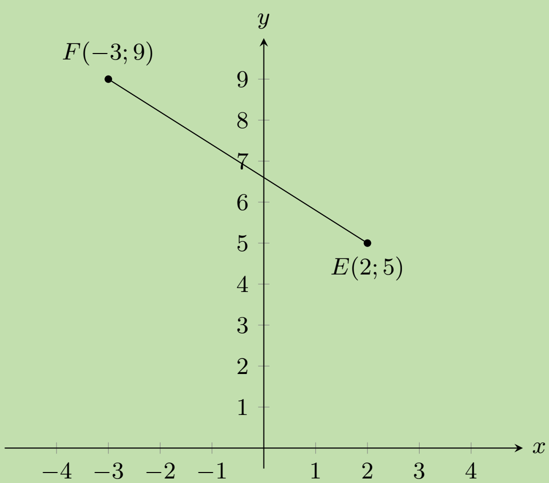
Worked example 3: Gradient between two points

QUESTION

Find the gradient of the line between points $E(2; 5)$ and $F(-3; 9)$.

SOLUTION

Step 1: Draw a sketch



Step 2: Assign values to $(x_1; y_1)$ and $(x_2; y_2)$

Let the coordinates of E be $(x_1; y_1)$ and the coordinates of F be $(x_2; y_2)$.

$$x_1 = 2 \quad y_1 = 5 \quad x_2 = -3 \quad y_2 = 9$$

Step 3: Write down the formula for gradient

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Step 4: Substitute known values

$$\begin{aligned} m_{EF} &= \frac{9-5}{-3-2} \\ &= \frac{4}{-5} \end{aligned}$$

Step 5: Write the final answer

The gradient of $EF = -\frac{4}{5}$

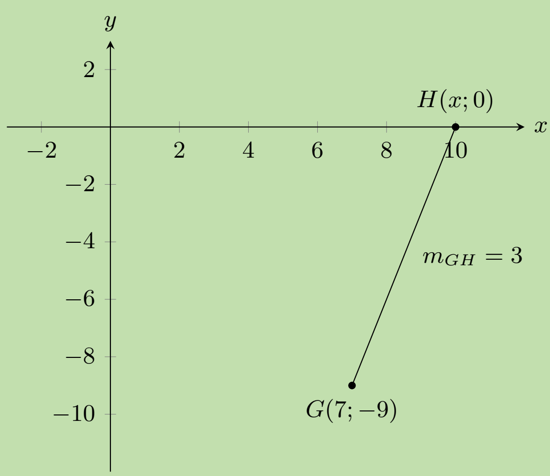
Worked example 4: Gradient between two points

QUESTION

Given $G\,(7; -9)$ and $H\,(x; 0)$, with $m_{GH} = 3$, find x.

SOLUTION

Step 1: Draw a sketch



Step 2: Assign values to $(x_1; y_1)$ and $(x_2; y_2)$

Let the coordinates of G be $(x_1; y_1)$ and the coordinates of H be $(x_2; y_2)$

$$x_1 = 7 \quad y_1 = -9 \quad x_2 = x \quad y_2 = 0$$

Step 3: Write down the formula for gradient

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Step 4: Substitute values and solve for x

$$\begin{aligned}
3 &= \frac{0 - (-9)}{x - 7} \\
3(x - 7) &= 9 \\
x - 7 &= \frac{9}{3} \\
x - 7 &= 3 \\
x &= 3 + 7 \\
&= 10
\end{aligned}$$

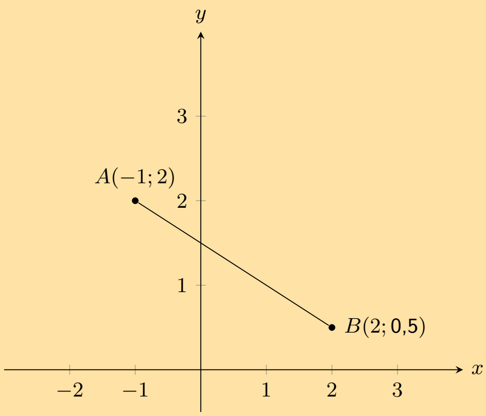
Step 5: Write the final answer

The coordinates of H are $(10; 0)$.

Therefore $x = 10$.

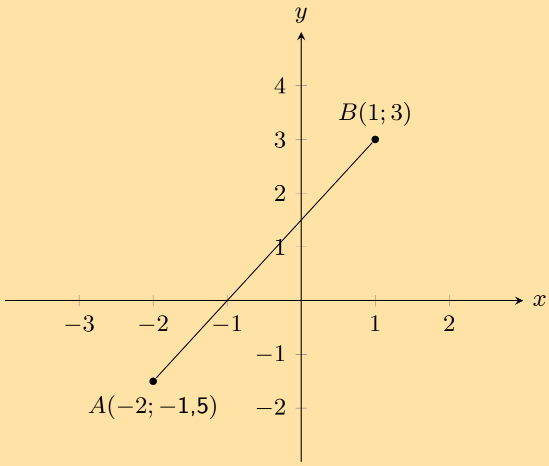
Exercise 8 – 3:

1. Find the gradient of AB if:
   a) $A(7; 10)$ and $B(-4; 1)$
   b) $A(-5; -9)$ and $B(3; 2)$
   c) $A(x - 3; y)$ and $B(x; y + 4)$
2. You are given the following diagram:

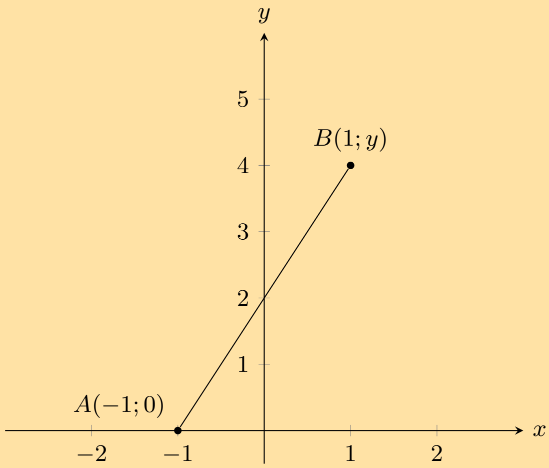


   Calculate the gradient (m) of line AB.

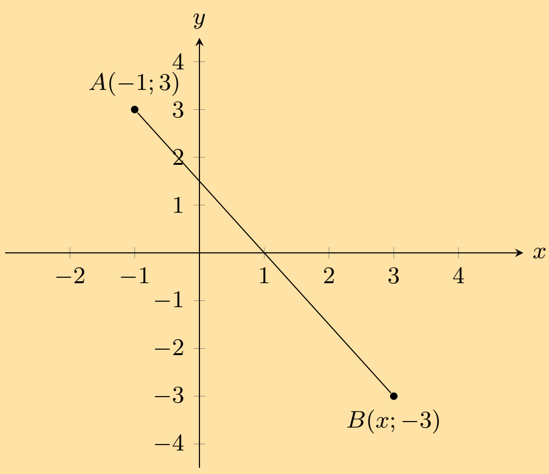
3. Calculate the gradient (m) of line AB in the following diagram:



4. If the gradient of $CD = \frac{2}{3}$, find p given:
   a) $C(16;2)$ and $D(8;p)$.
   b) $C(3;2p)$ and $D(9;14)$.
5. In the following diagram line AB has a gradient (m) of 2. Calculate the missing co-ordinate of point B.



6. You are given the following diagram:



You are also told that line AB has a gradient (m) of $-1{,}5$.
Calculate the missing co-ordinate of point B.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1a. 2GC7 1b. 2GC8 1c. 2GC9 2. 2GCB 3. 2GCC 4a. 2GCD
4b. 2GCF 5. 2GCG 6. 2GCH

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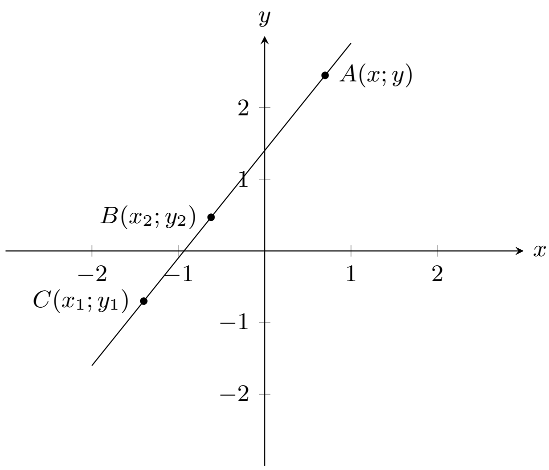
Straight lines EMA6C

DEFINITION: *Straight line*

A straight line is a set of points with a constant gradient between any two of the points.

Consider the diagram below with points $A\,(x;y)$, $B\,(x_2;y_2)$ and $C\,(x_1;y_1)$.



We have $m_{AB} = m_{BC} = m_{AC}$ and $m = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{y_1 - y_2}{x_1 - x_2}$

The general formula for finding the equation of a straight line is $\dfrac{y - y_1}{x - x_1} = \dfrac{y_2 - y_1}{x_2 - x_1}$ where $(x;y)$ is any point on the line.

This formula can also be written as $y - y_1 = m\,(x - x_1)$.

The standard form of the straight line equation is $y = mx + c$ where m is the gradient and c is the y-intercept.

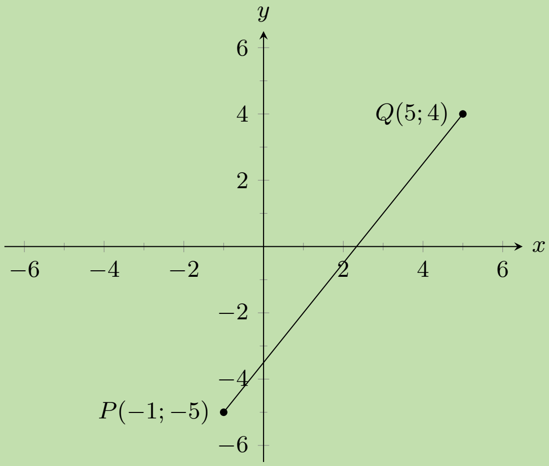
Worked example 5: Finding the equation of a straight line

QUESTION

Find the equation of the straight line through $P\,(-1;-5)$ and $Q\,(5;4)$.

SOLUTION

Step 1: Draw a sketch



Step 2: Assign values

Let the coordinates of P be $(x_1; y_1)$ and the coordinates of Q be $(x_2; y_2)$.

$$x_1 = -1 \quad y_1 = -5 \quad x_2 = 5 \quad y_2 = 4$$

Step 3: Write down the general formula of the line

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

Step 4: Substitute values and make y the subject of the equation

$$\begin{aligned}
\frac{y - (-5)}{x - (-1)} &= \frac{4 - (-5)}{5 - (-1)} \\
\frac{y + 5}{x + 1} &= \frac{3}{2} \\
2(y + 5) &= 3(x + 1) \\
2y + 10 &= 3x + 3 \\
2y &= 3x - 7 \\
y &= \frac{3}{2}x - \frac{7}{2}
\end{aligned}$$

Step 5: Write the final answer

The equation of the straight line is $y = \frac{3}{2}x - \frac{7}{2}$.

Parallel and perpendicular lines EMA6D

Two lines that run parallel to each other are always the same distance apart and have equal gradients.

If two lines intersect perpendicularly, then the product of their gradients is equal to -1.

If line $WX \perp YZ$, then $m_{WX} \times m_{YZ} = -1$. Perpendicular lines have gradients that are the negative inverses of each other.

VISIT:
The following video shows some examples of calculating the slope of a line and determining if two lines are perpendicular or parallel.
See video: 2GCJ at www.everythingmaths.co.za

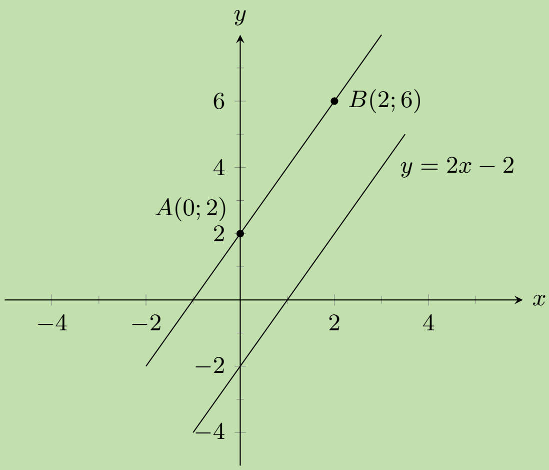
Worked example 6: Parallel lines

QUESTION

Prove that line AB with $A\,(0;2)$ and $B\,(2;6)$ is parallel to line CD with equation $2x - y = 2$.

SOLUTION

Step 1: Draw a sketch



(Be careful - some lines may look parallel but are not!)

Step 2: Write down the formula for gradient

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Step 3: Substitute values to find the gradient for line AB

$$\begin{aligned} m_{AB} &= \frac{6-2}{2-0} \\ &= \frac{4}{2} \\ &= 2 \end{aligned}$$

Step 4: Check that the equation of CD is in the standard form $y = mx + c$

$$\begin{aligned} 2x - y &= 2 \\ y &= 2x - 2 \\ \therefore m_{CD} &= 2 \end{aligned}$$

Step 5: Write the final answer

$$m_{AB} = m_{CD}$$

therefore line AB is parallel to line CD.

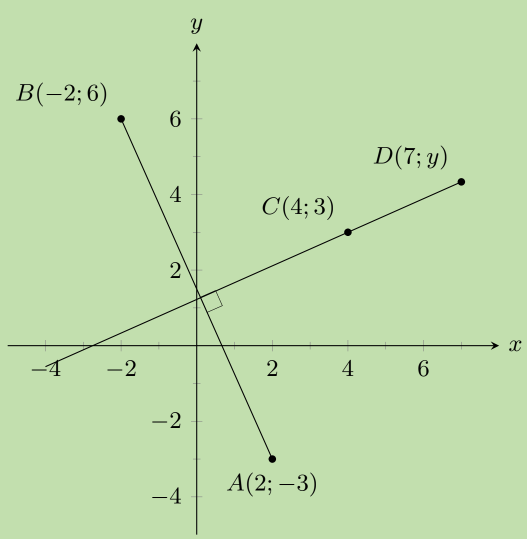
Worked example 7: Perpendicular lines

QUESTION

Line AB is perpendicular to line CD. Find y given $A(2; -3)$, $B(-2; 6)$, $C(4; 3)$ and $D(7; y)$.

SOLUTION

Step 1: Draw a sketch



Step 2: Write down the relationship between the gradients of the perpendicular lines $AB \perp CD$

$$\begin{aligned} m_{AB} \times m_{CD} &= -1 \\ \frac{y_B - y_A}{x_B - x_A} \times \frac{y_D - y_C}{x_D - x_C} &= -1 \end{aligned}$$

Step 3: Substitute values and solve for y

$$\begin{aligned}
\frac{6-(-3)}{-2-2} \times \frac{y-3}{7-4} &= -1 \\
\frac{9}{-4} \times \frac{y-3}{3} &= -1 \\
\frac{y-3}{3} &= -1 \times \frac{-4}{9} \\
\frac{y-3}{3} &= \frac{4}{9} \\
y-3 &= \frac{4}{9} \times 3 \\
y-3 &= \frac{4}{3} \\
y &= \frac{4}{3} + 3 \\
&= \frac{4+9}{3} \\
&= \frac{13}{3} \\
&= 4\frac{1}{3}
\end{aligned}$$

Step 4: Write the final answer

Therefore the coordinates of D are $\left(7; 4\frac{1}{3}\right)$.

Horizontal and vertical lines EMA6F

A line that runs parallel to the x-axis is called a horizontal line and has a gradient of zero. This is because there is no vertical change:

$$m = \frac{\text{change in } y}{\text{change in } x} = \frac{0}{\text{change in } x} = 0$$

A line that runs parallel to the y-axis is called a vertical line and its gradient is undefined. This is because there is no horizontal change:

$$m = \frac{\text{change in } y}{\text{change in } x} = \frac{\text{change in } y}{0} = \text{ undefined}$$

Points on a line EMA6G

A straight line is a set of points with a constant gradient between any of the two points. There are two methods to prove that points lie on the same line: the gradient method and a method using the distance formula.

NOTE:
If two points lie on the same line then the two points are said to be collinear.

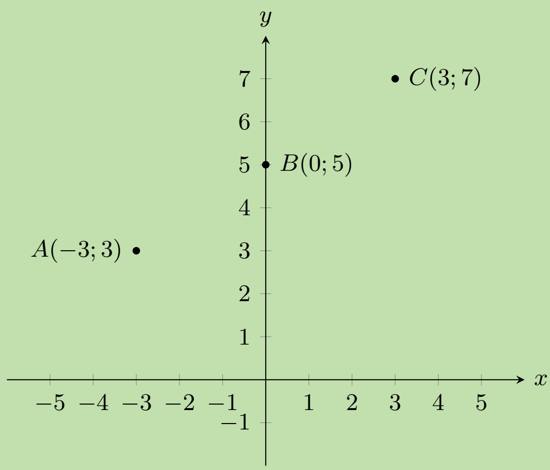
Worked example 8: Points on a line

QUESTION

Prove that $A(-3;3)$, $B(0;5)$ and $C(3;7)$ are on a straight line.

SOLUTION

Step 1: Draw a sketch



Step 2: Calculate two gradients between any of the three points

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$
$$m_{AB} = \frac{5-3}{0-(-3)} = \frac{2}{3}$$

and

$$m_{BC} = \frac{7-5}{3-0} = \frac{2}{3}$$

OR

$$m_{AC} = \frac{3-7}{3-3} = \frac{-4}{-6} = \frac{2}{3}$$

and

$$m_{BC} = \frac{7-5}{3-0} = \frac{2}{3}$$

Step 3: Explain your answer

$$m_{AB} = m_{BC} = m_{AC}$$

Therefore the points A, B and C are on a straight line.

To prove that three points are on a straight line using the distance formula, we must calculate the distances between each pair of points and then prove that the sum of the two smaller distances equals the longest distance.

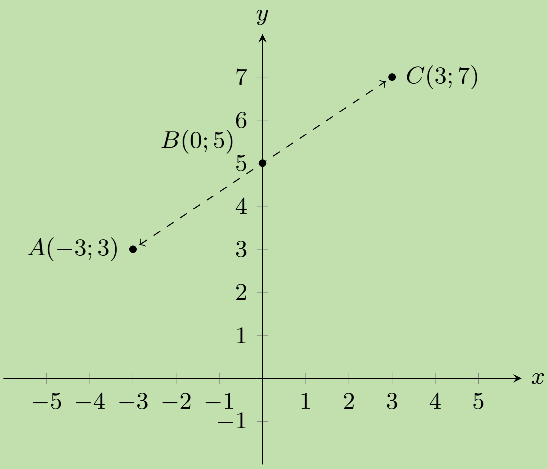
Worked example 9: Points on a straight line

QUESTION

Prove that $A\left(-3;3\right)$, $B\left(0;5\right)$ and $C\left(3;7\right)$ are on a straight line.

SOLUTION

Step 1: Draw a sketch



Step 2: Calculate the three distances AB, BC and AC

$$\begin{aligned} d_{AB} &= \sqrt{(-3-0)^2+(3-5)^2} \\ &= \sqrt{(-3)^2+(-2)^2} \\ &= \sqrt{9+4} \\ &= \sqrt{13} \end{aligned} \qquad \begin{aligned} d_{BC} &= \sqrt{(0-3)^2+(5-7)^2} \\ &= \sqrt{(-3)^2+(-2)^2} \\ &= \sqrt{9+4} \\ &= \sqrt{13} \end{aligned} \qquad \begin{aligned} d_{AC} &= \sqrt{((-3)-3)^2+(3-7)^2} \\ &= \sqrt{(-6)^2+(-4)^2} \\ &= \sqrt{36+16} \\ &= \sqrt{52} \end{aligned}$$

Step 3: Find the sum of the two shorter distances

$$d_{AB}+d_{BC}=\sqrt{13}+\sqrt{13}=2\sqrt{13}=\sqrt{4\times 13}=\sqrt{52}$$

Step 4: Explain your answer

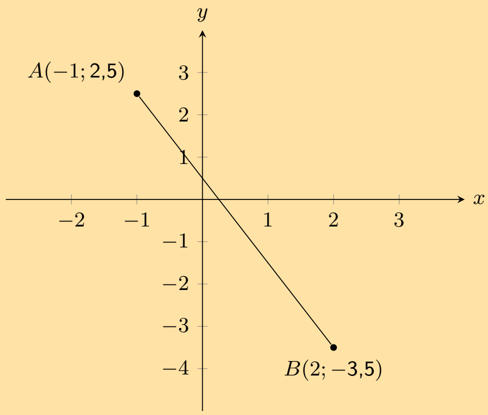
$$d_{AB}+d_{BC}=d_{AC}$$

therefore points A, B and C lie on the same straight line.

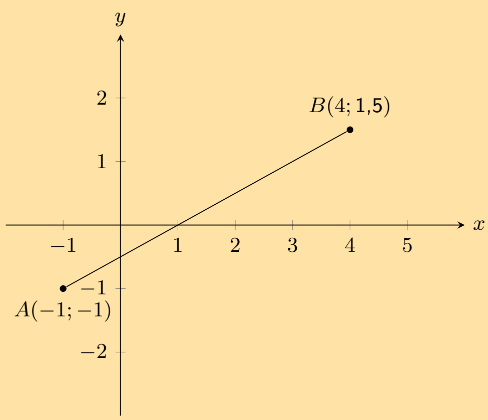
Exercise 8 – 4:

1. Determine whether AB and CD are parallel, perpendicular or neither if:
   a) $A(3;-4)$, $B(5;2)$, $C(-1;-1)$, $D(7;23)$
   b) $A(3;-4)$, $B(5;2)$, $C(-1;-1)$, $D(0;-4)$
   c) $A(3;-4)$, $B(5;2)$, $C(-1;3)$, $D(-2;2)$

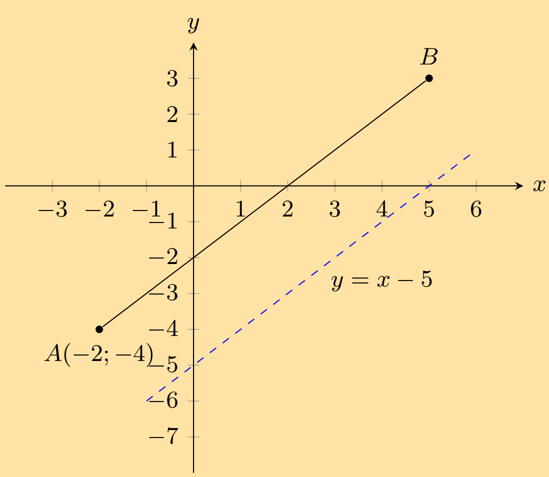
2. Determine whether the following points lie on the same straight line:
   a) $E(0;3)$, $F(-2;5)$, $G(2;1)$ b) $H(-3;-5)$, $I(0;0)$, $J(6;10)$ c) $K(-6;2)$, $L(-3;1)$, $M(1;-1)$
3. Calculate the equation of the line AB in the following diagram:



4. Calculate the equation of the line AB in the following diagram:

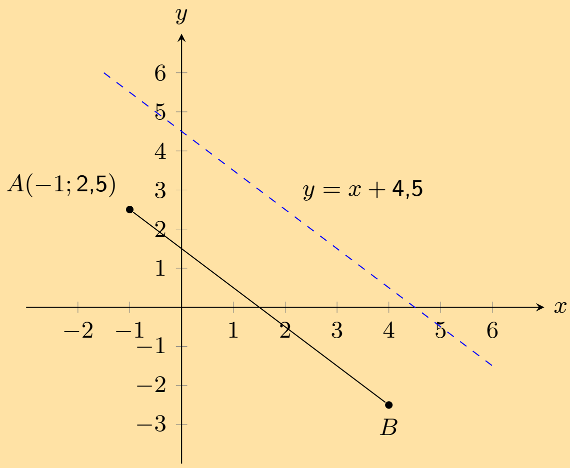


5. Points $P(-6;2)$, $Q(2;-2)$ and $R(-3;r)$ lie on a straight line. Find the value of r.
6. Line PQ with $P(-1;-7)$ and $Q(q;0)$ has a gradient of 1. Find q.
7. You are given the following diagram:



You are also told that line AB runs parallel to the following line: $y = x - 5$. Point A is at $(-2; -4)$. Find the equation of the line AB.

8. You are given the following diagram:

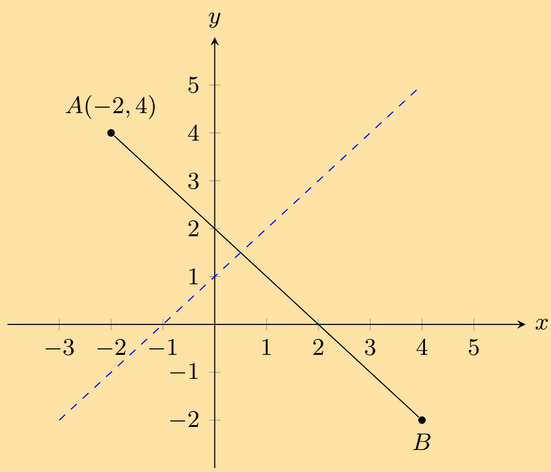


You are also told that line AB runs parallel to the following line: $y = -x + 4{,}5$. Point A is at $(-1; 2{,}5)$. Find the equation of the line AB.

9. Given line AB which runs parallel to $y = 0{,}5x - 6$. Points $A(-1; -2{,}5)$ and $B(x; 0)$ are also given. Calculate the missing co-ordinate of point B.

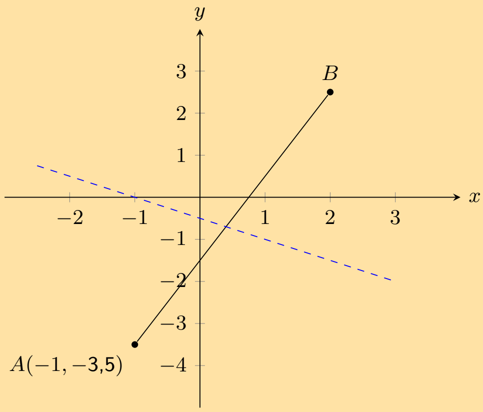
10. Given line AB which runs parallel to $y = -1{,}5x + 4$. Points $A(-2; 4)$ and $B(2; y)$ are also given. Calculate the missing co-ordinate of point $B(2; y)$.

11. The graph here shows line AB. The blue dashed line is perpendicular to AB.



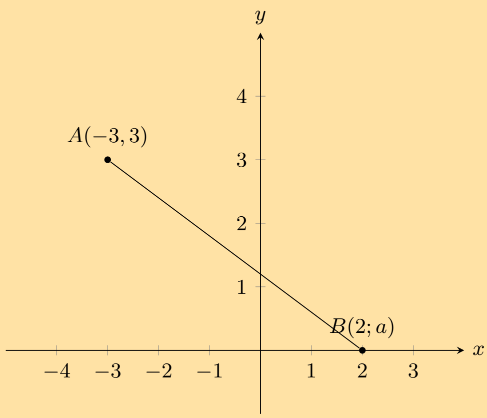
The equation of the blue dashed line is $y = x + 1$. Point A is at $(-2; 4)$. Determine the equation of line AB.

12. The graph here shows line AB. The blue dashed line is perpendicular to AB.



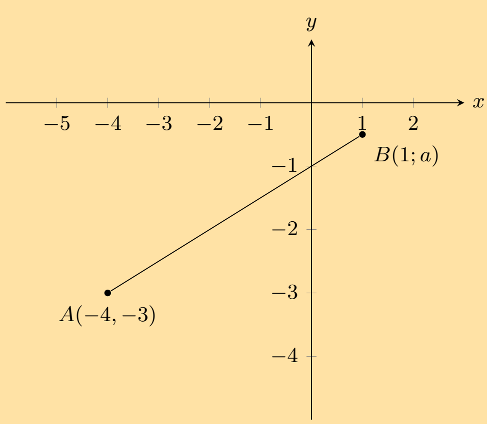
The equation of the blue dashed line is $y = -0{,}5x - 0{,}5$. Point A is at $(-1; -3{,}5)$. Determine the equation of line AB.

13. Given line AB which runs perpendicular to line CD with equation $y = -2x + 1$. Points $A(-5; -1)$ and $B(3; a)$ are also given. Calculate the missing co-ordinate of point B.
14. Given line AB which runs perpendicular to line CD with equation $y = 2x - 0{,}75$. Points $A(-5; 1)$ and $B(a; -2{,}5)$ are also given. Calculate the missing co-ordinate of point B.
15. You are given the following diagram:



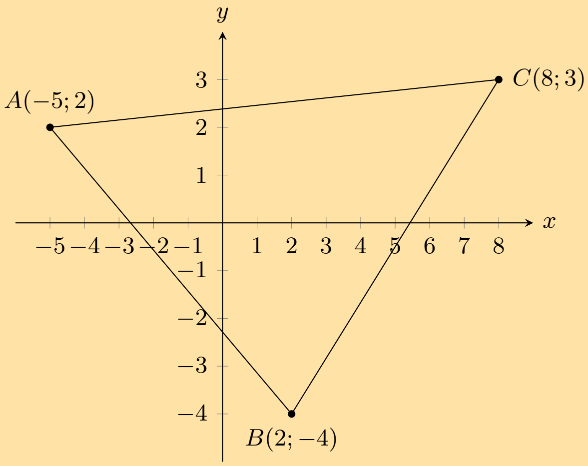
You are also told that line AB has the following equation: $y = -0{,}5x + 1{,}5$. Calculate the missing co-ordinate of point B.

16. You are given the following diagram:



You are also told that line AB has the following equation: $y = 0{,}5x - 1$.
Calculate the missing coordinate of point B.

17. A is the point $(-3;-5)$ and B is the point $(n;-11)$. AB is perpendicular to line CD with equation $y = \frac{3}{2}x - 5$. Find the value of n.
18. The points $A(4;-3)$, $B(-5;0)$ and $C(-3;p)$ are given. Determine the value of p if A, B and C are collinear.
19. Refer to the diagram below:



   a) Show that $\triangle ABC$ is right-angled. Show your working.
   b) Find the area of $\triangle ABC$.

20. The points $A(-3;1)$, $B(3;-2)$ and $C(9;10)$ are given.
   a) Prove that triangle ABC is a right-angled triangle.
   b) Find the coordinates of D, if $ABCD$ is a parallelogram.
   c) Find the equation of a line parallel to the line BC, which passes through the point A.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1a. 2GCK	1b. 2GCM	1c. 2GCN	2a. 2GCP	2b. 2GCQ	2c. 2GCR	3. 2GCS	4. 2GCT
5. 2GCV	6. 2GCW	7. 2GCX	8. 2GCY	9. 2GCZ	10. 2GD2	11. 2GD3	12. 2GD4
13. 2GD5	14. 2GD6	15. 2GD7	16. 2GD8	17. 2GD9	18. 2GDB	19. 2GDC	20. 2GDD

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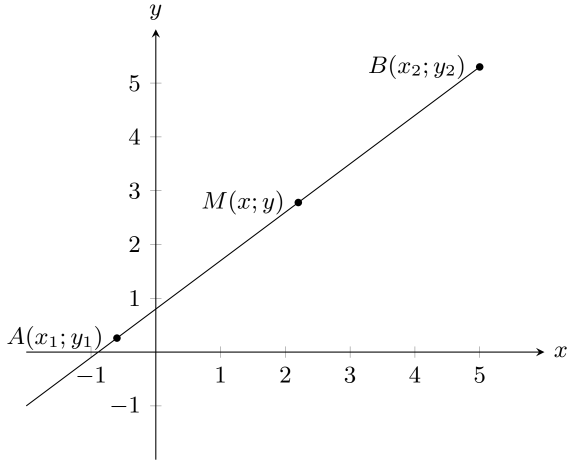
8.4 Mid-point of a line EMA6H

Investigation: Finding the mid-point of a line

On graph paper, accurately plot the points $P\,(2;1)$ and $Q\,(-2;2)$ and draw the line PQ.

- Fold the piece of paper so that point P is exactly on top of point Q.
- Where the folded line intersects with line PQ, label point S
- Count the blocks and find the exact position of S.
- Write down the coordinates of S.

To calculate the coordinates of the mid-point $M(x;y)$ of any line between the points $A(x_1;y_1)$ and $B(x_2;y_2)$, we use the following formulae:



$$x = \frac{x_1 + x_2}{2}$$
$$y = \frac{y_1 + y_2}{2}$$

From this we obtain the mid-point of a line:

Mid-point $M(x;y) = \left(\frac{x_1 + x_2}{2} ; \frac{y_1 + y_2}{2}\right)$

VISIT:
This video shows some examples of finding the mid-point of a line.
See video: 2GDF at www.everythingmaths.co.za

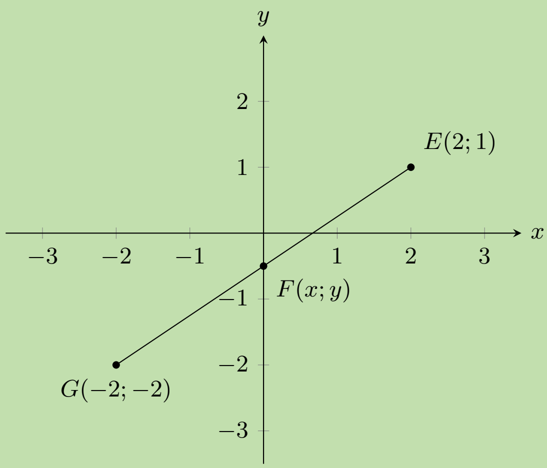
Worked example 10: Calculating the mid-point

QUESTION

Calculate the coordinates of the mid-point $F(x;y)$ of the line between point $E(2;1)$ and point $G(-2;-2)$.

SOLUTION

Step 1: Draw a sketch



From the sketch, we can estimate that F will lie on the y-axis, with a negative y-coordinate.

Step 2: Assign values to $(x_1; y_1)$ and $(x_2; y_2)$

$$x_1 = -2 \quad y_1 = -2 \quad x_1 = 2 \quad y_2 = 1$$

Step 3: Write down the mid-point formula

$$F\,(x; y) = \left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2} \right)$$

Step 4: Substitute values into the mid-point formula

$$\begin{aligned} x &= \frac{x_1 + x_2}{2} \\ &= \frac{-2 + 2}{2} \\ &= 0 \end{aligned}$$

$$\begin{aligned} y &= \frac{y_1 + y_2}{2} \\ &= \frac{-2 + 1}{2} \\ &= -\frac{1}{2} \end{aligned}$$

Step 5: Write the answer

The mid-point is at $F\left(0; -\frac{1}{2}\right)$.

Looking at the sketch we see that this is what we expect for the coordinates of F.

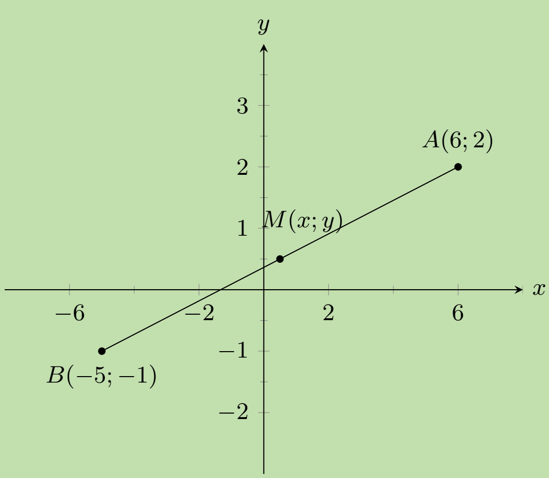
Worked example 11: Calculating the mid-point

QUESTION

Find the mid-point of line AB, given $A\,(6; 2)$ and $B\,(-5; -1)$.

SOLUTION

Step 1: Draw a sketch



From the sketch, we can estimate that M will lie in quadrant I, with positive x- and y-coordinates.

Step 2: Assign values to $(x_1; y_1)$ and $(x_2; y_2)$

Let the mid-point be $M\left(x; y\right)$

$$x_1 = 6 \quad y_1 = 2 \quad x_2 = -5 \quad y_2 = -1$$

Step 3: Write down the mid-point formula

$$M\left(x; y\right) = \left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$$

Step 4: Substitute values and simplify

$$M\left(x; y\right) = \left(\frac{6 - 5}{2}; \frac{2 - 1}{2}\right) = \left(\frac{1}{2}; \frac{1}{2}\right)$$

Step 5: Write the final answer

$M\left(\frac{1}{2}; \frac{1}{2}\right)$ is the mid-point of line AB.

We expected M to have a positive x- and y-coordinate and this is indeed what we have found by calculation.

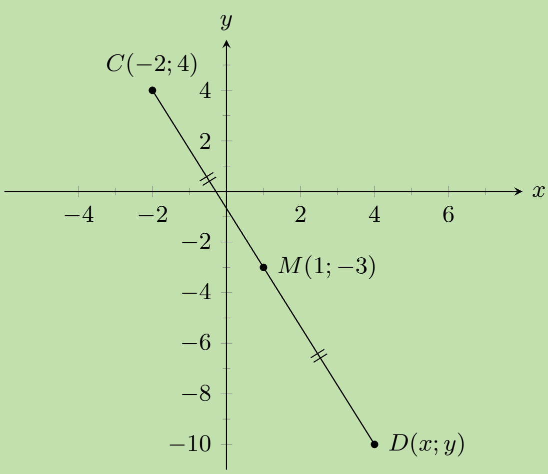
Worked example 12: Using the mid-point formula

QUESTION

The line joining $C\left(-2; 4\right)$ and $D\left(x; y\right)$ has the mid-point $M\left(1; -3\right)$. Find point D.

SOLUTION

Step 1: Draw a sketch



From the sketch, we can estimate that D will lie in Quadrant IV, with a positive x- and negative y-coordinate.

Step 2: Assign values to $(x_1; y_1)$ and $(x_2; y_2)$

Let the coordinates of C be $(x_1; y_1)$ and the coordinates of D be $(x_2; y_2)$.

$$x_1 = -2 \quad y_1 = 4 \quad x_2 = x \quad y_2 = y$$

Step 3: Write down the mid-point formula

$$M\,(x; y) = \left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$$

Step 4: Substitute values and solve for x_2 and y_2

$$\begin{aligned} 1 &= \frac{-2 + x_2}{2} \\ 1 \times 2 &= -2 + x_2 \\ 2 &= -2 + x_2 \\ x_2 &= 2 + 2 \\ x_2 &= 4 \end{aligned} \qquad \begin{aligned} -3 &= \frac{4 + y_2}{2} \\ -3 \times 2 &= 4 + y_2 \\ -6 &= 4 + y_2 \\ y_2 &= -6 - 4 \\ y_2 &= -10 \end{aligned}$$

Step 5: Write the final answer

The coordinates of point D are $(4; -10)$.

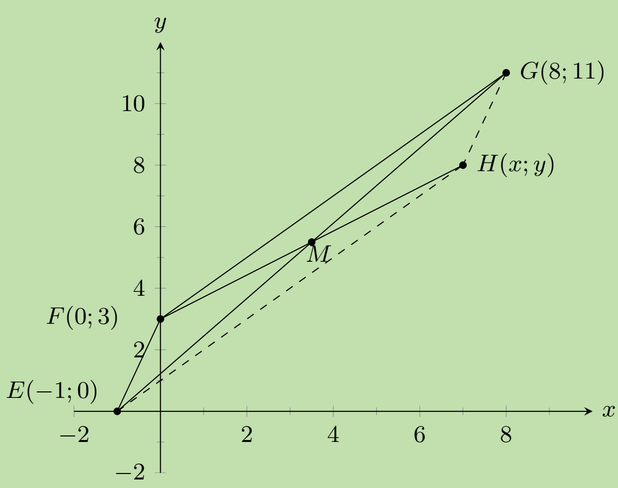
Worked example 13: Using the mid-point formula

QUESTION

Points $E\,(-1; 0)$, $F\,(0; 3)$, $G\,(8; 11)$ and $H\,(x; y)$ are points on the Cartesian plane. Find $H\,(x; y)$ if $EFGH$ is a parallelogram.

SOLUTION

Step 1: Draw a sketch



Method: the diagonals of a parallelogram bisect each other, therefore the mid-point of EG will be the same as the mid-point of FH. We must first find the mid-point of EG. We can then use it to determine the coordinates of point H.

Step 2: Assign values to $(x_1;y_1)$ and $(x_2;y_2)$

Let the mid-point of EG be $M\,(x;y)$

$$x_1=-1 \quad y_1=0 \quad x_2=8 \quad y_2=11$$

Step 3: Write down the mid-point formula

$$M\,(x;y)=\left(\frac{x_1+x_2}{2};\frac{y_1+y_2}{2}\right)$$

Step 4: Substitute values calculate the coordinates of M

$$M\,(x;y)=\left(\frac{-1+8}{2};\frac{0+11}{2}\right)=\left(\frac{7}{2};\frac{11}{2}\right)$$

Step 5: Use the coordinates of M to determine H

M is also the mid-point of FH so we use $M\left(\frac{7}{2};\frac{11}{2}\right)$ and $F\,(0;3)$ to solve for $H\,(x;y)$.

Step 6: Substitute values and solve for x and y

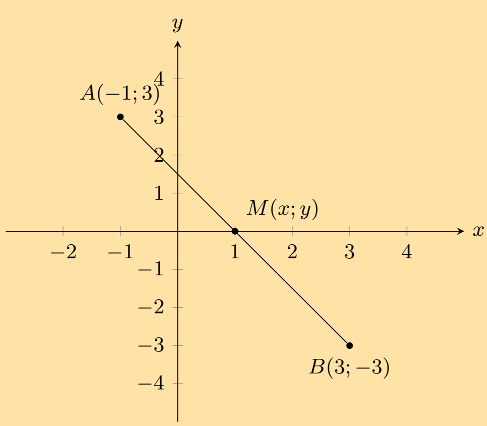
$$\begin{aligned}\frac{7}{2}&=\frac{0+x}{2} & \qquad \frac{11}{2}&=\frac{3+y}{2}\\ 7&=x+0 & 11&=3+y\\ x&=7 & y&=8\end{aligned}$$

Step 7: Write the final answer

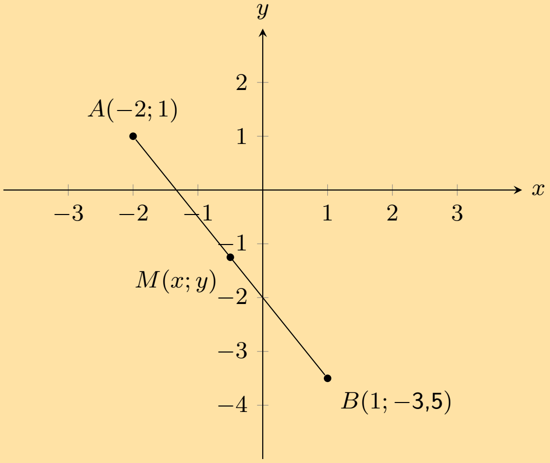
The coordinates of H are $(7;8)$.

Exercise 8 – 5:

1. Calculate the coordinates of the mid-point (M) between point $A(-1;3)$ and point $B(3;-3)$ in the following diagram:



2. Calculate the coordinates of the mid-point (M) between point $A(-2;1)$ and point $B(1;-3{,}5)$ in the following diagram:



3. Find the mid-points of the following lines:
   a) $A(2;5)$, $B(-4;7)$
   b) $C(5;9)$, $D(23;55)$
   c) $E(x+2;y-1)$, $F(x-5;y-4)$
4. The mid-point M of PQ is $(3;9)$. Find P if Q is $(-2;5)$.
5. $PQRS$ is a parallelogram with the points $P(5;3)$, $Q(2;1)$ and $R(7;-3)$. Find point S.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GDG 2. 2GDH 3a. 2GDJ 3b. 2GDK 3c. 2GDM 4. 2GDN 5. 2GDP

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8.5 Chapter summary EMA6J

See presentation: 2GDQ at www.everythingmaths.co.za

- A point is an ordered pair of numbers written as $(x;y)$.
- Distance is a measure of the length between two points.
- The formula for finding the distance between any two points is:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

- The gradient between two points is determined by the ratio of vertical change to horizontal change.
- The formula for finding the gradient of a line is:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

- A straight line is a set of points with a constant gradient between any two of the points.
- The standard form of the straight line equation is $y = mx + c$.
- The equation of a straight line can also be written as

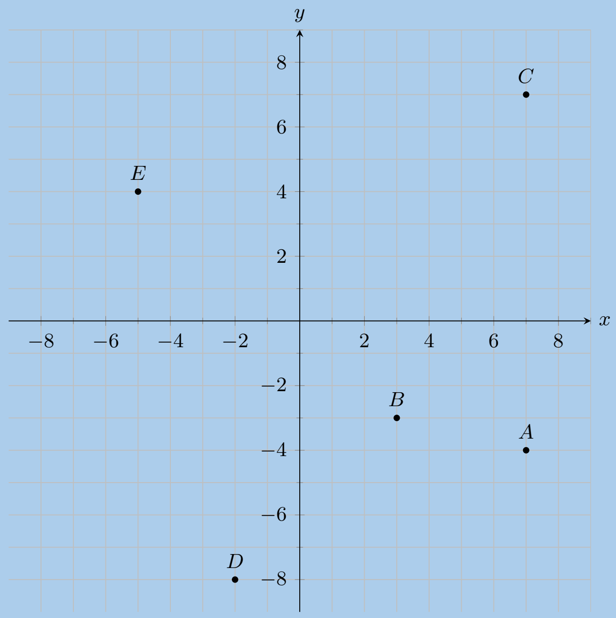
$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

- If two lines are parallel, their gradients are equal.
- If two lines are perpendicular, the product of their gradients is equal to -1.
- For horizontal lines the gradient is equal to 0.
- For vertical lines the gradient is undefined.
- The formula for finding the mid-point between two points is:

$$M\left(x;y\right)=\left(\frac{x_1+x_2}{2};\frac{y_1+y_2}{2}\right)$$

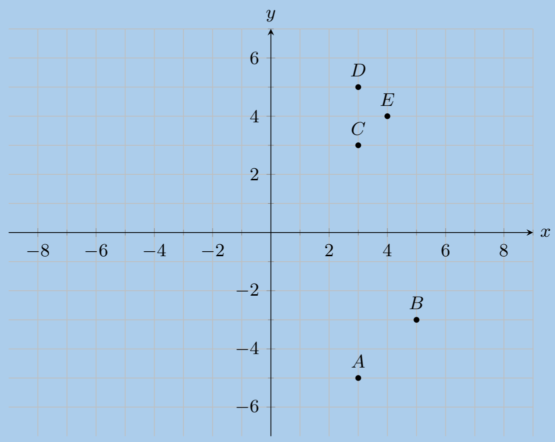
End of chapter Exercise 8 – 6:

1. You are given the following diagram, with various points shown:



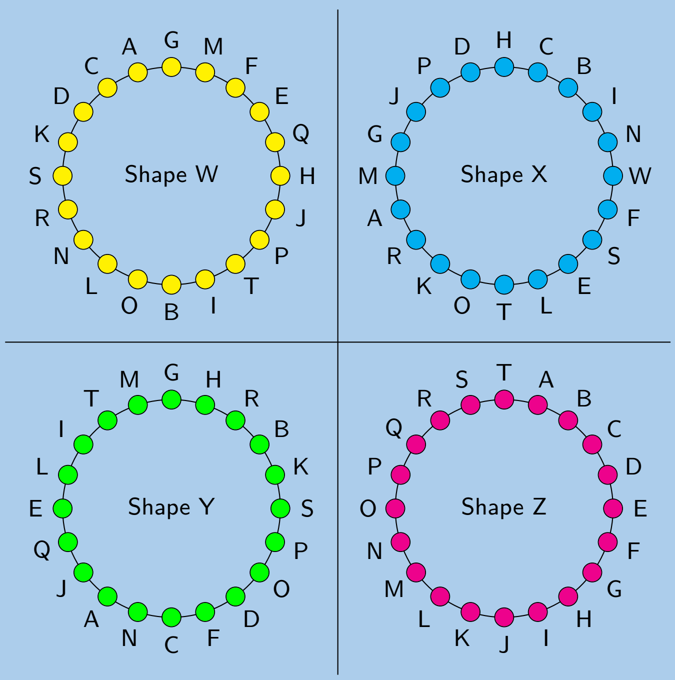
   Find the coordinates of points A, B, C, D and E.

2. You are given the following diagram, with various points shown:



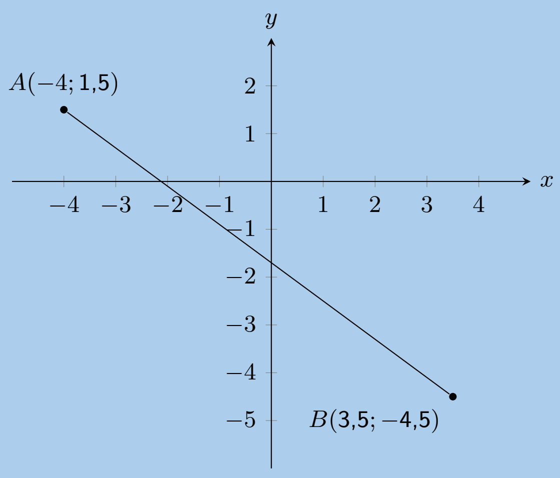
   Which point lies at the coordinates $(3;-5)$?

3. You are given the following diagram, with 4 shapes drawn.
All the shapes are identical, but each shape uses a different naming convention:



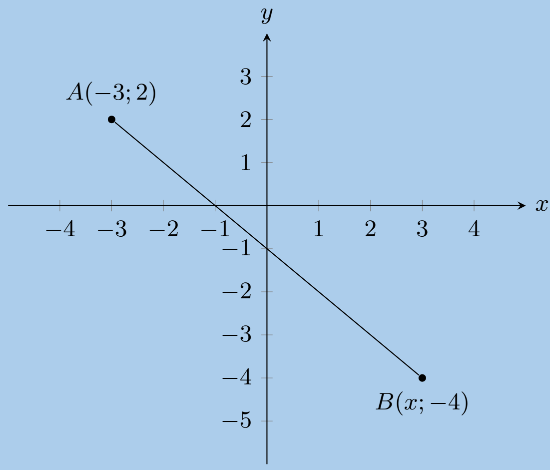
Which shape uses the correct naming convention?

4. Represent the following figures in the Cartesian plane:
   a) Triangle DEF with $D(1;2)$, $E(3;2)$ and $F(2;4)$.
   b) Quadrilateral $GHIJ$ with $G(2;-1)$, $H(0;2)$, $I(-2;-2)$ and $J(1;-3)$.
   c) Quadrilateral $MNOP$ with $M(1;1)$, $N(-1;3)$, $O(-2;3)$ and $P(-4;1)$.
   d) Quadrilateral $WXYZ$ with $W(1;-2)$, $X(-1;-3)$, $Y(2;-4)$ and $Z(3;-2)$.
5. You are given the following diagram:



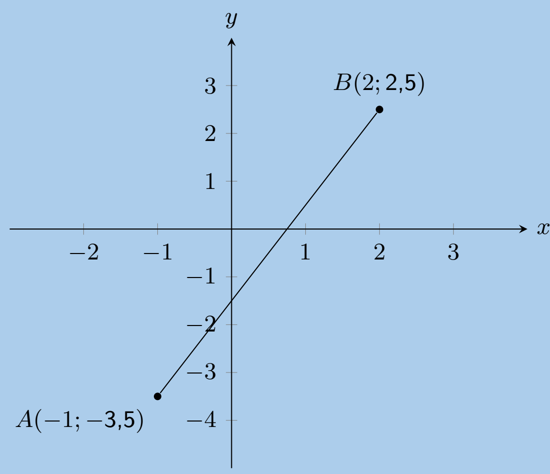
Calculate the length of line AB, correct to 2 decimal places.

6. The following picture shows two points on the Cartesian plane, A and B.



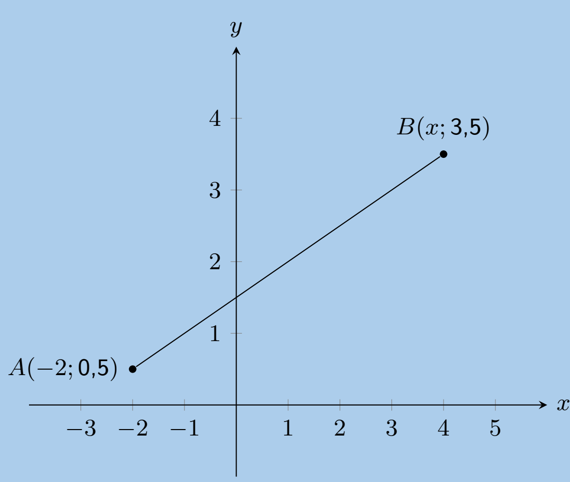
The distance between the points is 8,4853. Calculate the missing coordinate of point B.

7. You are given the following diagram:



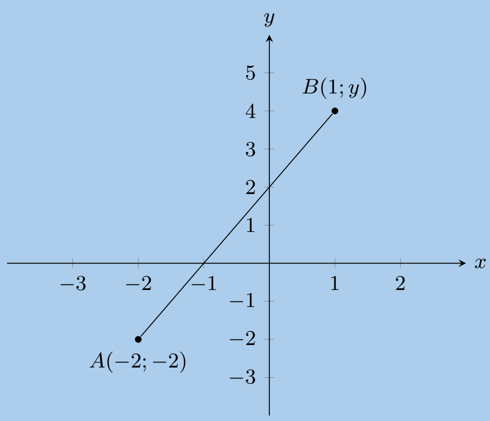
Calculate the gradient (m) of line AB. The coordinates are $A(-1;-3{,}5)$ and $B(2;2{,}5)$ respectively.

8. You are given the following diagram:



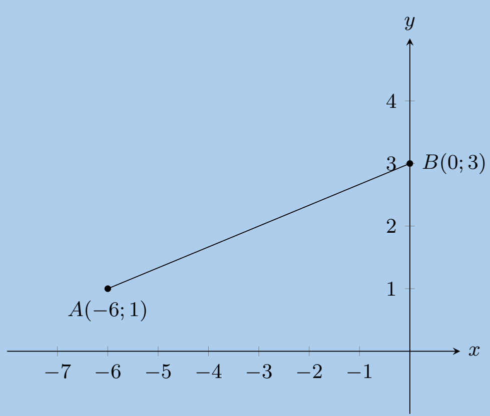
You are also told that line AB has a gradient, m, of 0,5. Calculate the missing co-ordinate of point B.

9. You are given the following diagram:



You are also told that line AB has a gradient, m, of 2. Calculate the missing co-ordinate of point B.

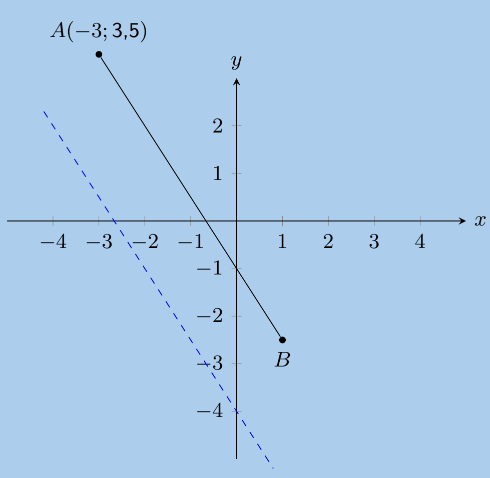
10. In the diagram, A is the point $(-6; 1)$ and B is the point $(0; 3)$.



a) Find the equation of line AB.

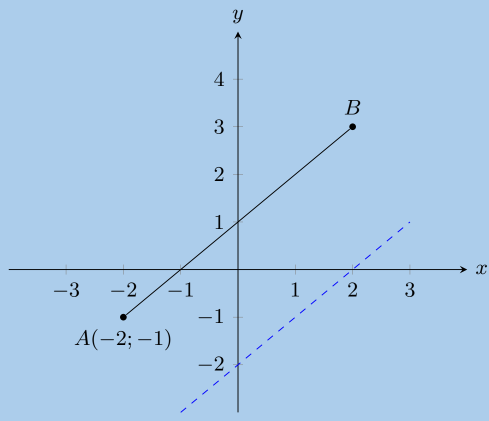
b) Calculate the length of AB.

11. You are given the following diagram:



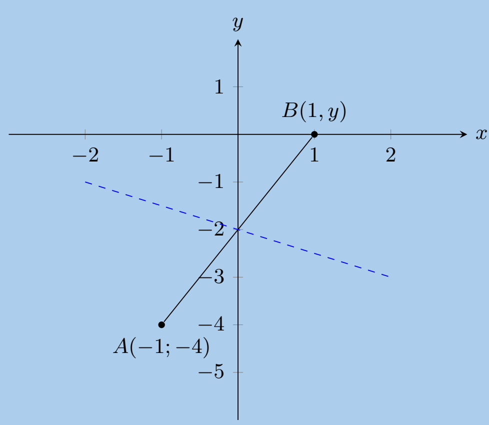
You are also told that line AB runs parallel to the following line: $y = -1{,}5x - 4$. Point A is at $(-3; 3{,}5)$. Find the equation of the line AB.

12. You are given the following diagram:



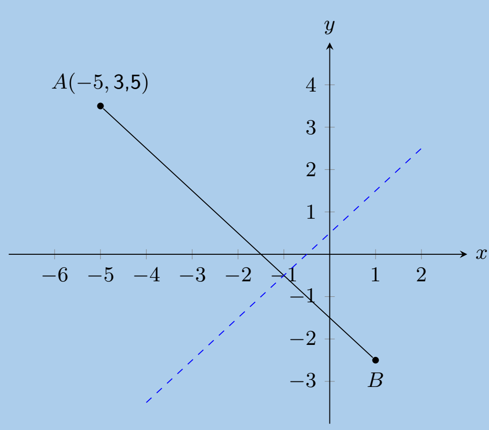
You are also told that line AB runs parallel to the following line: $y = x - 2$. Calculate the missing co-ordinate of point $B(x; 3)$.

13. You are given the following diagram:



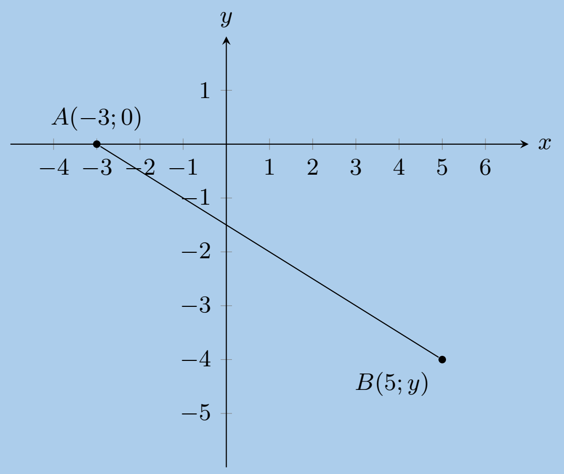
You are also told that line AB runs perpendicular to the following line: $y = -0{,}5x - 2$. Calculate the missing co-ordinate of point B.

14. The graph here shows line, AB. The blue dashed line is perpendicular to AB.



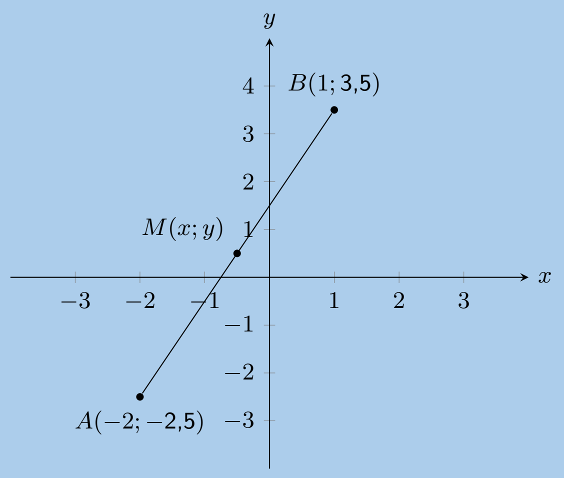
The equation of the blue dashed line is $y = x + 0{,}5$. Point A is at $(-5; 3{,}5)$. Determine the equation of line AB.

15. You are given the following diagram:



You are also told that line AB has the following equation: $y = -0{,}5x - 1{,}5$. Calculate the missing co-ordinate of point B.

16. You are given the following diagram:

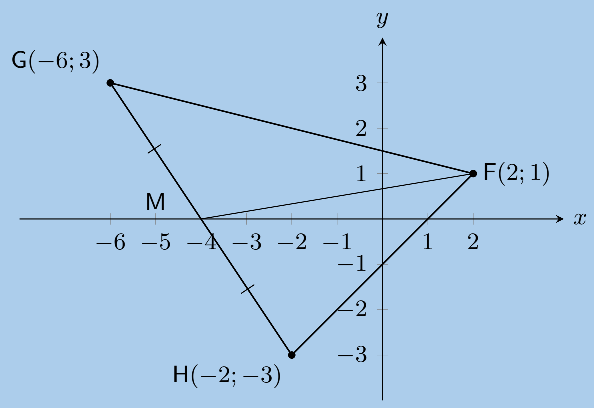


Calculate the coordinates of the mid-point (M) between point $A(-2;-2{,}5)$ and point $B(1;3{,}5)$ correct to 1 decimal place.

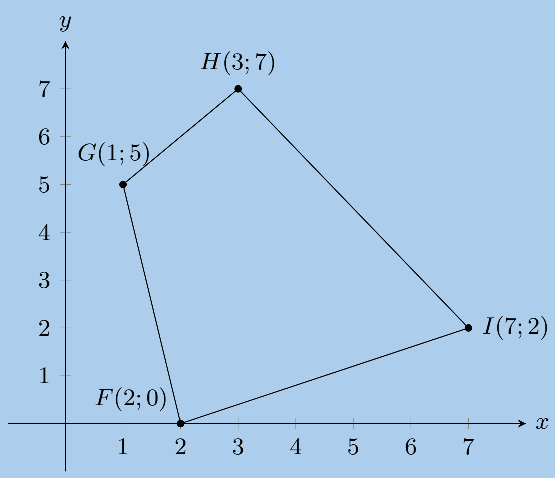
17. $A(-2;3)$and $B(2;6)$ are points in the Cartesian plane. $C(a;b)$ is the mid-point of AB. Find the values of a and b.

18. Determine the equations of the following straight lines:

    a) passing through $P(5;5)$ and $Q(-2;12)$.
    b) parallel to $y = 3x + 4$ and passing through $(4;0)$.
    c) passing through $F(2;1)$ and the mid-point of GH where $G(-6;3)$ and $H(-2;-3)$.



19. In the diagram below, the vertices of the quadrilateral are $F(2;0)$, $G(1;5)$, $H(3;7)$ and $I(7;2)$.



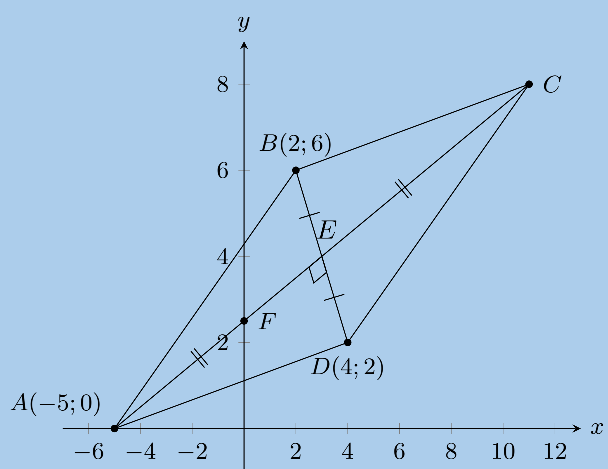
   a) Calculate the lengths of the sides of $FGHI$.
   b) Are the opposite sides of $FGHI$ parallel?
   c) Do the diagonals of $FGHI$ bisect each other?
   d) Can you state what type of quadrilateral $FGHI$ is? Give reasons for your answer.

20. Consider a quadrilateral $ABCD$ with vertices $A(3;2)$, $B(4;5)$, $C(1;7)$ and $D(1;3)$.
   a) Draw the quadrilateral.
   b) Find the lengths of the sides of the quadrilateral.

21. $ABCD$ is a quadrilateral with vertices $A(0;3)$, $B(4;3)$, $C(5;-1)$ and $D(-1;-1)$.
   a) Show by calculation that:
      i. $AD = BC$
      ii. $AB \parallel DC$
   b) What type of quadrilateral is $ABCD$?
   c) Show that the diagonals AC and BD do not bisect each other.

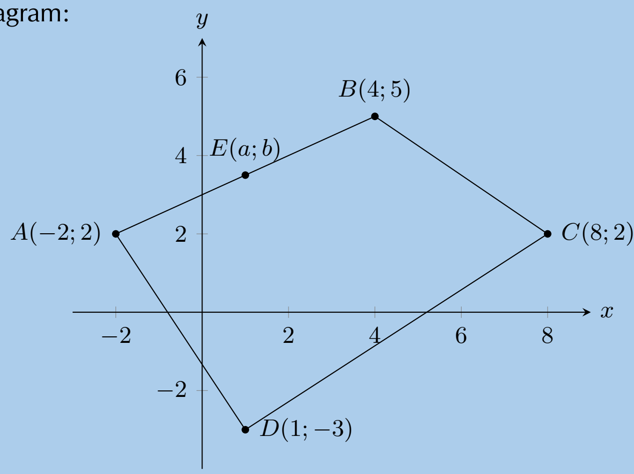
22. P, Q, R and S are the points $(-2;0)$, $(2;3)$, $(5;3)$ and $(-3;-3)$ respectively.
   a) Show that:
      i. $SR = 2PQ$
      ii. $SR \parallel PQ$
   b) Calculate:
      i. PS
      ii. QR
   c) What kind of quadrilateral is $PQRS$? Give reasons for your answer.

23. $EFGH$ is a parallelogram with vertices $E(-1;2)$, $F(-2;-1)$ and $G(2;0)$. Find the coordinates of H by using the fact that the diagonals of a parallelogram bisect each other.
24. $PQRS$ is a quadrilateral with points $P(0;-3)$, $Q(-2;5)$, $R(3;2)$ and $S(3;-2)$ in the Cartesian plane.
    a) Find the length of QR.
    b) Find the gradient of PS.
    c) Find the mid-point of PR.
    d) Is $PQRS$ a parallelogram? Give reasons for your answer.
25. Consider triangle ABC with vertices $A(1;3)$, $B(4;1)$ and $C(6;4)$.
    a) Sketch triangle ABC on the Cartesian plane.
    b) Show that ABC is an isosceles triangle.
    c) Determine the coordinates of M, the mid-point of AC.
    d) Determine the gradient of AB.
    e) Show that $D(7;-1)$ lies on the line that goes through A and B.
26. $\triangle PQR$ has vertices $P(1;8)$, $Q(8;7)$ and $R(7;0)$. Show through calculation that $\triangle PQR$ is a right angled isosceles triangle.
27. $\triangle ABC$ has vertices $A(-3;4)$, $B(3;-2)$ and $C(-5;-2)$. M is the mid-point of AC and N is the mid-point of BC. Use $\triangle ABC$ to prove the mid-point theorem using analytical geometry methods.
28. a) List two properties of a parallelogram.
    b) The points $A(-2;-4)$, $B(-4;1)$, $C(2;4)$ and $D(4;-1)$ are the vertices of a quadrilateral. Show that the quadrilateral is a parallelogram.
29. The diagram shows a quadrilateral. The points B and D have the coordinates $(2;6)$ and $(4;2)$ respectively. The diagonals of $ABCD$ bisect each other at right angles. F is the point of intersection of line AC with the y-axis.



    a) Determine the gradient of AC.
    b) Show that the equation of AC is given by $2y = x + 5$.
    c) Determine the coordinates of C.
30. $A(4;-1)$, $B(-6;-3)$ and $C(-2;3)$ are the vertices of $\triangle ABC$.
    a) Find the length of BC, correct to 1 decimal place.
    b) Calculate the gradient of AC.
    c) If P has coordinates $(-26;19)$, show that A, C and P are collinear.
    d) Determine the equation of line BC.
    e) Show that $\triangle ABC$ is right-angled.

31. Given the following diagram:



a) If E is the mid-point of AB, find the values of a and b.
b) Find the equation of the line perpendicular to BC, which passes through the origin.
c) Find the coordinates of the mid-point of diagonal BD.
d) Hence show that $ABCD$ is not a parallelogram.
e) If C could be moved, give its new coordinates so that $ABCD$ would be a parallelogram.

32. A triangle has vertices $A(-1;7)$, $B(8;4)$ and $C(5;-5)$.

a) Calculate the gradient of AB.
b) Prove that the triangle is right-angled at B.
c) Determine the length of AB.
d) Determine the equation of the line from A to the mid-point of BC.
e) Find the area of the triangle ABC.

33. A quadrilateral has vertices $A(0;5)$, $B(-3;-4)$, $C(0;-5)$ and $D(4;k)$ where $k \geq 0$.

a) What should k be so that AD is parallel to CD?
b) What should k be so that $CD = \sqrt{52}$?

34. On the Cartesian plane, the three points $P(-3;4)$, $Q(7;-1)$ and $R(3;b)$ are collinear.

a) Find the length of PQ.
b) Find the gradient of PQ.
c) Find the equation of PQ.
d) Find the value of b.

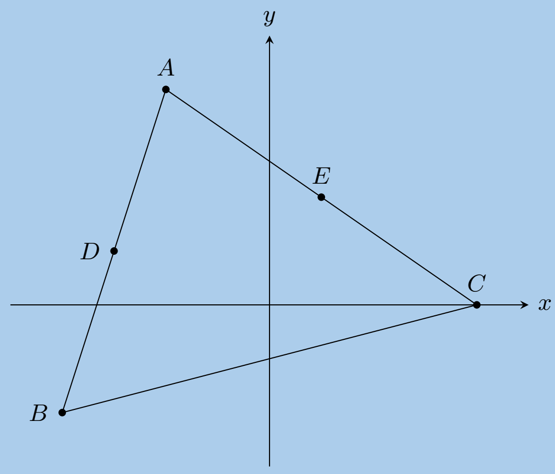
35. Given $A(4;9)$ and $B(-2;-3)$.

a) Find the mid-point M of AB.
b) Find the gradient of AB.
c) Find the gradient of the line perpendicular to AB.
d) Find the equation of the perpendicular bisector of AB.
e) Find the equation of the line parallel to AB, passing through $(0;6)$.

36. $L(-1;-1)$, $M(-2;4)$, $N(x;y)$ and $P(4;0)$ are the vertices of parallelogram $LMNP$.

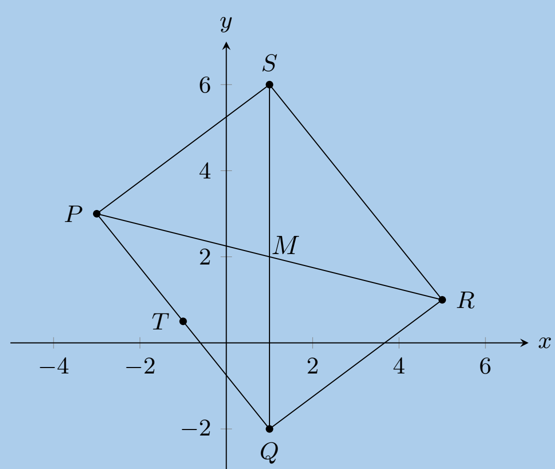
a) Determine the coordinates of N.
b) Show that MP is perpendicular to LN and state what type of quadrilateral $LMNP$ is, other than a parallelogram.
c) Show that $LMNP$ is a square.

37. $A(-2;4)$, $B(-4;-2)$ and $C(4;0)$ are the vertices of $\triangle ABC$. D and $E(1;2)$ are the mid-points of AB and AC respectively.



a) Find the gradient of BC.
b) Show that the coordinates of D, the mid-point of AB are $(-3;1)$.
c) Find the length of DE.
d) Find the gradient of DE. Make a conjecture regarding lines BC and DE.
e) Determine the equation of BC.

38. In the diagram points $P(-3;3)$, $Q(1;-2)$, $R(5;1)$ and $S(x;y)$ are the vertices of a parallelogram.

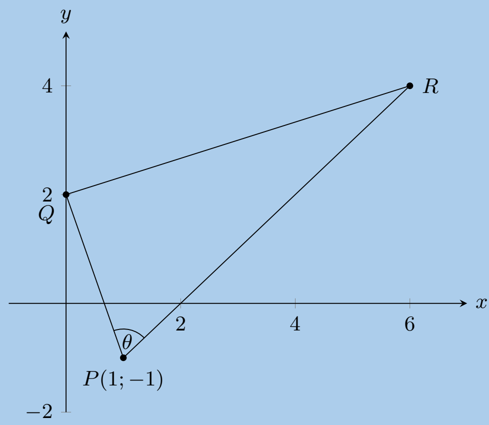


a) Calculate the length of PQ.
b) Find the coordinates of M where the diagonals meet.
c) Find T, the mid-point of PQ.
d) Show that $MT \parallel QR$.
e) Calculate the coordinates of S.

39. The coordinates of $\triangle PQR$ are as follows: $P(5;1)$, $Q(1;3)$ and $R(1;-2)$.

a) Through calculation, determine whether the triangle is equilateral, isosceles or scalene. Be sure to show all your working.
b) Find the coordinates of points S and T, the mid-points of PQ and QR.
c) Determine the gradient of the line ST.
d) Prove that $ST \parallel PR$.

40. The following diagram shows $\triangle PQR$ with $P(-1;1)$. The equation of QR is $x - 3y = -6$ and the equation of PR is $x - y - 2 = 0$. $R\hat{P}Q = \theta$.



a) Write down the coordinates of Q.
b) Prove that $PQ \perp QR$.
c) Write down the gradient of PR.
d) If the y-coordinate of R is 4, calculate the x-coordinate.
e) Find the equation of the line from P to S (the mid-point of QR).

41. The points $E(-3;0)$, $L(3;5)$ and $S(t+1, 2{,}5)$ are collinear.

a) Determine the value of t.
b) Determine the values of a and b if the equation of the line passing through E, L and S is $\dfrac{x}{a} + \dfrac{y}{b} = 1$.

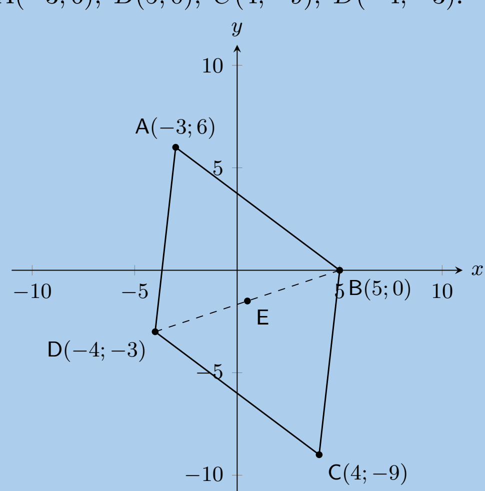
42. Given: $A(-3;-4), B(-1;-7), C(2;-5)$ and $D(0;-2)$.

a) Calculate the distance AC and the distance BD. Leave your answers in surd form.
b) Determine the coordinates of M, the mid-point of BD.
c) Prove that $AM \perp BD$.
d) Prove that A, M and C are collinear.
e) What type of quadrilateral is $ABCD$?

43. $M(2;-2)$ is the mid-point of AB with point $A(3;1)$. Determine:

a) the coordinates of B.
b) the gradient of AM.
c) the equation of the line AM.
d) the perpendicular bisector of AB.

44. ABCD is a quadrilateral with $A(-3;6),\ B(5;0),\ C(4;-9),\ D(-4;-3)$.



a) Determine the coordinates of E, the mid-point of BD.

b) Prove that $ABCD$ is a parallelogram.

c) Find the equation of diagonal BD.

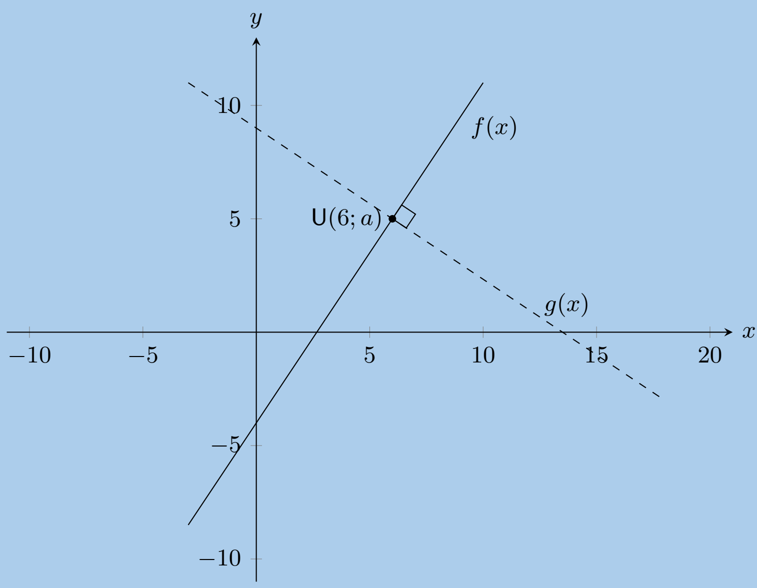
d) Determine the equation of the perpendicular bisector of BD.

e) Determine the gradient of AC.

f) Is $ABCD$ a rhombus? Explain why or why not.

g) Find the length of AB.

45. In the diagram below, $f(x) = \frac{3}{2}x - 4$ is sketched with $U(6; a)$ on $f(x)$.

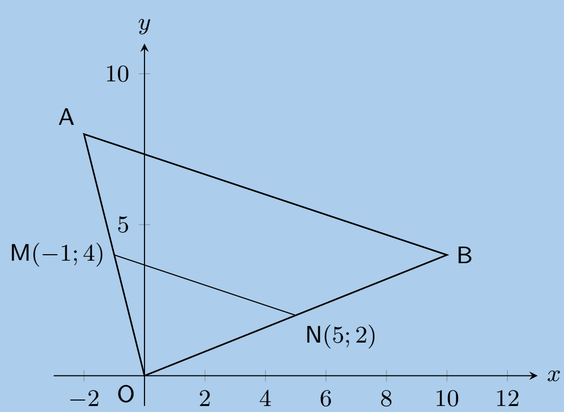


a) Determine the value of a in $U(6; a)$.

b) A line, $g(x)$, passing through U, is perpendicular to $f(x)$. $V(b; 4)$ lies on $g(x)$. Determine the value of b.

c) If $U(6; 5)$, $V(7\frac{1}{2}, 4)$ and $W(1; c)$ are collinear, determine the value of c.

46. In the diagram below, M and N are the mid-points of OA and OB respectively.



a) Calculate the gradient of MN.

b) Find the equation of the line through M and N in the form $y = mx + c$.

c) Show that $AB \parallel MN$.

d) Write down the value of the ratio: $\dfrac{\text{area } \triangle OAB}{\text{area } \triangle OMN}$.

e) Write down the coordinates of P such that $OAPB$ is a parallelogram.

47. $A(6;-4)$, $B(8;2)$, $C(3;a)$ and $D(b;c)$ are points on the Cartesian plane. Determine the value of:

a) a if A, B and C are collinear.

b) b and c if B is the mid-point of A and D.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GDR	2. 2GDS	3. 2GDT	4a. 2GDV	4b. 2GDW	4c. 2GDX
4d. 2GDY	5. 2GDZ	6. 2GF2	7. 2GF3	8. 2GF4	9. 2GF5
10. 2GF6	11. 2GF7	12. 2GF8	13. 2GF9	14. 2GFB	15. 2GFC
16. 2GFD	17. 2GFF	18. 2GFG	19. 2GFH	20. 2GFJ	21. 2GFK
22. 2GFM	23. 2GFN	24. 2GFP	25. 2GFQ	26. 2GFR	27. 2GFS
28. 2GFT	29. 2GFV	30. 2GFW	31. 2GFX	32. 2GFY	33. 2GFZ
34. 2GG2	35. 2GG3	36. 2GG4	37. 2GG5	38. 2GG6	39. 2GG7
40. 2GG8	41. 2GG9	42. 2GGB	43. 2GGC	44. 2GGD	45. 2GGF
46. 2GGG	47. 2GGH				

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CHAPTER 9

Finance and growth

9.1 Introduction EMA6K

In this chapter, we apply mathematical skills to everyday financial situations.

If you had R 1000, you could either keep it in your piggy bank, or deposit it into a bank account. If you deposit the money into a bank account, you are effectively lending money to the bank. Because you are lending the bank money, you can expect some extra money back. This is known as interest. Similarly, if you borrow money from a bank, then you can expect to pay interest on the loan. Interest is charged at a percentage of the money owed over the period of time it takes to pay back the loan. This means that the longer the loan exists, the more interest will have to be paid on it.

Figure 9.1: The entrance to the Johannesburg Stock Exchange (JSE) located in Sandton, Johannesburg - the financial centre of South Africa. The JSE is Africa's largest stock exchange and the 19^{th} largest in the world. Each month, more than 60 billion rand worth of shares are traded on the JSE.

The concept is simple, yet it is core to the world of finance. Accountants, actuaries and bankers can spend their entire working career dealing with the effects of interest on financial matters.

DEFINITION: *Interest*

In finance, interest is the money charged for borrowing money. It is usually expressed as a percentage of the borrowed amount.

9.2 Simple interest EMA6M

DEFINITION: *Simple interest*

Simple interest is interest calculated only on the initial amount that you invested.

As an easy example of simple interest, consider how much we will get by investing R 1000 for 1 year with a bank that pays 5% p.a. simple interest.

At the end of the year we have:

$$\begin{aligned}\text{Interest} &= \text{R } 1000 \times 5\% \\ &= \text{R } 1000 \times \frac{5}{100} \\ &= \text{R } 1000 \times 0{,}05 \\ &= \text{R } 50\end{aligned}$$

With an opening balance of R 1000 at the start of the year, the closing balance at the end of the year will therefore be:

$$\begin{aligned}\text{Closing balance} &= \text{Opening balance} + \text{Interest}\\ &= \text{R } 1000 + \text{R } 50\\ &= \text{R } 1050\end{aligned}$$

The opening balance in financial calculations is often called the principal, denoted as P (R 1000 in the example). The interest rate is usually labelled i (5% p.a. in the example and "p.a." means per annum or per year). The interest amount is labelled I (R 50 in the example).

So we can see that:

$$I = P \times i$$

and:

$$\begin{aligned}\text{Closing balance} &= \text{Opening balance} + \text{Interest}\\ &= P + I\\ &= P + P \times i\\ &= P(1+i)\end{aligned}$$

The above calculations give a good idea of what the simple interest formula looks like. However, the example shows an investment that lasts for only one year. If the investment or loan is over a longer period, we need to take this into account. We use the symbol n to indicate time period, which must be given in years.

The general formula for calculating simple interest is

$$A = P(1 + in)$$

Where:

$$\begin{aligned}A &= \text{accumulated amount (final)}\\ P &= \text{principal amount (initial)}\\ i &= \text{interest written as decimal}\\ n &= \text{number of years}\end{aligned}$$

Worked example 1: Calculating interest on a deposit

QUESTION

Carine deposits R 1000 into a special bank account which pays a simple interest rate of 7% p.a. for 3 years. How much will be in her account at the end of the investment term?

SOLUTION

Step 1: Write down known values

$$\begin{aligned}P &= 1000\\ i &= 0{,}07\\ n &= 3\end{aligned}$$

Step 2: Write down the formula

$$A = P(1 + in)$$

Step 3: Substitute the values

$$\begin{aligned} A &= 1000(1 + 0{,}07 \times 3) \\ &= 1210 \end{aligned}$$

Step 4: Write the final answer

At the end of 3 years, Carine will have R 1210 in her bank account.

Worked example 2: Calculating interest on a loan

QUESTION

Sarah borrows R 5000 from her neighbour at an agreed simple interest rate of 12,5% p.a. She will pay back the loan in one lump sum at the end of 2 years. How much will she have to pay her neighbour?

SOLUTION

Step 1: Write down the known variables

$$\begin{aligned} P &= 5000 \\ i &= 0{,}125 \\ n &= 2 \end{aligned}$$

Step 2: Write down the formula

$$A = P(1 + in)$$

Step 3: Substitute the values

$$\begin{aligned} A &= 5000(1 + 0{,}125 \times 2) \\ &= 6250 \end{aligned}$$

Step 4: Write the final answer

At the end of 2 years, Sarah will pay her neighbour R 6250.

We can use the simple interest formula to find pieces of missing information. For example, if we have an amount of money that we want to invest for a set amount of time to achieve a goal amount, we can rearrange the variables to solve for the required interest rate. The same principles apply to finding the length of time we would need to invest the money, if we knew the principal and accumulated amounts and the interest rate.

Important: to get a more accurate answer, try to do all your calculations on the calculator in one go. This will prevent rounding off errors from influencing your final answer.

Worked example 3: Determining the investment period to achieve a goal amount

QUESTION

Prashant deposits R 30 000 into a bank account that pays a simple interest rate of 7,5% p.a.. How many years must he invest for to generate R 45 000?

SOLUTION

Step 1: Write down the known variables

$$\begin{aligned} A &= 45\ 000 \\ P &= 30\ 000 \\ i &= 0{,}075 \end{aligned}$$

Step 2: Write down the formula

$$A = P(1 + in)$$

Step 3: Substitute the values and solve for n

$$\begin{aligned} 45\ 000 &= 30\ 000\,(1 + 0{,}075 \times n) \\ \frac{45\ 000}{30\ 000} &= 1 + 0{,}075 \times n \\ \frac{45\ 000}{30\ 000} - 1 &= 0{,}075 \times n \\ \frac{\left(\frac{45\ 000}{30\ 000}\right) - 1}{0{,}075} &= n \\ n &= 6\frac{2}{3} \end{aligned}$$

Step 4: Write the final answer

It will take 6 years and 8 months to make R 45 000 from R 30 000 at a simple interest rate of 7,5% p.a.

Worked example 4: Calculating the simple interest rate to achieve the desired growth

QUESTION

At what simple interest rate should Fritha invest if she wants to grow R 2500 to R 4000 in 5 years?

SOLUTION

Step 1: Write down the known variables

$$\begin{aligned} A &= 4000 \\ P &= 2500 \\ n &= 5 \end{aligned}$$

Step 2: Write down the formula

$$A = P(1 + in)$$

Step 3: Substitute the values and solve for i

$$\begin{aligned} 4000 &= 2500(1 + i \times 5) \\ \frac{4000}{2500} &= 1 + i \times 5 \\ \frac{4000}{2500} - 1 &= i \times 5 \\ \frac{\left(\frac{4000}{2500}\right) - 1}{5} &= i \\ i &= 0{,}12 \end{aligned}$$

Step 4: Write the final answer

A simple interest rate of 12% p.a. will be needed when investing R 2500 for 5 years to become R 4000.

Exercise 9 – 1:

1. An amount of R 3500 is invested in a savings account which pays simple interest at a rate of 7,5% per annum. Calculate the balance accumulated by the end of 2 years.
2. An amount of R 4090 is invested in a savings account which pays simple interest at a rate of 8% per annum. Calculate the balance accumulated by the end of 4 years.
3. An amount of R 1250 is invested in a savings account which pays simple interest at a rate of 6% per annum. Calculate the balance accumulated by the end of 6 years.
4. An amount of R 5670 is invested in a savings account which pays simple interest at a rate of 8% per annum. Calculate the balance accumulated by the end of 3 years.
5. Calculate the accumulated amount in the following situations:
   a) A loan of R 300 at a rate of 8% for 1 year.
   b) An investment of R 2250 at a rate of 12,5% p.a. for 6 years.
6. A bank offers a savings account which pays simple interest at a rate of 6% per annum. If you want to accumulate R 15 000 in 5 years, how much should you invest now?

7. Sally wanted to calculate the number of years she needed to invest R 1000 for in order to accumulate R 2500. She has been offered a simple interest rate of 8,2% p.a. How many years will it take for the money to grow to R 2500?
8. Joseph deposited R 5000 into a savings account on his son's fifth birthday. When his son turned 21, the balance in the account had grown to R 18 000. If simple interest was used, calculate the rate at which the money was invested.
9. When his son was 6 years old, Methuli made a deposit of R 6610 in the bank. The investment grew at a simple interest rate and when Methuli's son was 18 years old, the value of the investment was R 11 131,24.

   At what rate was the money invested? Give the answer correct to one decimal place.
10. When his son was 6 years old, Phillip made a deposit of R 5040 in the bank. The investment grew at a simple interest rate and when Phillip's son was 18 years old, the value of the investment was R 7338,24.

    At what rate was the money invested? Give your answer correct to one decimal place.
11. When his son was 10 years old, Lefu made a deposit of R 2580 in the bank. The investment grew at a simple interest rate and when Lefu's son was 20 years old, the value of the investment was R 3689,40.

    At what rate was the money invested? Give your answer correct to one decimal place.
12. Abdoul wants to invest R 1080 at a simple interest rate of 10,9% p.a.

    How many years will it take for the money to grow to R 3348? Round **up** your answer to the nearest year.
13. Andrew wants to invest R 3010 at a simple interest rate of 11,9% p.a.

    How many years will it take for the money to grow to R 14 448? Round **up** your answer to the nearest year.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GGJ 2. 2GGK 3. 2GGM 4. 2GGN 5a. 2GGP 5b. 2GGQ 6. 2GGR 7. 2GGS
8. 2GGT 9. 2GGV 10. 2GGW 11. 2GGX 12. 2GGY 13. 2GGZ

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9.3 Compound interest EMA6N

Compound interest allows interest to be earned on interest. With simple interest, only the original investment earns interest, but with compound interest, the original investment and the interest earned on it, both earn interest. Compound interest is advantageous for investing money but not for taking out a loan.

DEFINITION: *Compound interest*

Compound interest is the interest earned on the principal amount and on its accumulated interest.

Consider the example of R 1000 invested for 3 years with a bank that pays 5% p.a. compound interest.

At the end of the first year, the accumulated amount is

$$\begin{aligned} A_1 &= P(1+i) \\ &= 1000(1+0{,}05) \\ &= 1050 \end{aligned}$$

The amount A_1 becomes the new principal amount for calculating the accumulated amount at the end of the second year.

$$\begin{aligned} A_2 &= P\left(1+i\right) \\ &= 1050\left(1+0{,}05\right) \\ &= 1000\left(1+0{,}05\right)\left(1+0{,}05\right) \\ &= 1000(1+0{,}05)^2 \end{aligned}$$

Similarly, we use the amount A_2 as the new principal amount for calculating the accumulated amount at the end of the third year.

$$\begin{aligned} A_3 &= P\left(1+i\right) \\ &= 1000(1+0{,}05)^2\left(1+0{,}05\right) \\ &= 1000(1+0{,}05)^3 \end{aligned}$$

Do you see a pattern?

Using the formula for simple interest, we can develop a similar formula for compound interest.

With an opening balance P and an interest rate of i, the closing balanced at the end of the first year is:

$$\text{Closing balance after } 1 \text{ year } = P\left(1+i\right)$$

This is the same as simple interest because it only covers a single year. This closing balance becomes the opening balance for the second year of investment.

$$\begin{aligned} \text{Closing balance after } 2 \text{ years } &= \left[P\left(1+i\right)\right] \times \left(1+i\right) \\ &= P(1+i)^2 \end{aligned}$$

And similarly, for the third year

$$\begin{aligned} \text{Closing balance after } 3 \text{ years } &= \left[P(1+i)^2\right] \times \left(1+i\right) \\ &= P(1+i)^3 \end{aligned}$$

We see that the power of the term $(1+i)$ is the same as the number of years. Therefore the general formula for calculating compound interest is:

$$A = P(1+i)^n$$

Where:

$$\begin{aligned} A &= \text{accumulated amount} \\ P &= \text{principal amount} \\ i &= \text{interest written as a decimal} \\ n &= \text{number of years} \end{aligned}$$

Worked example 5: Compound interest

QUESTION

Mpho wants to invest R 30 000 into an account that offers a compound interest rate of 6% p.a. How much money will be in the account at the end of 4 years?

SOLUTION

Step 1: Write down the known variables

$$\begin{aligned} P &= 30\ 000 \\ i &= 0{,}06 \\ n &= 4 \end{aligned}$$

Step 2: Write down the formula

$$A = P(1+i)^n$$

Step 3: Substitute the values

$$\begin{aligned} A &= 30\ 000(1+0{,}06)^4 \\ &= 37\ 874{,}31 \end{aligned}$$

Step 4: Write the final answer

Mpho will have R 37 874,31 in the account at the end of 4 years.

Worked example 6: Calculating the compound interest rate to achieve the desired growth

QUESTION

Charlie has been given R 5000 for his sixteenth birthday. Rather than spending it, he has decided to invest it so that he can put down a deposit of R 10 000 on a car on his eighteenth birthday. What compound interest rate does he need to achieve this growth? Comment on your answer.

SOLUTION

Step 1: Write down the known variables

$$\begin{aligned} A &= 10\ 000 \\ P &= 5000 \\ n &= 2 \end{aligned}$$

Step 2: Write down the formula

$$A = P(1+i)^n$$

Step 3: Substitute the values and solve for i

$$\begin{aligned}
10\ 000 &= 5000(1+i)^2 \\
\frac{10\ 000}{5000} &= (1+i)^2 \\
\sqrt{\frac{10\ 000}{5000}} &= 1+i \\
\sqrt{\frac{10\ 000}{5000}} - 1 &= i \\
i &= 0{,}4142
\end{aligned}$$

Step 4: Write the final answer and comment

Charlie needs to find an account that offers a compound interest rate of 41,42% p.a. to achieve the desired growth. A typical savings account gives a return of approximately 2% p.a. and an aggressive investment portfolio gives a return of approximately 13% p.a. It therefore seems unlikely that Charlie will be able to invest his money at an interest rate of 41,42% p.a.

The power of compound interest EMA6P

To illustrate how important "interest on interest" is, we compare the difference in closing balances for an investment earning simple interest and an investment earning compound interest. Consider an amount of R 10 000 invested for 10 years, at an interest rate of 9% p.a.

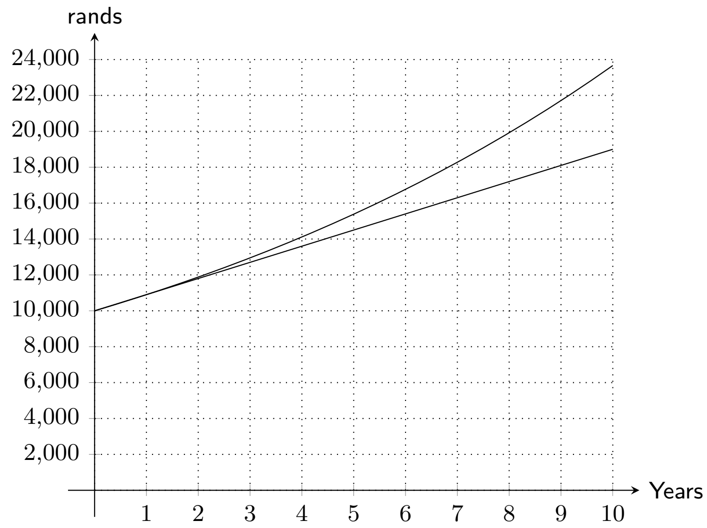
The closing balance for the investment earning simple interest is

$$\begin{aligned}
A &= P\left(1+in\right) \\
&= 10\ 000(1+0{,}09\times 10) \\
&= \text{R } 19\ 000
\end{aligned}$$

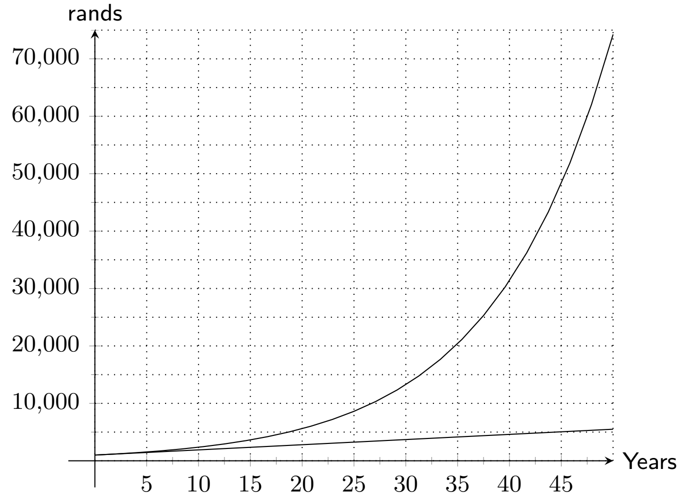
The closing balance for the investment earning compound interest is

$$\begin{aligned}
A &= P(1+i)^n \\
&= 10\ 000(1+0{,}09)^{10} \\
&= \text{R } 23\ 673{,}64
\end{aligned}$$

We plot the growth of the two investments on the same set of axes and note the significant different in their rate of change: simple interest is a straight line graph and compound interest is an exponential graph.



It is easier to see the vast difference in growth if we extend the time period to 50 years:



Keep in mind that this is good news and bad news. When earning interest on money invested, compound interest helps that amount to grow exponentially. But if money is borrowed the accumulated amount of money owed will increase exponentially too.

VISIT:
This video explains the difference between simple and compound interest. Note that the video uses dollars but the calculation is the same for rands.
See video: 2GH3 at www.everythingmaths.co.za

Exercise 9 – 2:

1. An amount of R 3500 is invested in a savings account which pays a compound interest rate of 7,5% p.a. Calculate the balance accumulated by the end of 2 years.
2. An amount of R 3070 is invested in a savings account which pays a compound interest rate of 11,6% p.a. Calculate the balance accumulated by the end of 6 years. As usual with financial calculations, round your answer to two decimal places, but do not round off until you have reached the solution.

3. An amount of R 6970 is invested in a savings account which pays a compound interest rate of 10,2% p.a. Calculate the balance accumulated by the end of 3 years. As usual with financial calculations, round your answer to two decimal places, but do not round off until you have reached the solution.
4. Nicola wants to invest some money at a compound interest rate of 11% p.a. How much money (to the nearest rand) should be invested if she wants to reach a sum of R 100 000 in five years time?
5. Thobeka wants to invest some money at a compound interest rate of 11,8% p.a. How much money should be invested if she wants to reach a sum of R 30 000 in 2 years' time? Round up your answer to the nearest rand.
6. Likengkeng wants to invest some money at a compound interest rate of 11,4% p.a. How much money should be invested if she wants to reach a sum of R 38 200 in 7 years' time? Round up your answer to the nearest rand.
7. Morgan invests R 5000 into an account which pays out a lump sum at the end of 5 years. If he gets R 7500 at the end of the period, what compound interest rate did the bank offer him?
8. Kabir invests R 1790 into an account which pays out a lump sum at the end of 9 years. If he gets R 2613,40 at the end of the period, what compound interest rate did the bank offer him? Give the answer correct to one decimal place.
9. Bongani invests R 6110 into an account which pays out a lump sum at the end of 7 years. If he gets R 6904,30 at the end of the period, what compound interest rate did the bank offer him? Give the answer correct to one decimal place.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GH4 2. 2GH5 3. 2GH6 4. 2GH7 5. 2GH8 6. 2GH9 7. 2GHB 8. 2GHC 9. 2GHD

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9.4 Calculations using simple and compound interest EMA6Q

Hire purchase EMA6R

As a general rule, it is not wise to buy items on credit. When buying on credit you have to borrow money to pay for the object, meaning you will have to pay more for it due to the interest on the loan. That being said, occasionally there are appliances, such as a fridge, that are very difficult to live without. Most people don't have the cash up front to purchase such items, so they buy it on a hire purchase agreement.

A hire purchase agreement is a financial agreement between the shop and the customer about how the customer will pay for the desired product. The interest on a hire purchase loan is always charged at a simple interest rate and only charged on the amount owing. Most agreements require that a deposit is paid before the product can be taken by the customer. The principal amount of the loan is therefore the cash price minus the deposit. The accumulated loan will be worked out using the number of years the loan is needed for. The total loan amount is then divided into monthly payments over the period of the loan.

IMPORTANT!

Hire purchase is charged at a simple interest rate. When you are asked a hire purchase question, don't forget to always use the simple interest formula.

VISIT:
This video explains hire purchase and shows some examples of hire purchase calculations.
See video: 2GHF at www.everythingmaths.co.za

Worked example 7: Hire purchase

QUESTION

Troy wants to buy an additional screen for his computer which he saw advertised for R 2500 on the internet. There is an option of paying a 10% deposit and then making 24 monthly payments using a hire purchase agreement, where interest is calculated at 7,5% p.a. simple interest. Calculate what Troy's monthly payments will be.

SOLUTION

Step 1: Write down the known variables

A new opening balance is required, as the 10% deposit is paid in cash.

$$\begin{aligned} \text{10\% of 2500} &= 250 \\ \therefore P &= 2500 - 250 = 2250 \\ i &= 0{,}075 \\ n &= \frac{24}{12} = 2 \end{aligned}$$

Step 2: Write down the formula

$$A = P\left(1 + in\right)$$

Step 3: Substitute the values

$$\begin{aligned} A &= 2250\left(1 + 0{,}075 \times 2\right) \\ &= 2587{,}50 \end{aligned}$$

Step 4: Calculate the monthly repayments on the hire purchase agreement

$$\begin{aligned} \text{Monthly payment } &= \frac{2587{,}50}{24} \\ &= 107{,}81 \end{aligned}$$

Step 5: Write the final answer

Troy's monthly payment is R 107,81.

A shop can also add a monthly insurance premium to the monthly instalments. This insurance premium will be an amount of money paid monthly and gives the customer more time between a missed payment and possible repossession of the product.

NOTE:
The monthly payment is also called the monthly instalment.

Worked example 8: Hire purchase with extra conditions

QUESTION

Cassidy wants to buy a TV and decides to buy one on a hire purchase agreement. The TV's cash price is R 5500. She will pay it off over 54 months at an interest rate of 21% p.a. An insurance premium of R 12,50 is added to every monthly payment. How much are her monthly payments?

SOLUTION

Step 1: Write down the known variables

$$\begin{aligned} P &= 5500 \\ i &= 0{,}21 \\ n &= \frac{54}{12} = 4{,}5 \end{aligned}$$

The question does not mention a deposit, therefore we assume that Cassidy did not pay one.

Step 2: Write down the formula

$$A = P\,(1 + in)$$

Step 3: Substitute the values

$$\begin{aligned} A &= 5500\,(1 + 0{,}21 \times 4{,}5) \\ &= 10\ 697{,}50 \end{aligned}$$

Step 4: Calculate the monthly repayments on the hire purchase agreement

$$\begin{aligned} \text{Monthly payment } &= \frac{10\ 697{,}50}{54} \\ &= 198{,}10 \end{aligned}$$

Step 5: Add the insurance premium

$198{,}10 + 12{,}50 = 210{,}60$

Step 6: Write the final answer

Cassidy will pay R 210,60 per month for 54 months until her TV is paid off.

Exercise 9 – 3:

1. Angelique wants to buy a microwave on a hire purchase agreement. The cash price of the microwave is R 4400. She is required to pay a deposit of 10% and pay the remaining loan amount off over 12 months at an interest rate of 9% p.a.
   a) What is the principal loan amount?
   b) What is the accumulated loan amount?
   c) What are Angelique's monthly repayments?
   d) What is the total amount she has paid for the microwave?
2. Nyakallo wants to buy a television on a hire purchase agreement. The cash price of the television is R 5600. She is required to pay a deposit of 15% and pay the remaining loan amount off over 24 months at an interest rate of 14% p.a.
   a) What is the principal loan amount?
   b) What is the accumulated loan amount?
   c) What are Nyakallo's monthly repayments?
   d) What is the total amount she has paid for the television?
3. A company wants to purchase a printer. The cash price of the printer is R 4500. A deposit of 15% is required on the printer. The remaining loan amount will be paid off over 24 months at an interest rate of 12% p.a.
   a) What is the principal loan amount?
   b) What is the accumulated loan amount?
   c) How much will the company pay each month?
   d) What is the total amount the company paid for the printer?
4. Sandile buys a dining room table costing R 8500 on a hire purchase agreement. He is charged an interest rate of 17,5% p.a. over 3 years.
   a) How much will Sandile pay in total?
   b) How much interest does he pay?
   c) What is his monthly instalment?
5. Mike buys a table costing R 6400 on a hire purchase agreement. He is charged an interest rate of 15% p.a. over 4 years.
   a) How much will Mike pay in total?
   b) How much interest does he pay?
   c) What is his monthly instalment?
6. Talwar buys a cupboard costing R 5100 on a hire purchase agreement. He is charged an interest rate of 12% p.a. over 2 years.
   a) How much will Talwar pay in total?
   b) How much interest does he pay?
   c) What is his monthly instalment?
7. A lounge suite is advertised for sale on TV, to be paid off over 36 months at R 150 per month.
   a) Assuming that no deposit is needed, how much will the buyer pay for the lounge suite once it has been paid off?
   b) If the interest rate is 9% p.a., what is the cash price of the suite?
8. Two stores are offering a fridge and washing machine combo package. Store A offers a monthly payment of R 350 over 24 months. Store B offers a monthly payment of R 175 over 48 months.

If both stores offer 7,5% interest, which store should you purchase the fridge and washing machine from if you want to pay the least amount of interest?

9. Tlali wants to buy a new computer and decides to buy one on a hire purchase agreement. The computers cash price is R 4250. He will pay it off over 30 months at an interest rate of 9,5% p.a. An insurance premium of R 10,75 is added to every monthly payment. How much are his monthly payments?
10. Richard is planning to buy a new stove on hire purchase. The cash price of the stove is R 6420. He has to pay a 10% deposit and then pay the remaining amount off over 36 months at an interest rate of 8% p.a. An insurance premium of R 11,20 is added to every monthly payment. Calculate Richard's monthly payments.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GHG 2. 2GHH 3. 2GHJ 4. 2GHK 5. 2GHM 6. 2GHN
7. 2GHP 8. 2GHQ 9. 2GHR 10. 2GHS

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Inflation EMA6S

There are many factors that influence the change in price of an item, one of them is inflation. Inflation is the average increase in the price of goods each year and is given as a percentage. Since the rate of inflation increases year on year, it is calculated using the compound interest formula.

Worked example 9: Calculating future cost based on inflation

QUESTION

Milk costs R 14 for two litres. How much will it cost in 4 years time if the inflation rate is 9% p.a.?

SOLUTION

Step 1: Write down the known variables

$$\begin{aligned} P &= 14 \\ i &= 0{,}09 \\ n &= 4 \end{aligned}$$

Step 2: Write down the formula

$$A = P(1+i)^n$$

Step 3: Substitute the values

$$\begin{aligned} A &= 14(1+0{,}09)^4 \\ &= 19{,}76 \end{aligned}$$

Step 4: Write the final answer

In four years time, two litres of milk will cost R 19,76.

Worked example 10: Calculating past cost based on inflation

QUESTION

A box of chocolates costs R 55 today. How much did it cost 3 years ago if the average rate of inflation was 11% p.a.?

SOLUTION

Step 1: Write down the known variables

$$A = 55$$
$$i = 0{,}11$$
$$n = 3$$

Step 2: Write down the formula

$$A = P(1+i)^n$$

Step 3: Substitute the values and solve for P

$$55 = P(1+0{,}11)^3$$
$$\frac{55}{(1+0{,}11)^3} = P$$
$$\therefore P = 40{,}22$$

Step 4: Write the final answer

Three years ago, the box of chocolates would have cost R 40,22.

Exercise 9 – 4:

1. The price of a bag of apples is R 12. How much will it cost in 9 years time if the inflation rate is 12% p.a.?
2. The price of a bag of potatoes is R 15.
   How much will it cost in 6 years time if the inflation rate is 12% p.a.?
3. The price of a box of popcorn is R 15. How much will it cost in 4 years time if the inflation rate is 11% p.a.?
4. A box of raisins costs R 24 today. How much did it cost 4 years ago if the average rate of inflation was 13% p.a.? Round your answer to 2 decimal places.
5. A box of biscuits costs R 24 today. How much did it cost 5 years ago if the average rate of inflation was 11% p.a.? Round your answer to 2 decimal places.
6. If the average rate of inflation for the past few years was 7,3% p.a. and your water and electricity account is R 1425 on average, what would you expect to pay in 6 years time?

7. The price of popcorn and a cooldrink at the movies is now R 60. If the average rate of inflation is 9,2% p.a. what was the price of popcorn and cooldrink 5 years ago?

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GHT 2. 2GHV 3. 2GHW 4. 2GHX 5. 2GHY 6. 2GHZ 7. 2GJ2

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Population growth EMA6T

Family trees increase exponentially as every person born has the ability to start another family. For this reason we calculate population growth using the compound interest formula.

Worked example 11: Population growth

QUESTION

If the current population of Johannesburg is 3 888 180, and the average rate of population growth in South Africa is 2,1% p.a., what can city planners expect the population of Johannesburg to be in 10 years?

SOLUTION

Step 1: Write down the known variables

$$\begin{aligned} P &= 3\ 888\ 180 \\ i &= 0{,}021 \\ n &= 10 \end{aligned}$$

Step 2: Write down the formula

$$A = P(1+i)^n$$

Step 3: Substitute the values

$$\begin{aligned} A &= 3\ 888\ 180(1+0{,}021)^{10} \\ &= 4\ 786\ 343 \end{aligned}$$

Step 4: Write the final answer

City planners can expect Johannesburg's population to be 4 786 343 in ten years time.

Exercise 9 – 5:

1. The current population of Durban is 3 879 090 and the average rate of population growth in South Africa is 1,1% p.a.

   What can city planners expect the population of Durban to be in 6 years time? Round your answer to the nearest integer.
2. The current population of Polokwane is 3 878 970 and the average rate of population growth in South Africa is 0,7% p.a.

   What can city planners expect the population of Polokwane to be in 12 years time? Round your answer to the nearest integer.
3. A small town in Ohio, USA is experiencing a huge increase in births. If the average growth rate of the population is 16% p.a., how many babies will be born to the 1600 residents in the next 2 years?

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GJ3 2. 2GJ4 3. 2GJ5

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9.5 Foreign exchange rates EMA6V

Different countries have their own currencies. In England, a Big Mac from McDonald's costs £ 4, in South Africa it costs R 20 and in Norway it costs 48 kr. The meal is the same in all three countries but in some places it costs more than in others. If £ 1 = R 12,41 and 1 kr = R 1,37, this means that a Big Mac in England costs R 49,64 and a Big Mac in Norway costs R 65,76.

Exchange rates affect a lot more than just the price of a Big Mac. The price of oil increases when the South African rand weakens. This is because when the rand is weaker, we can buy less of other currencies with the same amount of money.

A currency gets stronger when money is invested in the country. When we buy products that are made in South Africa, we are investing in South African business and keeping the money in the country. When we buy products imported from other countries, we are investing money in those countries and as a result, the rand will weaken. The more South African products we buy, the greater the demand for them will be and more jobs will become available for South Africans. Local is lekker!

NOTE:
The three currencies you are most likely to see are the British pound (£), the American dollar ($) and the euro (€).

VISIT:
This video explains exchange rates and shows some examples of exchange rate calculations.
See video: 2GJ6 at www.everythingmaths.co.za

Worked example 12: Foreign exchange rates

QUESTION

Saba wants to travel to see her family in Spain. She has been given R 10 000 spending money. How many euros can she buy if the exchange rate is currently € 1 =R 10,68?

SOLUTION

Step 1: Write down the equation

Let the equivalent amount in euros be x

$$\begin{aligned} x &= \frac{10\ 000}{10{,}68} \\ &= 936{,}33 \end{aligned}$$

Step 2: Write the final answer

Saba can buy € 936,33 with R 10 000.

Exercise 9 – 6:

1. Bridget wants to buy an iPod that costs £ 100, with the exchange rate currently at £ 1 = R 14. She estimates that the exchange rate will drop to R 12 in a month.
   a) How much will the iPod cost in rands, if she buys it now?
   b) How much will she save if the exchange rate drops to R 12?
   c) How much will she lose if the exchange rate moves to R 15?
2. Mthuli wants to buy a television that costs £ 130, with the exchange rate currently at £ 1 = R 11. He estimates that the exchange rate will drop to R 9 in a month.
   a) How much will the television cost in rands, if he buys it now?
   b) How much will he save if the exchange rate drops to R 9?
   c) How much will he lose if the exchange rate moves to R 19?
3. Nthabiseng wants to buy an iPad that costs £ 120, with the exchange rate currently at £ 1 = R 14. She estimates that the exchange rate will drop to R 9 in a month.
   a) How much will the iPad cost, in rands, if she buys it now?
   b) How much will she save if the exchange rate drops to R 9?
   c) How much will she lose if the exchange rate moves to R 18?
4. Study the following exchange rate table:

Country	Currency	Exchange Rate
United Kingdom (UK)	Pounds (£)	R 14,13
United States (USA)	Dollars ($)	R 7,04

   a) In South Africa the cost of a new Honda Civic is R 173 400. In England the same vehicle costs £ 12 200 and in the USA $21 900. In which country is the car the cheapest?
   b) Sollie and Arinda are waiters in a South African restaurant attracting many tourists from abroad. Sollie gets a £ 6 tip from a tourist and Arinda gets $12. Who got the better tip?
5. Yaseen wants to buy a book online. He finds a publisher in London selling the book for £ 7,19. This publisher is offering free shipping on the product.

   He then finds the same book from a publisher in New York for $8,49 with a shipping fee of $2.

   Next he looks up the exchange rates to see which publisher has the better deal. If $1 = R 11,48 and £ 1 = R 17,36, which publisher should he buy the book from?
6. Mathe is saving up to go visit her friend in Germany. She estimates the total cost of her trip to be R 50 000. The exchange rate is currently € 1 = R 13,22.

   Her friend decides to help Mathe out by giving her € 1000. How much (in rand) does Mathe now need to save up?

7. Lulamile and Jacob give tours over the weekends. They do not charge for these tours but instead accept tips from the group. The table below shows the total amount of tips they receive from various tour groups.

Group	Total tips
British tourists	£ 5,50
Japanese tourists	¥ 85,50
American tourists	\$ 7,00
Dutch tourists	€ 9,70
Brazilian tourists	40,50 BRL
Australian tourists	9,20 AUD
South African tourists	R 55,00

The current exchange rates are:

$$\begin{aligned} £\ 1 &= \text{R } 17{,}12 \\ ¥\ 1 &= \text{R } 0{,}10 \\ \$\ 1 &= \text{R } 11{,}42 \\ €\ 1 &= \text{R } 12{,}97 \\ 1 \text{ BRL} &= \text{R } 4{,}43 \\ 1 \text{ AUD} &= \text{R } 9{,}12 \end{aligned}$$

a) Which group of tourists tipped the most? How much did they tip (give your answer in rand)?

b) Which group of tourists tipped the least? How much did they tip (give your answer in rand)?

8. Kayla is planning a trip to visit her family in Malawi followed by spending some time in Tanzania at the Serengeti. She will first need to convert her South African rands into the Malawian kwacha. After that she will convert her remaining Malawian kwacha into Tanzanian shilling.

   She looks up the current exchange rates and finds the following information:

$$\begin{aligned} \text{R } 1 &= 39{,}46 \text{ MWK} \\ 1 \text{ MWK} &= 4{,}01 \text{ TZS} \end{aligned}$$

   She starts off with R 5000 in South Africa. In Malawi she spends 65 000 MWK. When she converts the remaining Malawian kwacha to Tanzanian shilling, how much money does she have (in Tanzanian shilling)?

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GJ7 2. 2GJ8 3. 2GJ9 4. 2GJB 5. 2GJC 6. 2GJD 7. 2GJF 8. 2GJG

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9.6 Chapter summary EMA6W

See presentation: 2GJH at www.everythingmaths.co.za

- There are two types of interest rates:

Simple interest	Compound interest
$A = P(1 + in)$	$A = P(1 + i)^n$

  Where:

$$\begin{aligned} A &= \text{accumulated amount} \\ P &= \text{principal amount} \\ i &= \text{interest written as decimal} \\ n &= \text{number of years} \end{aligned}$$

- Hire purchase loan repayments are calculated using the simple interest formula on the cash price minus the deposit. Monthly repayments are calculated by dividing the accumulated amount by the number of months for the repayment.
- Population growth and inflation are calculated using the compound interest formula.
- Foreign exchange rate is the price of one currency in terms of another.

End of chapter Exercise 9 – 7:

1. An amount of R 6330 is invested in a savings account which pays simple interest at a rate of 11% p.a.. Calculate the balance accumulated by the end of 7 years.
2. An amount of R 1740 is invested in a savings account which pays simple interest at a rate of 7% p.a.. Calculate the balance accumulated by the end of 6 years.
3. Adam opens a savings account when he is 13. He would like to have R 50 000 by the time he is 18. If the savings account offers simple interest at a rate of 8,5% per annum, how much money should he invest now to reach his goal?
4. When his son was 4 years old, Dumile made a deposit of R 6700 in the bank. The investment grew at a simple interest rate and when Dumile's son was 24 years old, the value of the investment was R 11 524.

   At what rate was the money invested? Give your answer correct to one decimal place.
5. When his son was 7 years old, Jared made a deposit of R 5850 in the bank. The investment grew at a simple interest rate and when Jared's son was 35 years old, the value of the investment was R 11 746,80.

   At what rate was the money invested? Give your answer correct to one decimal place.
6. Sehlolo wants to invest R 6360 at a simple interest rate of 12,4% p.a.

   How many years will it take for the money to grow to R 26 075? Round **up** your answer to the nearest year.
7. Mphikeleli wants to invest R 5540 at a simple interest rate of 9,1% p.a.

   How many years will it take for the money to grow to R 16 620? Round **up** your answer to the nearest year.
8. An amount of R 3500 is invested in an account which pays simple interest at a rate of 6,7% per annum. Calculate the amount of interest accumulated at the end of 4 years.
9. An amount of R 3270 is invested in a savings account which pays a compound interest rate of 12,2% p.a.

   Calculate the balance accumulated by the end of 7 years. As usual with financial calculations, round your answer to two decimal places, but do not round off until you have reached the solution.
10. An amount of R 2380 is invested in a savings account which pays a compound interest rate of 8,3% p.a.

    Calculate the balance accumulated by the end of 7 years. As usual with financial calculations, round your answer to two decimal places, but do not round off until you have reached the solution.
11. Emma wants to invest some money at a compound interest rate of 8,2% p.a.

    How much money should be invested if she wants to reach a sum of R 61 500 in 4 years' time? Round up your answer to the nearest rand.
12. Limpho wants to invest some money at a compound interest rate of 13,9% p.a.

    How much money should be invested if she wants to reach a sum of R 24 300 in 2 years' time? Round up your answer to the nearest rand.
13. Calculate the compound interest for the following problems.
    a) A R 2000 loan for 2 years at 5% p.a.
    b) A R 1500 investment for 3 years at 6% p.a.
    c) A R 800 loan for 1 year at 16% p.a.
14. Ali invests R 1110 into an account which pays out a lump sum at the end of 12 years.

    If he gets R 1642,80 at the end of the period, what compound interest rate did the bank offer him? Give your answer correct to one decimal place.
15. Christopher invests R 4480 into an account which pays out a lump sum at the end of 7 years.

    If he gets R 6496,00 at the end of the period, what compound interest rate did the bank offer him? Give your answer correct to one decimal place.
16. Calculate how much you will earn if you invested R 500 for 1 year at the following interest rates:
    a) 6,85% simple interest
    b) 4,00% compound interest

17. Bianca has R 1450 to invest for 3 years. Bank A offers a savings account which pays simple interest at a rate of 11% per annum, whereas Bank B offers a savings account paying compound interest at a rate of 10,5% per annum. Which account would leave Bianca with the highest accumulated balance at the end of the 3 year period?
18. Given:
A loan of R 2000 for a year at an interest rate of 10% p.a.
    a) How much simple interest is payable on the loan?
    b) How much compound interest is payable on the loan?
19. R 2250 is invested at an interest rate of 5,25% per annum.
Complete the following table.

Number of years	Simple interest	Compound interest
1		
2		
3		
4		
20		

20. Discuss:
    a) Which type of interest would you like to use if you are the borrower?
    b) Which type of interest would you like to use if you were the banker?
21. Portia wants to buy a television on a hire purchase agreement. The cash price of the television is R 6000. She is required to pay a deposit of 20% and pay the remaining loan amount off over 12 months at an interest rate of 9% p.a.
    a) What is the principal loan amount?
    b) What is the accumulated loan amount?
    c) What are Portia's monthly repayments?
    d) What is the total amount she has paid for the television?
22. Gabisile wants to buy a heater on a hire purchase agreement. The cash price of the heater is R 4800. She is required to pay a deposit of 10% and pay the remaining loan amount off over 12 months at an interest rate of 12% p.a.
    a) What is the principal loan amount?
    b) What is the accumulated loan amount?
    c) What are Gabisile's monthly repayments?
    d) What is the total amount she has paid for the heater?
23. Khayalethu buys a couch costing R 8000 on a hire purchase agreement. He is charged an interest rate of 12% p.a. over 3 years.
    a) How much will Khayalethu pay in total?
    b) How much interest does he pay?
    c) What is his monthly instalment?
24. Jwayelani buys a sofa costing R 7700 on a hire purchase agreement. He is charged an interest rate of 16% p.a. over 5 years.
    a) How much will Jwayelani pay in total?
    b) How much interest does he pay?
    c) What is his monthly instalment?

25. Bonnie bought a stove for R 3750. After 3 years she had finished paying for it and the R 956,25 interest that was charged for hire purchase. Determine the rate of simple interest that was charged.

26. A new furniture store has just opened in town and is offering the following special:

    Purchase a lounge suite, a bedroom suite and kitchen appliances (fridge, stove, washing machine) for just R 50 000 and receive a free microwave. No deposit required, 5 year payment plan available. Interest charged at just 6,5% p.a.

    Babelwa purchases all the items on hire purchase. She decides to pay a R 1500 deposit. The store adds in an insurance premium of R 35,00 per month.

    What is Babelwa's monthly payment on the items?

27. The price of 2 litres of milk is R 17. How much will it cost in 3 years time if the inflation rate is 13% p.a.?

28. The price of a 2 l bottle of juice is R 16. How much will the juice cost in 8 years time if the inflation rate is 7% p.a.?

29. A box of fruity-chews costs R 27 today. How much did it cost 8 years ago if the average rate of inflation was 10% p.a.? Round your answer to 2 decimal places.

30. A box of smarties costs R 23 today. How much did the same box cost 8 years ago if the average rate of inflation was 14% p.a.? Round your answer to 2 decimal places.

31. According to the latest census, South Africa currently has a population of 57 000 000.

    a) If the annual growth rate is expected to be 0,9%, calculate how many South Africans there will be in 10 years time (correct to the nearest hundred thousand).

    b) If it is found after 10 years that the population has actually increased by 10 million to 67 million, what was the growth rate?

32. The current population of Cape Town is 3 875 190 and the average rate of population growth in South Africa is 0,4% p.a.

    What can city planners expect the population of Cape Town to be in 12 years time?
    Note: Round your answer to the nearest integer.

33. The current population of Pretoria is 3 888 420 and the average rate of population growth in South Africa is 0,7% p.a.

    What will the population of Pretoria be in 7 years time?
    Note: Round your answer to the nearest integer.

34. Monique wants to buy an iPad that costs £ 140, with the exchange rate currently at £ 1 = R 15. She estimates that the exchange rate will drop to R 9 in a month.

    a) How much will the iPad cost in rands, if she buys it now?

    b) How much will she save if the exchange rate drops to R 9?

    c) How much will she lose if the exchange rate moves to R 20?

35. Xolile wants to buy a CD player that costs £ 140, with the exchange rate currently at £ 1 = R 14. She estimates that the exchange rate will drop to R 10 in a month.

    a) How much will the CD player cost in rands, if she buys it now?

    b) How much will she save if the exchange rate drops to R 10?

    c) How much will she lose if the exchange rate moves to R 20?

36. Alison is going on holiday to Europe. Her hotel will cost € 200 per night. How much will she need, in rands, to cover her hotel bill, if the exchange rate is € 1 = R 9,20?

37. Jennifer is buying some books online. She finds a publisher in the UK selling the books for £ 16,99.

    She then finds the same books from a publisher in the USA for $23,50.

    Next she looks up the exchange rates to see which publisher has the better deal. If $1 = R 12,43 and £ 1 = R 16,89, which publisher should she buy the books from?

38. Bonani won a trip to see Machu Picchu in Peru followed by a trip to Brazil for the carnival. He is given R 25 000 to spend while on the trip.

He then looks up the current exchange rates and finds the following information:

$$\text{R } 1 = 0{,}26 \text{ PEN}$$
$$1 \text{ BRL} = 1{,}17 \text{ PEN}$$

In Peru he spends 2380 PEN. When he converts the remaining Peruvian sol to Brazilian real, how much money does he have (in Brazilian real)?

39. If the exchange rate to the rand for the Japanese yen is ¥ 100 = R 6,23 and for the Australian dollar is 1 AUD = R 5,11, determine the exchange rate between the Australian dollar and the Japanese yen.
40. Khetang has just been to Europe to work for a few months. He returns to South Africa with € 2850 to invest in a savings account.
    His bank offers him a savings account which pays 5,3% compound interest per annum. The bank converts Khetang's Euros to rands at an exchange rate of € 1 = R 12,89.
    If Khetang invests his money for 6 years, how much interest does he earn on his investment?

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GJJ	2. 2GJK	3. 2GJM	4. 2GJN	5. 2GJP	6. 2GJQ
7. 2GJR	8. 2GJS	9. 2GJT	10. 2GJV	11. 2GJW	12. 2GJX
13a. 2GJY	13b. 2GJZ	13c. 2GK2	14. 2GK3	15. 2GK4	16. 2GK5
17. 2GK6	18. 2GK7	19. 2GK8	20. 2GK9	21. 2GKB	22. 2GKC
23. 2GKD	24. 2GKF	25. 2GKG	26. 2GKH	27. 2GKJ	28. 2GKK
29. 2GKM	30. 2GKN	31. 2GKP	32. 2GKQ	33. 2GKR	34. 2GKS
35. 2GKT	36. 2GKV	37. 2GKW	38. 2GKX	39. 2GKY	40. 2GKZ

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CHAPTER 10

Statistics

When running an experiment or conducting a survey we can potentially end up with many hundreds, thousands or even millions of values in the resulting data set. Too much data can be overwhelming and we need to reduce them or represent them in a way that is easier to understand and communicate.

Statistics is about summarising data. The methods of statistics allow us to represent the essential information in a data set while disregarding the unimportant information. We have to be careful to make sure that we do not accidentally throw away some of the important aspects of a data set.

By applying statistics properly we can highlight the important aspects of data and make the data easier to interpret. By applying statistics poorly or dishonestly we can also hide important information and let people draw the wrong conclusions.

In this chapter we will look at a few numerical and graphical ways in which data sets can be represented, to make them easier to interpret.

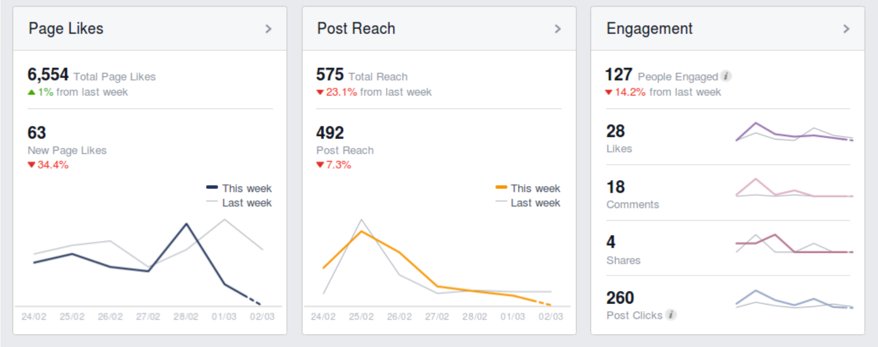

Figure 10.1: Statistics is used by various websites to show users who is viewing their content.

10.1 Collecting data EMA6X

DEFINITION: *Data*

Data refers to the pieces of information that have been observed and recorded, from an experiment or a survey.

NOTE:
The word **data** is the plural of the word **datum**, and therefore one should say, “the data are” and not “the data is”.

We distinguish between two main types of data: quantitative and qualitative.

DEFINITION: *Quantitative data*

Quantitative data are data that can be written as numbers.

Quantitative data can be discrete or continuous.

Discrete quantitative data can be represented by integers and usually occur when we count things, for example, the number of learners in a class, the number of molecules in a chemical solution, or the number of SMS messages sent in one day.

Continuous quantitative data can be represented by real numbers, for example, the height or mass of a person, the distance travelled by a car, or the duration of a phone call.

DEFINITION: *Qualitative data*

Qualitative data are data that cannot be written as numbers.

Two common types of qualitative data are categorical and anecdotal data. Categorical data can come from one of a limited number of possibilities, for example, your favourite cooldrink, the colour of your cell phone, or the language that you learnt to speak at home.

Anecdotal data take the form of an interview or a story, for example, when you ask someone what their personal experience was when using a product, or what they think of someone else's behaviour.

Categorical qualitative data are sometimes turned into quantitative data by counting the number of times that each category appears. For example, in a class with 30 learners, we ask everyone what the colours of their cell phones are and get the following responses:

black	black	black	white	purple	red	red	black	black	black
white	white	black	black	black	black	purple	black	black	white
purple	black	red	red	white	black	orange	orange	black	white

This is a categorical qualitative data set since each of the responses comes from one of a small number of possible colours.

We can represent exactly the same data in a different way, by counting how many times each colour appears.

Colour	Count
black	15
white	6
red	4
purple	3
orange	2

This is a discrete quantitative data set since each count is an integer.

Worked example 1: Qualitative and quantitative data

QUESTION

Thembisile is interested in becoming an airtime reseller to his classmates. He would like to know how much business he can expect from them. He asked each of his 20 classmates how many SMS messages they sent during the previous day. The results were:

20	3	0	14	30	9	11	13	13	15
9	13	16	12	13	7	17	14	9	13

Is this data set qualitative or quantitative? Explain your answer.

SOLUTION

The number of SMS messages is a count represented by an integer, which means that it is quantitative and discrete.

Worked example 2: Qualitative and quantitative data

QUESTION

Thembisile would like to know who the most popular cellular provider is among learners in his school. This time Thembisile randomly selects 20 learners from the entire school and asks them which cellular provider they currently use. The results were:

Cell C	Vodacom	Vodacom	MTN	Vodacom
MTN	MTN	Virgin Mobile	Cell C	8-ta
Vodacom	MTN	Vodacom	Vodacom	MTN
Vodacom	Vodacom	Vodacom	Virgin Mobile	MTN

Is this data set qualitative or quantitative? Explain your answer.

SOLUTION

Since each response is not a number, but one of a small number of possibilities, these are categorical qualitative data.

Exercise 10 – 1:

1. The following data set of dreams that learners have was collected from Grade 12 learners just after their final exams: {"I want to build a bridge!"; "I want to help the sick."; "I want running water!"}
   Categorise the data set.
2. Categorise the following data set of sweets in a packet that was collected from visitors to a sweet shop.
   $\{23; 25; 22; 26; 27; 25; 21; 28\}$
3. Categorise the following data set of questions answered correctly that was collected from a class of maths learners.
   $\{3; 5; 2; 6; 7; 5; 1; 2\}$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GM2 2. 2GM3 3. 2GM4

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10.2 Measures of central tendency EMA6Y

Mean EMA6Z

DEFINITION: *Mean*

The mean is the sum of a set of values, divided by the number of values in the set. The notation for the mean of a set of values is a horizontal bar over the variable used to represent the set, for example $\bar{x}$. The formula for the mean of a data set $\{x_1; x_2; ...; x_n\}$ is:

$$\begin{aligned}\bar{x} &= \frac{1}{n}\sum_{i=1}^{n} x_i \\ &= \frac{x_1 + x_2 + \ldots + x_n}{n}\end{aligned}$$

The mean is sometimes also called the average or the arithmetic mean.

Worked example 3: Calculating the mean

QUESTION

What is the mean of the data set $\{10; 20; 30; 40; 50\}$?

SOLUTION

Step 1: Calculate the sum of the data

$$10 + 20 + 30 + 40 + 50 = 150$$

Step 2: Divide by the number of values in the data set to get the mean

Since there are 5 values in the data set, the mean is:

$$\text{mean} = \frac{150}{5} = 30$$

Median

EMA72

DEFINITION: *Median*

The median of a data set is the value in the central position, when the data set has been arranged from the lowest to the highest value.

Note that exactly half of the values from the data set are less than the median and the other half are greater than the median.

To calculate the median of a quantitative data set, first sort the data from the smallest to the largest value and then find the value in the middle. If there is an odd number of values in the data set, the median will be equal to one of the values in the data set. If there is an even number of values in the data set, the median will lie halfway between two values in the data set.

Worked example 4: Median for an odd number of values

QUESTION

What is the median of $\{10; 14; 86; 2; 68; 99; 1\}$?

SOLUTION

Step 1: Sort the values

The values in the data set, arranged from the smallest to the largest, are

$$1; 2; 10; 14; 68; 86; 99$$

Step 2: Find the number in the middle

There are 7 values in the data set. Since there are an odd number of values, the median will be equal to the value in the middle, namely, in the fourth position. Therefore the median of the data set is 14.

Worked example 5: Median for an even number of values

QUESTION

What is the median of $\{11; 10; 14; 86; 2; 68; 99; 1\}$?

SOLUTION

Step 1: Sort the values

The values in the data set, arranged from the smallest to the largest, are

$$1; 2; 10; 11; 14; 68; 86; 99$$

Step 2: Find the number in the middle

There are 8 values in the data set. Since there are an even number of values, the median will be halfway between the two values in the middle, namely, between the fourth and fifth positions. The value in the fourth position is 11 and the value in the fifth position is 14. The median lies halfway between these two values and is therefore

$$\text{median} = \frac{11 + 14}{2} = 12{,}5$$

Mode EMA73

DEFINITION: *Mode*

The mode of a data set is the value that occurs most often in the set. The mode can also be described as the most frequent or most common value in the data set.

To calculate the mode, we simply count the number of times that each value appears in the data set and then find the value that appears most often.

A data set can have more than one mode if there is more than one value with the highest count. For example, both 2 and 3 are modes in the data set $\{1; 2; 2; 3; 3\}$. If all points in a data set occur with equal frequency, it is equally accurate to describe the data set as having many modes or no mode.

VISIT:
The following video explains how to calculate the mean, median and mode of a data set.
See video: 2GM5 at www.everythingmaths.co.za

Worked example 6: Finding the mode

QUESTION

Find the mode of the data set $\{2; 2; 3; 4; 4; 4; 6; 6; 7; 8; 8; 10; 10\}$.

SOLUTION

Step 1: Count the number of times that each value appears in the data set

Value	Count
2	2
3	1
4	3
6	2
7	1
8	2
10	2

Step 2: Find the value that appears most often

From the table above we can see that 4 is the only value that appears 3 times. All the other values appear less than 3 times. Therefore the mode of the data set is 4.

One problem with using the mode as a measure of central tendency is that we can usually not compute the mode of a continuous data set. Since continuous values can lie anywhere on the real line, any particular value will almost never repeat. This means that the frequency of each value in the data set will be 1 and that there will be no mode. We will look at one way of addressing this problem in the section on grouping data.

Worked example 7: Comparison of measures of central tendency

QUESTION

There are regulations in South Africa related to bread production to protect consumers. By law, if a loaf of bread is not labelled, it must weigh 800 g, with the leeway of 5 percent under or 10 percent over. Vishnu is interested in how a well-known, national retailer measures up to this standard. He visited his local branch of the supplier and recorded the masses of 10 different loaves of bread for one week. The results, in grams, are given below:

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
802,4	787,8	815,7	807,4	801,5	786,6	799,0
796,8	798,9	809,7	798,7	818,3	789,1	806,0
802,5	793,6	785,4	809,3	787,7	801,5	799,4
819,6	812,6	809,1	791,1	805,3	817,8	801,0
801,2	795,9	795,2	820,4	806,6	819,5	796,7
789,0	796,3	787,9	799,8	789,5	802,1	802,2
789,0	797,7	776,7	790,7	803,2	801,2	807,3
808,8	780,4	812,6	801,8	784,7	792,2	809,8
802,4	790,8	792,4	789,2	815,6	799,4	791,2
796,2	817,6	799,1	826,0	807,9	806,7	780,2

1. Is this data set qualitative or quantitative? Explain your answer.
2. Determine the mean, median and mode of the mass of a loaf of bread for each day of the week. Give your answer correct to 1 decimal place.
3. Based on the data, do you think that this supplier is providing bread within the South African regulations?

SOLUTION

Step 1: Qualitative or quantitative?

Since each mass can be represented by a number, the data set is quantitative. Furthermore, since a mass can be any real number, the data are continuous.

Step 2: Calculate the mean

In each column (for each day of the week), we add up the measurements and divide by the number of measurements, 10.

For Monday, the sum of the measured values is 8007,9 and so the mean for Monday is

$$\frac{8007{,}9}{10} = 800{,}8 \text{ g}$$

In the same way, we can compute the mean for each day of the week. See the table below for the results.

Step 3: Calculate the median

In each column we sort the numbers from lowest to highest and find the value in the middle. Since there are an even number of measurements (10), the median is halfway between the two numbers in the middle.

For Monday, the sorted list of numbers is

$$789{,}0; 789{,}0; 796{,}2; 796{,}7; 801{,}2;$$
$$802{,}3; 802{,}3; 802{,}5; 808{,}7; 819{,}6$$

The two numbers in the middle are 801,2 and 802,3 and so the median is

$$\frac{801{,}2 + 802{,}3}{2} = 801{,}8 \text{ g}$$

In the same way, we can compute the median for each day of the week:

Day	Mean	Median
Monday	800,8 g	801,8 g
Tuesday	797,2 g	796,1 g
Wednesday	798,4 g	797,2 g
Thursday	803,4 g	800,8 g
Friday	802,0 g	804,3 g
Saturday	801,6 g	801,4 g
Sunday	799,3 g	800,2 g

From the above calculations we can see that the means and medians are close to one another, but not quite equal. In the next worked example we will see that the mean and median are not always close to each other.

Step 4: Determine the mode

Since the data are continuous we cannot compute the mode. In the next section we will see how we can group data in order to make it possible to compute an approximation for the mode.

Step 5: Conclusion: Is the supplier reliable?

From the question, the requirements are that the mass of a loaf of bread be between 800 g minus 5%, which is 760 g, and plus 10%, which is 880 g. Since every one of the measurements made by Vishnu lies within this range and since the means and medians are all close to 800 g, we can conclude that the supplier is reliable.

DEFINITION: *Outlier*

An outlier is a value in the data set that is not typical of the rest of the set. It is usually a value that is much greater or much less than all the other values in the data set.

Worked example 8: Effect of outliers on mean and median

QUESTION

The heights of 10 learners are measured in centimetres to obtain the following data set:

$$\{150; 172; 153; 156; 146; 157; 157; 143; 168; 157\}$$

Afterwards, we include one more learner in the group, who is exceptionally tall at 181 cm.

Compare the mean and median of the heights of the learners before and after the eleventh learner was included.

SOLUTION

Step 1: Calculate the mean of the first 10 learners

$$\begin{aligned} \text{mean} &= \frac{150 + 172 + 153 + 156 + 146 + 157 + 157 + 143 + 168 + 157}{10} \\ &= 155{,}9 \text{ cm} \end{aligned}$$

Step 2: Calculate the mean of all 11 learners

$$\begin{aligned} \text{mean} &= \frac{150 + 172 + 153 + 156 + 146 + 157 + 157 + 143 + 168 + 157 + 181}{11} \\ &= 158{,}2 \text{ cm} \end{aligned}$$

From this we see that the average height changes by $158{,}2 - 155{,}9 = 2{,}3$ cm when we introduce the outlier value (the tall person) to the data set.

Step 3: Calculate the median of the first 10 learners

To find the median, we need to sort the data set:

$$\{143; 146; 150; 153; 156; 157; 157; 157; 168; 172\}$$

Since there are an even number of values, 10, the median lies halfway between the fifth and sixth values:

$$\text{median} = \frac{156 + 157}{2} = 156{,}5 \text{ cm}$$

Step 4: Calculate the median of all 11 learners

After adding the tall learner, the sorted data set is

$$\{143; 146; 150; 153; 156; 157; 157; 157; 168; 172; 181\}$$

Now, with 11 values, the median is the sixth value: 157 cm. So, the median changes by only 0,5 cm when we add the outlier value to the data set.

In general, the median is less affected by the addition of outliers to a data set than the mean is. This is important because it is quite common that outliers are measured during an experiment, because of problems with the equipment or unexpected interference.

Exercise 10 – 2:

1. Calculate the **mean** of the following data set: $\{9; 14; 9; 14; 8; 8; 9; 8; 9; 9\}$. Round your answer to 1 decimal place.
2. Calculate the **median** of the following data set: $\{4; 13; 10; 13; 13; 4; 2; 13; 13; 13\}$.
3. Calculate the **mode** of the following data set: $\{6; 10; 6; 6; 13; 12; 12; 7; 13; 6\}$
4. Calculate the mean, median and mode of the following data sets:
   a) $\{2; 5; 8; 8; 11; 13; 22; 23; 27\}$
   b) $\{15; 17; 24; 24; 26; 28; 31; 43\}$
   c) $\{4; 11; 3; 15; 11; 13; 25; 17; 2; 11\}$
   d) $\{24; 35; 28; 41; 31; 49; 31\}$
5. The ages of 15 runners of the Comrades Marathon were recorded:
$$\{31; 42; 28; 38; 45; 51; 33; 29; 42; 26; 34; 56; 33; 46; 41\}$$
Calculate the mean, median and modal age.
6. A group of 10 friends each have some stones. They work out that the **mean** number of stones they have is 6. Then 7 friends leave with an unknown number (x) of stones. The remaining 3 friends work out that the **mean** number of stones they have left is 12,33.
When the 7 friends left, how many stones did they take with them?
7. A group of 9 friends each have some coins. They work out that the **mean** number of coins they have is 4. Then 5 friends leave with an unknown number (x) of coins. The remaining 4 friends work out that the **mean** number of coins they have left is 2,5.
When the 5 friends left, how many coins did they take with them?
8. A group of 9 friends each have some marbles. They work out that the **mean** number of marbles they have is 3. Then 3 friends leave with an unknown number (x) of marbles. The remaining 6 friends work out that the **mean** number of marbles they have left is 1,17.
When the 3 friends left, how many marbles did they take with them?
9. In the first of a series of jars, there is 1 sweet. In the second jar, there are 3 sweets. The mean number of sweets in the first two jars is 2.
   a) If the mean number of sweets in the first three jars is 3, how many sweets are there in the third jar?
   b) If the mean number of sweets in the first four jars is 4, how many sweets are there in the fourth jar?
10. Find a set of five ages for which the mean age is 5, the modal age is 2 and the median age is 3 years.
11. Four friends each have some marbles. They work out that the mean number of marbles they have is 10. One friend leaves with 4 marbles. How many marbles do the remaining friends have together?

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GM6 2. 2GM7 3. 2GM8 4a. 2GM9 4b. 2GMB 4c. 2GMC 4d. 2GMD
5. 2GMF 6. 2GMG 7. 2GMH 8. 2GMJ 9. 2GMK 10. 2GMM 11. 2GMN

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10.3 Grouping data EMA74

A common way of handling continuous quantitative data is to subdivide the full range of values into a few sub-ranges. By assigning each continuous value to the sub-range or class within which it falls, the data set changes from continuous to discrete.

Grouping is done by defining a set of ranges and then counting how many of the data fall inside each range. The sub-ranges must not overlap and must cover the entire range of the data set.

One way of visualising grouped data is as a histogram. A histogram is a collection of rectangles, where the base of a rectangle (on the x-axis) covers the values in the range associated with it, and the height of a rectangle corresponds to the number of values in its range.

VISIT:
The following video explains how to group data.
See video: 2GMP at www.everythingmaths.co.za

Worked example 9: Groups and histograms

QUESTION

The heights in centimetres of 30 learners are given below.

142	163	169	132	139	140	152	168	139	150
161	132	162	172	146	152	150	132	157	133
141	170	156	155	169	138	142	160	164	168

Group the data into the following ranges and draw a histogram of the grouped data:

$$130 \leq h < 140$$
$$140 \leq h < 150$$
$$150 \leq h < 160$$
$$160 \leq h < 170$$
$$170 \leq h < 180$$

(Note that the ranges do not overlap since each one starts where the previous one ended.)

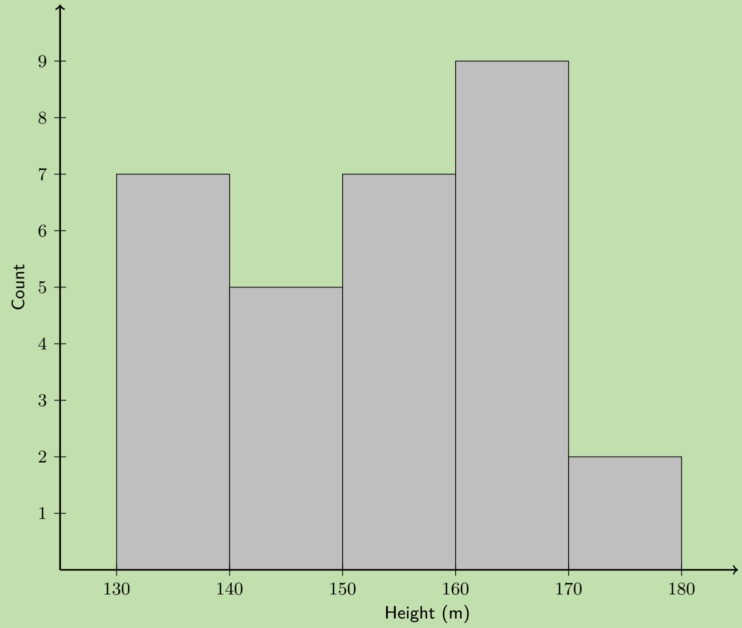
SOLUTION

Step 1: Count the number of values in each range

Range	Count
$130 \leq h < 140$	7
$140 \leq h < 150$	5
$150 \leq h < 160$	7
$160 \leq h < 170$	9
$170 \leq h < 180$	2

Step 2: Draw the histogram

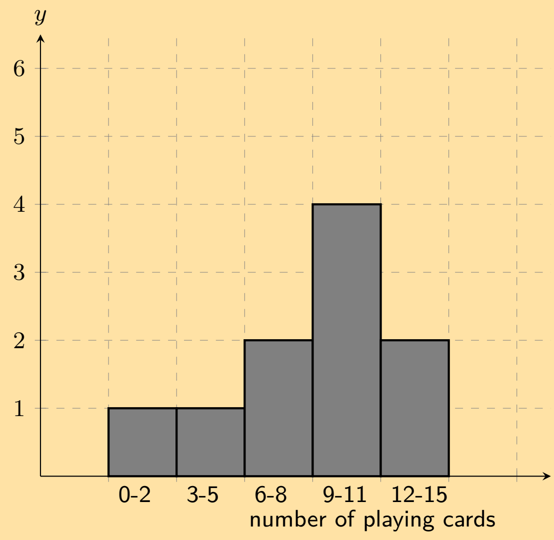
Since there are 5 ranges, the histogram will have 5 rectangles. The base of each rectangle is defined by its range. The height of each rectangle is determined by the count in its range.



The histogram makes it easy to see in which range most of the heights are located and provides an overview of the distribution of the values in the data set.

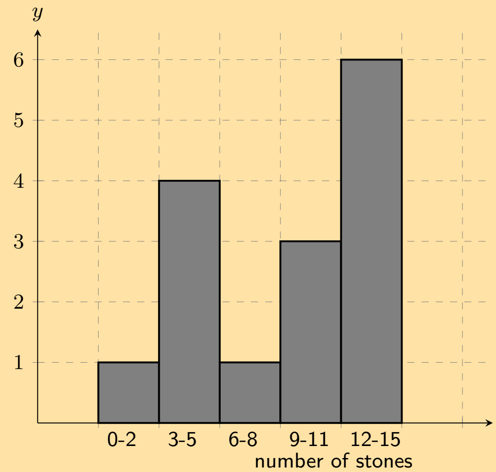
Exercise 10 – 3:

1. A group of 10 learners count the number of playing cards they each have. This is a histogram describing the data they collected:



Count the number of playing cards in the following range: $0 \leq$ number of playing cards ≤ 2

2. A group of 15 learners count the number of stones they each have. This is a histogram describing the data they collected:



Count the number of stones in the following range: $0 \leq$ number of stones ≤ 2

3. A group of **20** learners count the number of playing cards they each have. This is the data they collect:

14	9	11	8	13	2	3	4	16	17
9	19	10	14	4	16	16	11	2	17

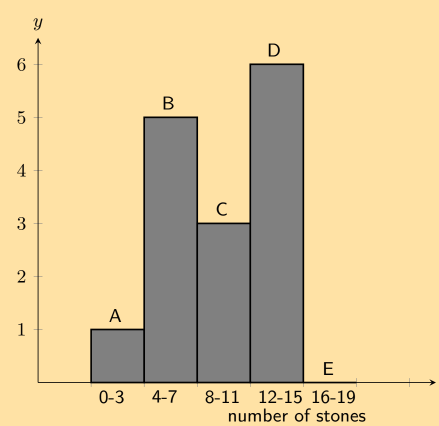
Count the number of learners who have from 12 up to 15 playing cards. In other words, how many learners have playing cards in the following range: $12 \leq$ number of playing cards ≤ 15? It may be helpful for you to draw a histogram in order to answer the question.

4. A group of **20** learners count the number of stones they each have. This is the data they collect:

16	6	11	19	20	17	13	1	5	12
5	2	16	11	16	6	10	13	6	17

Count the number of learners who have from 4 up to 7 stones. In other words, how many learners have stones in the following range: $4 \leq$ number of stones ≤ 7? It may be helpful for you to draw a histogram in order to answer the question.

5. A group of 20 learners count the number of stones they each have. The learners draw a histogram describing the data they collected. However, they have made a mistake in drawing the histogram.

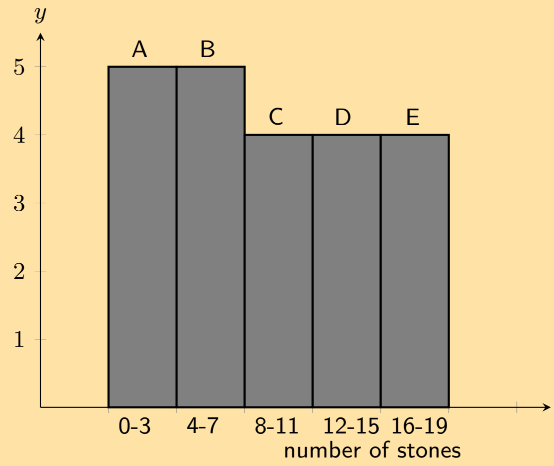


The data set below shows the correct information for the number of stones the learners have. Each value represents the number of stones for one learner.

$$\{4; 12; 15; 14; 18; 12; 17; 15; 1; 6; 6; 12; 6; 8; 6; 8; 17; 19; 16; 8\}$$

Help them figure out which **column** in the histogram is **incorrect**.

6. A group of 20 learners count the number of stones they each have. The learners draw a histogram describing the data they collected. However, they have made a mistake in drawing the histogram.

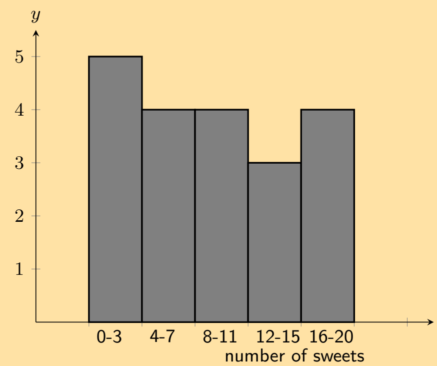


The data set below shows the correct information for the number of stones the learners have. Each value represents the number of stones for one learner.

$$\{19; 11; 5; 2; 3; 4; 14; 2; 12; 19; 11; 14; 2; 19; 11; 5; 17; 10; 1; 12\}$$

Help them figure out which **column** in the histogram is **incorrect**.

7. A group of learners count the number of sweets they each have. This is a histogram describing the data they collected:



A cat jumps onto the table, and all their notes land on the floor, mixed up, by accident!

Help them find which of the following data sets match the above histogram:

Data Set A

2	1	20	10	5	3	10	2	6	1
2	2	17	3	18	3	7	10	8	18

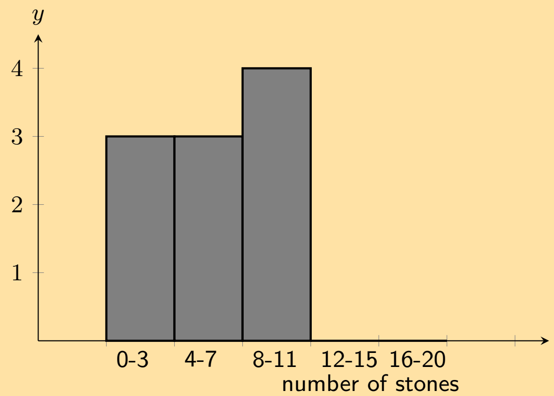
Data Set B

2	9	12	10	5	9	9	10
13	6	5	11	10	7	7	

Data Set C

3	12	16	10	15	17	18	2	3	7
11	12	8	2	7	17	3	11	4	4

8. A group of learners count the number of stones they each have. This is a histogram describing the data they collected:



A cleaner knocks over their table, and all their notes land on the floor, mixed up, by accident! Help them find which of the following data sets match the above histogram:

Data Set A

12	4	2	15	10	18	10	16	16	19
1	2	9	10	16	10	11	9	2	13

Data Set B

7	10	4	5	8	7	12	10
14	5	1	9	2	13	3	

Data Set C

9	3	8	5	8
5	8	1	4	3

9. A class experiment was conducted and 50 learners were asked to guess the number of sweets in a jar. The following guesses were recorded:

56	49	40	11	33	33	37	29	30	59
21	16	38	44	38	52	22	24	30	34
42	15	48	33	51	44	33	17	19	44
47	23	27	47	13	25	53	57	28	23
36	35	40	23	45	39	32	58	22	40

a) Draw up a grouped frequency table using the intervals $10 < x \leq 20$, $20 < x \leq 30$, $30 < x \leq 40$, $40 < x \leq 50$ and $50 < x \leq 60$.

b) Draw the histogram corresponding to the frequency table of the grouped data.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GMQ 2. 2GMR 3. 2GMS 4. 2GMT 5. 2GMV 6. 2GMW 7. 2GMX 8. 2GMY 9. 2GMZ

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With grouped data our estimates of central tendency will change because we lose some information when we place each value in a range. If all we have to work with is the grouped data, we do not know the measured values to the same accuracy as before. The best we can do is to assume that values are grouped at the centre of each range.

Looking back to the previous worked example, we started with this data set of learners' heights.

$$\{132; 132; 132; 133; 138; 139; 139; 140; 141; 142; 142; 146; 150; 150; 152;$$
$$152; 155; 156; 157; 160; 161; 162; 163; 164; 168; 168; 169; 169; 170; 172\}$$

Note that the data are sorted.

The mean of these data is 151,8 and the median is 152. The mode is 132, but remember that there are problems with computing the mode of continuous quantitative data.

After grouping the data, we now have the data set shown below. Note that each value is placed at the centre of its range and that the number of times that each value is repeated corresponds exactly to the counts in each range.

$$\{135; 135; 135; 135; 135; 135; 135; 145; 145; 145; 145; 145; 155; 155; 155;$$
$$155; 155; 155; 155; 165; 165; 165; 165; 165; 165; 165; 165; 165; 175; 175\}$$

The grouping changes the measures of central tendency since each datum is treated as if it occurred at the centre of the range in which it was placed.

The mean is now 153, the median 155 and the mode is 165. This is actually a better estimate of the mode, since the grouping showed in which range the learners' heights were clustered.

NOTE:
We can also just give the modal group and the median group for grouped data. The modal group is the group that has the highest number of data values. The median group is the central group when the groups are arranged in order.

Exercise 10 – 4:

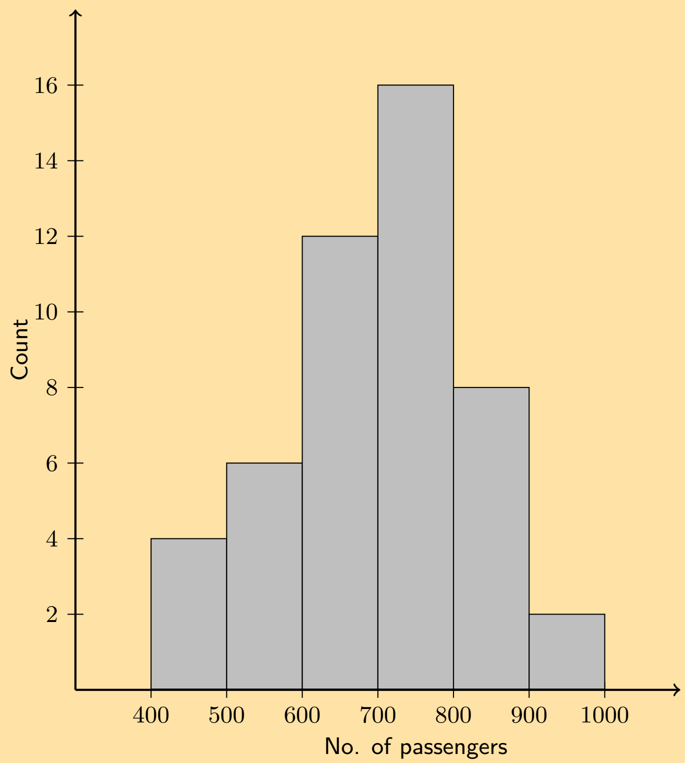
1. Consider the following grouped data and calculate the mean, the modal group and the median group.

Mass (kg)	Count
$40 < m \leq 45$	7
$45 < m \leq 50$	10
$50 < m \leq 55$	15
$55 < m \leq 60$	12
$60 < m \leq 65$	6

2. Find the mean, the modal group and the median group in this data set of how much time people needed to complete a game.

Time (s)	Count
$35 < t \leq 45$	5
$45 < t \leq 55$	11
$55 < t \leq 65$	15
$65 < t \leq 75$	26
$75 < t \leq 85$	19
$85 < t \leq 95$	13
$95 < t \leq 105$	6

3. The histogram below shows the number of passengers that travel in Alfred's minibus taxi per week.



Calculate:

a) the modal interval
b) the total number of passengers to travel in Alfred's taxi
c) an estimate of the mean
d) an estimate of the median
e) if it is estimated that every passenger travelled an average distance of 5 km, how much money would Alfred have made if he charged R 3,50 per km?

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1. 2GN2 2. 2GN3 3. 2GN4

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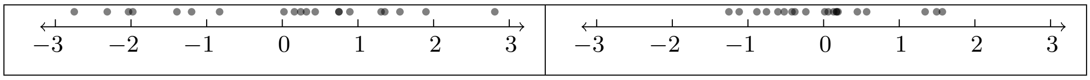
10.4 Measures of dispersion EMA76

The central tendency is not the only interesting or useful information about a data set. The two data sets illustrated below have the same mean (0), but have different spreads around the mean. Each circle represents one value from the data set (or one datum).



Dispersion is a general term for different statistics that describe how values are distributed around the centre. In this section we will look at measures of dispersion.

Range EMA77

DEFINITION: *Range*

The range of a data set is the difference between the maximum and minimum values in the set.

The most straightforward measure of dispersion is the range. The range simply tells us how far apart the largest and smallest values in a data set are. The range is very sensitive to outliers.

Worked example 10: Range

QUESTION

Find the range of the following data set:

$$\{1; 4; 5; 8; 6; 7; 5; 6; 7; 4; 10; 9; 10\}$$

What would happen if we removed the first value from the set?

SOLUTION

Step 1: Determine the range

The smallest value in the data set is 1 and the largest value is 10.

The range is $10 - 1 = 9$

Step 2: Remove the first value

If the first value, 1, were to be removed from the set, the minimum value would be 4. This means that the range would change to $10 - 4 = 6$. 1 is not typical of the other values. It is an outlier and has a big influence on the range.

Percentiles EMA78

DEFINITION: *Percentile*

The p^{th} percentile is the value, v, that divides a data set into two parts, such that p percent of the values in the data set are less than v and $100 - p$ percent of the values are greater than v. Percentiles can lie in the range $0 \leq p \leq 100$.

To understand percentiles properly, we need to distinguish between 3 different aspects of a datum: its value, its rank and its percentile:

- The value of a datum is what we measured and recorded during an experiment or survey.
- The rank of a datum is its position in the sorted data set (for example, first, second, third, and so on).
- The percentile at which a particular datum is, tells us what percentage of the values in the full data set are less than this datum.

The table below summarises the value, rank and percentile of the data set:

$$\{14{,}2;\ 13{,}9;\ 19{,}8;\ 10{,}3;\ 13{,}0;\ 11{,}1\}$$

Value	Rank	Percentile
10,3	1	0
11,1	2	20
13,0	3	40
13,9	4	60
14,2	5	80
19,8	6	100

As an example, 13,0 is at the 40^{th} percentile since there are 2 values less than 13,0 and 3 values greater than 13,0.

$$\frac{2}{2+3} = 0{,}4 = 40\%$$

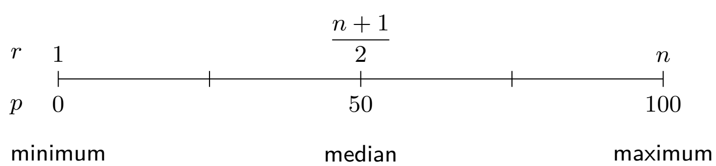
In general, the formula for finding the p^{th} percentile in an ordered data set with n values is

$$r = \frac{p}{100}(n-1) + 1$$

This gives us the rank, r, of the p^{th} percentile. To find the value of the p^{th} percentile, we have to count from the first value in the ordered data set up to the r^{th} value.

Sometimes the rank will not be an integer. This means that the percentile lies between two values in the data set. The convention is to take the value halfway between the two values indicated by the rank.

The figure below shows the relationship between rank and percentile graphically. We have already encountered three percentiles in this chapter: the median (50^{th} percentile), the minimum (0^{th} percentile) and the maximum (100^{th}). The median is defined as the value halfway in a sorted data set.



Worked example 11: Using the percentile formula

QUESTION

Determine the minimum, maximum and median values of the following data set using the percentile formula.

$$\{14; 17; 45; 20; 19; 36; 7; 30; 8\}$$

SOLUTION

Step 1: Sort the values in the data set

Before we can use the rank to find values in the data set, we always have to order the values from the smallest to the largest. The sorted data set is:

$$\{7; 8; 14; 17; 19; 20; 30; 36; 45\}$$

Step 2: Find the minimum

We already know that the minimum value is the first value in the ordered data set. We will now confirm that the percentile formula gives the same answer. The minimum is equivalent to the 0^{th} percentile. According to the percentile formula the rank, r, of the $p = 0^{\text{th}}$ percentile in a data set with $n = 9$ values is:

$$\begin{aligned} r &= \frac{p}{100}(n-1)+1 \\ &= \frac{0}{100}(9-1)+1 \\ &= 1 \end{aligned}$$

This confirms that the minimum value is the first value in the list, namely 7.

Step 3: Find the maximum

We already know that the maximum value is the last value in the ordered data set. The maximum is also equivalent to the 100^{th} percentile. Using the percentile formula with $p = 100$ and $n = 9$, we find the rank of the maximum value is:

$$\begin{aligned} r &= \frac{p}{100}(n-1)+1 \\ &= \frac{100}{100}(9-1)+1 \\ &= 9 \end{aligned}$$

This confirms that the maximum value is the last (the ninth) value in the list, namely 45.

Step 4: Find the median

The median is equivalent to the 50^{th} percentile. Using the percentile formula with $p = 50$ and $n = 9$, we find the rank of the median value is:

$$\begin{aligned} r &= \frac{50}{100}(n-1)+1 \\ &= \frac{50}{100}(9-1)+1 \\ &= \frac{1}{2}(8)+1 \\ &= 5 \end{aligned}$$

This shows that the median is in the middle (at the fifth position) of the ordered data set. Therefore the median value is 19.

DEFINITION: *Quartiles*

The quartiles are the three data values that divide an ordered data set into four groups, where each group contains an equal number of data values. The median (50^{th} percentile) is the second quartile ($Q2$). The 25^{th} percentile is also called the first or lower quartile ($Q1$). The 75^{th} percentile is also called the third or upper quartile ($Q3$).

Worked example 12: Quartiles

QUESTION

Determine the quartiles of the following data set:

$$\{7; 45; 11; 3; 9; 35; 31; 7; 16; 40; 12; 6\}$$

SOLUTION

Step 1: Sort the data set

$$\{3; 6; 7; 7; 9; 11; 12; 16; 31; 35; 40; 45\}$$

Step 2: Find the ranks of the quartiles

Using the percentile formula with $n = 12$, we can find the rank of the 25^{th}, 50^{th} and 75^{th} percentiles:

$$\begin{aligned} r_{25} &= \frac{25}{100}(12 - 1) + 1 \\ &= 3{,}75 \\ r_{50} &= \frac{50}{100}(12 - 1) + 1 \\ &= 6{,}5 \\ r_{75} &= \frac{75}{100}(12 - 1) + 1 \\ &= 9{,}25 \end{aligned}$$

Step 3: Find the values of the quartiles

Note that each of these ranks is a fraction, meaning that the value for each percentile is somewhere in between two values from the data set.

For the 25^{th} percentile the rank is 3,75, which is between the third and fourth values. Since both these values are equal to 7, the 25^{th} percentile is 7.

For the 50^{th} percentile (the median) the rank is 6,5, meaning halfway between the sixth and seventh values. The sixth value is 11 and the seventh value is 12, which means that the median is $\frac{11+12}{2} = 11{,}5$. For the 75^{th} percentile the rank is 9,25, meaning between the ninth and tenth values. Therefore the 75^{th} percentile is $\frac{31+35}{2} = 33$.

Deciles

The deciles are the nine data values that divide an ordered data set into ten groups, where each group contains an equal number of data values.

For example, consider the ordered data set:

$$28; 33; 35; 45; 57; 59; 61; 68; 69; 72; 75; 78; 80; 83; 86; 91;$$
$$92; 95; 101; 105; 111; 117; 118; 125; 127; 131; 137; 139; 141$$

The nine deciles are: $35; 59; 69; 78; 86; 95; 111; 125; 137$.

Percentiles for grouped data EMA79

In grouped data, the percentiles will lie somewhere inside a range, rather than at a specific value. To find the range in which a percentile lies, we still use the percentile formula to determine the rank of the percentile and then find the range within which that rank is.

Worked example 13: Percentiles in grouped data

QUESTION

The mathematics marks of 100 grade 10 learners at a school have been collected. The data are presented in the following table:

Percentage mark	Number of learners
$0 \leq x < 20$	2
$20 \leq x < 30$	5
$30 \leq x < 40$	18
$40 \leq x < 50$	22
$50 \leq x < 60$	18
$60 \leq x < 70$	13
$70 \leq x < 80$	12
$80 \leq x < 100$	10

1. Calculate the mean of this grouped data set.
2. In which intervals are the quartiles of the data set?
3. In which interval is the 30^{th} percentile of the data set?

SOLUTION

Step 1: Calculate the mean

Since we are given grouped data rather than the original ungrouped data, the best we can do is approximate the mean as if all the learners in each interval were located at the central value of the interval.

$$\begin{aligned}\text{Mean} &= \frac{2(10) + 5(25) + 18(35) + 22(45) + 18(55) + 13(65) + 12(75) + 10(90)}{100} \\ &= 54\%\end{aligned}$$

Step 2: Find the quartiles

Since the data have been grouped, they have also already been sorted. Using the percentile formula and the fact that there are 100 learners, we can find the rank of the 25^{th}, 50^{th} and 75^{th} percentiles as

$$\begin{aligned} r_{25} &= \frac{25}{100}(100-1)+1 \\ &= 24{,}75 \\ r_{50} &= \frac{50}{100}(100-1)+1 \\ &= 50{,}5 \\ r_{75} &= \frac{75}{100}(100-1)+1 \\ &= 75{,}25 \end{aligned}$$

Now we need to find in which ranges each of these ranks lie.

- For the lower quartile, we have that there are $2+5=7$ learners in the first two ranges combined and $2+5+18=25$ learners in the first three ranges combined. Since $7 < r_{25} < 25$, this means the lower quartile lies somewhere in the third range: $30 \leq x < 40$.
- For the second quartile (the median), we have that there are $2+5+18+22=47$ learners in the first four ranges combined. Since $47 < r_{50} < 65$, this means that the median lies somewhere in the fifth range: $50 \leq x < 60$.
- For the upper quartile, we have that there are 65 learners in the first five ranges combined and $65+13=78$ learners in the first six ranges combined. Since $65 < r_{75} < 78$, this means that the upper quartile lies somewhere in the sixth range: $60 \leq x < 70$.

Step 3: Find the 30^{th} percentile

Using the same method as for the quartiles, we first find the rank of the 30^{th} percentile.

$$\begin{aligned} r &= \frac{30}{100}(100-1)+1 \\ &= 30{,}7 \end{aligned}$$

Now we have to find the range in which this rank lies. Since there are 25 learners in the first 3 ranges combined and 47 learners in the first 4 ranges combined, the 30^{th} percentile lies in the fourth range: $40 \leq x < 50$

Ranges EMA7B

We define data ranges in terms of percentiles. We have already encountered the full data range, which is simply the difference between the 100^{th} and the 0^{th} percentile (that is, between the maximum and minimum values in the data set).

DEFINITION: *Interquartile range*

The interquartile range is a measure of dispersion, which is calculated by subtracting the first quartile ($Q1$) from the third quartile ($Q3$). This gives the range of the middle half of the data set.

DEFINITION: *Semi interquartile range*

The semi interquartile range is half of the interquartile range.

Exercise 10 – 5:

1. A group of **15** learners count the number of sweets they each have. This is the data they collect:

4	11	14	7	14	5	8	7
12	12	5	13	10	6	7	

   Calculate the **range** of values in the data set.

2. A group of **10** learners count the number of playing cards they each have. This is the data they collect:

$$5 \quad 1 \quad 3 \quad 1 \quad 4 \quad 10 \quad 1 \quad 3 \quad 3 \quad 4$$

   Calculate the **range** of values in the data set.

3. Find the range of the following data set:

$$\{1; 2; 3; 4; 4; 4; 5; 6; 7; 8; 8; 9; 10; 10\}$$

4. What are the quartiles of this data set?

$$\{3; 5; 1; 8; 9; 12; 25; 28; 24; 30; 41; 50\}$$

5. A class of 12 learners writes a test and the results are as follows:

$$\{20; 39; 40; 43; 43; 46; 53; 58; 63; 70; 75; 91\}$$

   Find the range, quartiles and the interquartile range.

6. Three sets of data are given:

   Data set 1: $\{9; 12; 12; 14; 16; 22; 24\}$
   Data set 2: $\{7; 7; 8; 11; 13; 15; 16\}$
   Data set 3: $\{11; 15; 16; 17; 19; 22; 24\}$

   For each data set find:

   a) the range
   b) the lower quartile
   c) the median
   d) the upper quartile
   e) the interquartile range
   f) the semi-interquartile range

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1. 2GN5 2. 2GN6 3. 2GN7 4. 2GN8 5. 2GN9 6. 2GNB

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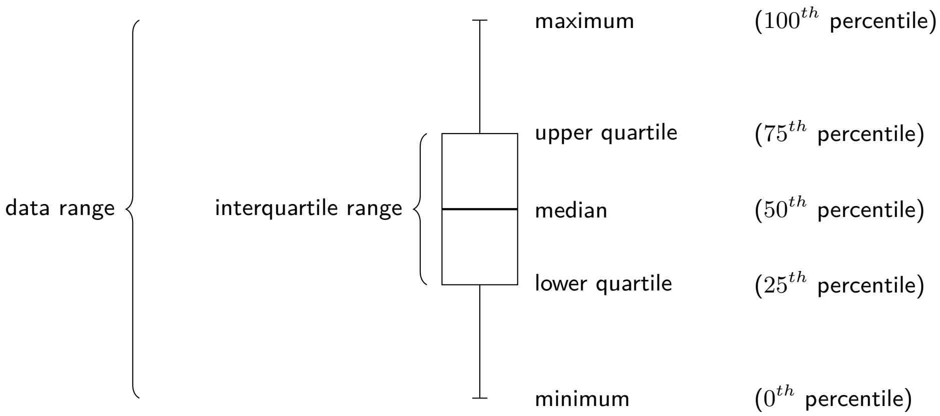
10.5 Five number summary EMA7C

A common way of summarising the overall data set is with the five number summary and the box-and-whisker plot. These two represent exactly the same information, numerically in the case of the five number summary and graphically in the case of the box-and-whisker plot.

The five number summary consists of the minimum value, the maximum value and the three quartiles. Another way of saying this is that the five number summary consists of the following percentiles: 0^{th}, 25^{th}, 50^{th}, 75^{th}, 100^{th}.

The box-and-whisker plot shows these five percentiles as in the figure below. The box shows the interquartile range (the distance between Q_1 and Q_3). A line inside the box shows the median. The lines extending outside the box (the whiskers) show where the minimum and maximum values lie. This graph can also be drawn horizontally.



VISIT:
This video explains how to draw a box-and-whisker plot for a data set.
See video: 2GNC at www.everythingmaths.co.za

Worked example 14: Five number summary

QUESTION

Draw a box and whisker diagram for the following data set:

$$\{1{,}25; 1{,}5; 2{,}5; 2{,}5; 3{,}1; 3{,}2; 4{,}1; 4{,}25; 4{,}75; 4{,}8; 4{,}95; 5{,}1\}$$

SOLUTION

Step 1: Determine the minimum and maximum

Since the data set is already sorted, we can read off the minimum as the first value (1,25) and the maximum as the last value (5,1).

Step 2: Determine the quartiles

There are 12 values in the data set. Using the percentile formula, we can determine that the median lies between the sixth and seventh values, making it:

$$\frac{3{,}2 + 4{,}1}{2} = 3{,}65$$

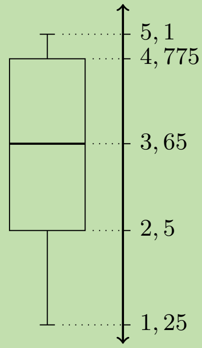
The first quartile lies between the third and fourth values, making it:

$$\frac{2{,}5 + 2{,}5}{2} = 2{,}5$$

The third quartile lies between the ninth and tenth values, making it:

$$\frac{4{,}75 + 4{,}8}{2} = 4{,}775$$

This provides the five number summary of the data set and allows us to draw the following box-and-whisker plot.



Exercise 10 – 6:

1. Lisa is working in a computer store. She sells the following number of computers each month:

$$\{27; 39; 3; 15; 43; 27; 19; 54; 65; 23; 45; 16\}$$

   Give the five number summary and box-and-whisker plot of Lisa's sales.

2. Zithulele works as a telesales person. He keeps a record of the number of sales he makes each month. The data below show how much he sells each month.

$$\{49; 12; 22; 35; 2; 45; 60; 48; 19; 1; 43; 12\}$$

   Give the five number summary and box-and-whisker plot of Zithulele's sales.

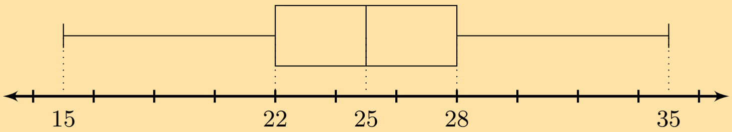
3. Nombusa has worked as a florist for nine months. She sold the following number of wedding bouquets:

$$\{16; 14; 8; 12; 6; 5; 3; 5; 7\}$$

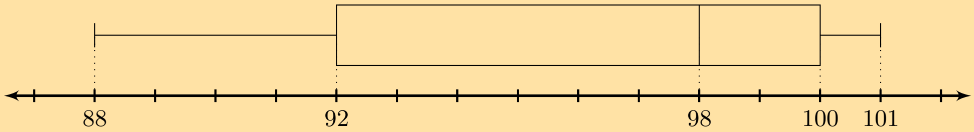
   Give the five number summary of Nombusa's sales.

4. Determine the five number summary for each of the box-and-whisker plots below.

   a)



   b)



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1. 2GND 2. 2GNF 3. 2GNG 4a. 2GNH 4b. 2GNJ

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10.6 Chapter summary EMA7D

See presentation: 2GNK at www.everythingmaths.co.za

- Data refer to the pieces of information that have been observed and recorded, from an experiment or a survey.
- Quantitative data are data that can be written as numbers. Quantitative data can be discrete or continuous.
- Qualitative data are data that cannot be written as numbers. There are two common types of qualitative data: categorical and anecdotal data.
- The mean is the sum of a set of values divided by the number of values in the set.

$$\begin{aligned}\overline{x} &= \frac{1}{n}\sum_{i=1}^{n} x_i \\ &= \frac{x_1 + x_2 + \cdots + x_n}{n}\end{aligned}$$

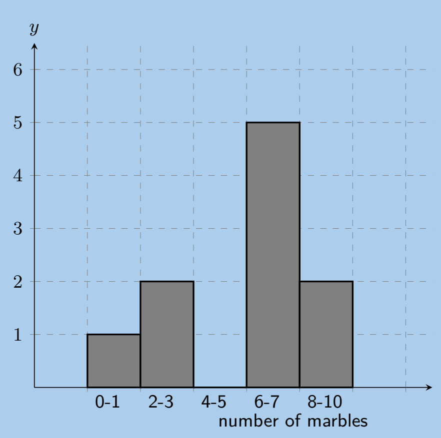
- The median of a data set is the value in the central position, when the data set has been arranged from the lowest to the highest value. If there are an odd number of data, the median will be equal to one of the values in the data set. If there are an even number of data, the median will lie half way between two values in the data set.
- The mode of a data set is the value that occurs most often in the set.
- An outlier is a value in the data set that is not typical of the rest of the set. It is usually a value that is much greater or much less than all the other values in the data set.
- Continuous quantitative data can be grouped by dividing the full range of values into a few sub-ranges. By assigning each continuous value to the sub-range or class within which it falls, the data set changes from continuous to discrete.
- Dispersion is a general term for different statistics that describe how values are distributed around the centre.
- The range of a data set is the difference between the maximum and minimum values in the set.
- The p^{th} percentile is the value, v, that divides a data set into two parts, such that $p\%$ of the values in the data set are less than v and $100 - p\%$ of the values are greater than v. The general formula for finding the p^{th} percentile in an ordered data set of n values is

$$r = \frac{p}{100}(n-1) + 1$$

- The quartiles are the three data values that divide an ordered data set into four groups, where each group contains an equal number of data values. The lower quartile is denoted $Q1$, the median is $Q2$ and the upper quartile is $Q3$.
- The interquartile range is a measure of dispersion, which is calculated by subtracting the lower (first) quartile from the upper (third) quartile. This gives the range of the middle half of the data set.
- The semi interquartile range is half of the interquartile range.
- The five number summary consists of the minimum value, the maximum value and the three quartiles ($Q1$, $Q2$ and $Q3$).
- The box-and-whisker plot is a graphical representation of the five number summary.

End of chapter Exercise 10 – 7:

1. The following data set of heights was collected from a class of learners.
   {1,70 m; 1,41 m; 1,60 m; 1,32 m; 1,80 m; 1,40 m}
   Categorise the data set.
2. The following data set of sandwich spreads was collected from learners at lunch.
   {cheese; peanut butter; jam; cheese; honey}
   Categorise the data set.
3. Calculate the **mode** of the following data set:
   $\{10; 10; 10; 18; 7; 10; 3; 10; 7; 10; 7\}$
4. Calculate the **median** of the following data set:
   $\{5; 5; 10; 7; 10; 2; 16; 10; 10; 10; 7\}$
5. In a park, the tallest 7 trees have heights (in metres):
   $$\{41; 60; 47; 42; 44; 42; 47\}$$
   Find the median of their heights.
6. The learners in Ndeme's class have the following ages:
   $$\{5; 6; 7; 5; 4; 6; 6; 6; 7; 4\}$$
   Find the mode of their ages.
7. A group of 7 friends each have some sweets. They work out that the **mean** number of sweets they have is 6. Then 4 friends leave with an unknown number (x) of sweets. The remaining 3 friends work out that the **mean** number of sweets they have left is 10,67.
   When the 4 friends left, how many sweets did they take with them?
8. A group of 10 friends each have some sweets. They work out that the **mean** number of sweets they have is 3. Then 5 friends leave with an unknown number (x) of sweets. The remaining 5 friends work out that the **mean** number of sweets they have left is 3.
   When the 5 friends left, how many sweets did they take with them?
9. Five data values are represented as follows: $3x; x + 2; x–3; x + 4; 2x–5$, with a mean of 30. Solve for x.
10. Five data values are represented as follows: $p + 1; p + 2; p + 9$. Find the mean in terms of p.
11. A group of 10 learners count the number of marbles they each have. This is a histogram describing the data they collected:



    Count the number of marbles in the following range: $0 \leq$ number of marbles ≤ 1

12. A group of **20** learners count the number of playing cards they each have. This is the data they collect:

12	1	5	4	17
14	7	5	1	3
9	4	12	17	5
19	1	19	7	15

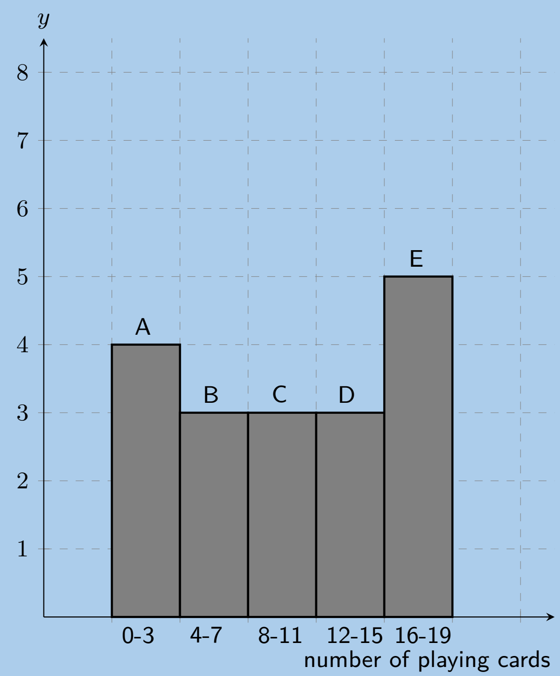
Count the number of learners who have from 0 up to 3 playing cards. In other words, how many learners have playing cards in the following range: $0 \leq$ number of playing cards ≤ 3? It may be helpful for you to draw a histogram in order to answer the question.

13. A group of **20** learners count the number of coins they each have. This is the data they collect:

17	11	1	15	14
3	4	18	5	14
18	19	4	18	15
16	13	20	8	18

Count the number of learners who have from 4 up to 7 coins. In other words, how many learners have coins in the following range: $4 \leq$ number of coins ≤ 7? It may be helpful for you to draw a histogram in order to answer the question.

14. A group of 20 learners count the number of playing cards they each have. The learners draw a histogram describing the data they collected. However, they have made a mistake in drawing the histogram.

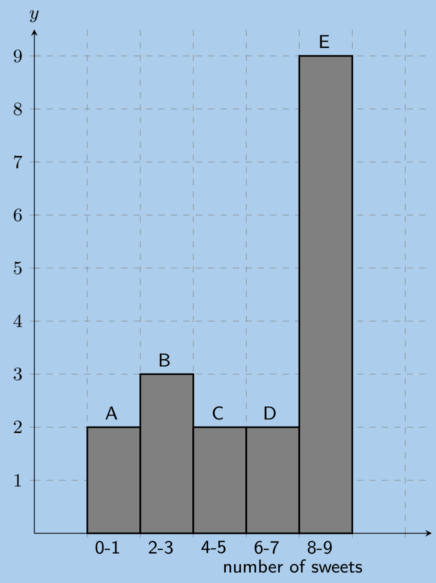


The data set below shows the correct information for the number of playing cards the learners have. Each value represents the number of playing cards for one learner.

$$\{18; 10; 3; 2; 19; 15; 2; 13; 11; 14; 10; 3; 5; 9; 4; 18; 11; 18; 16; 5\}$$

Help them figure out which **column** in the histogram is **incorrect**.

15. A group of 10 learners count the number of sweets they each have. The learners draw a histogram describing the data they collected. However, they have made a mistake in drawing the histogram.

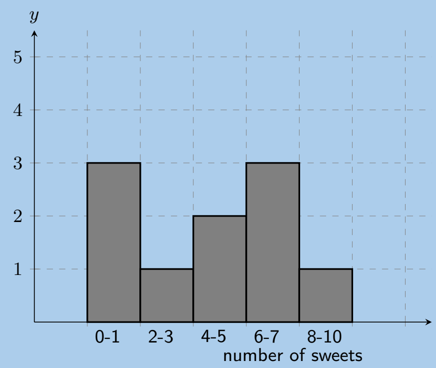


The data set below shows the correct information for the number of sweets the learners have. Each value represents the number of sweets for one learner.

$$\{1; 3; 7; 4; 5; 8; 2; 2; 1; 7\}$$

Help them figure out which **column** in the histogram is **incorrect**.

16. A group of learners count the number of sweets they each have. This is a histogram describing the data they collected:



A cleaner knocks over their table, and all their notes land on the floor, mixed up, by accident!

Help them find which of the following data sets match the above histogram:

Data set A

$$1 \quad 8 \quad 4 \quad 8 \quad 8 \quad 6 \quad 1 \quad 5 \quad 7 \quad 5$$

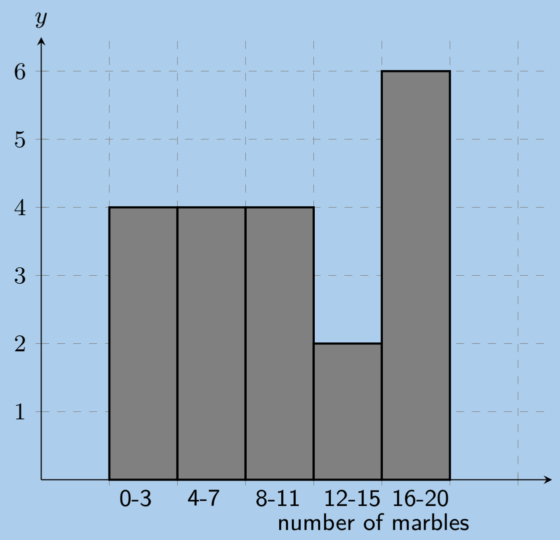
Data set B

$$5 \quad 6 \quad 9 \quad 2 \quad 1 \quad 6 \quad 6 \quad 4 \quad 4 \quad 6$$

Data set C

$$7 \quad 2 \quad 4 \quad 1 \quad 5 \quad 1 \quad 1 \quad 7 \quad 8 \quad 6$$

17. A group of learners count the number of marbles they each have. This is a histogram describing the data they collected:



A cat jumps onto the table, and all their notes land on the floor, mixed up, by accident!

Help them find which of the following data sets match the above histogram:

Data set A

7	13	15	13	12	13	8	14
3	15	1	7	4	11	1	

Data set B

17	1	5	4	11	13	6	19	6	20
19	1	14	9	17	3	16	3	10	10

Data set C

10	3	5	5	6	5	2	1	4	3

18. A group of **20** learners count the number of marbles they each have. This is the data they collect:

11	8	17	13	9	12	2	6	15	7
14	15	1	6	6	13	19	9	6	19

Calculate the **range** of values in the data set.

19. A group of **15** learners count the number of sweets they each have. This is the data they collect:

5	13	4	15	5	6	1	3
13	13	15	14	7	2	4	

Calculate the **range** of values in the data set.

20. An engineering company has designed two different types of engines for motorbikes. The two different motorbikes are tested for the time (in seconds) it takes for them to accelerate from 0 km·h^{-1} to 60 km·h^{-1}.

Test	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Bike 1	1,55	1,00	0,92	0,80	1,49	0,71	1,06	0,68	0,87	1,09
Bike 2	0,9	1,0	1,1	1,0	1,0	0,9	0,9	1,0	0,9	1,1

a) Which measure of central tendency should be used for this information?

b) Calculate the measure of central tendency that you chose in the previous question, for each motorbike.

c) Which motorbike would you choose based on this information? Take note of the accuracy of the numbers from each set of tests.

21. In a traffic survey, a random sample of 50 motorists were asked the distance they drove to work daily. This information is shown in the table below.

Distance (km)	Count
$0 < d \leq 5$	4
$5 < d \leq 10$	5
$10 < d \leq 15$	9
$15 < d \leq 20$	10
$20 < d \leq 25$	7
$25 < d \leq 30$	8
$30 < d \leq 35$	3
$35 < d \leq 40$	2
$40 < d \leq 45$	2

a) Find the approximate mean of the data.

b) What percentage of drivers had a distance of

i. less than or equal to 15 km?

ii. more than 30 km?

iii. between 16 km and 30 km?

c) Draw a histogram to represent the data.

22. A company wanted to evaluate the training programme in its factory. They gave the same task to trained and untrained employees and timed each one in seconds.

Trained	121	137	131	135	130
	128	130	126	132	127
	129	120	118	125	134
Untrained	135	142	126	148	145
	156	152	153	149	145
	144	134	139	140	142

a) Find the medians and quartiles for both sets of data.

b) Find the interquartile range for both sets of data.

c) Comment on the results.

d) Draw a box-and-whisker diagram for each data set to illustrate the five number summary.

23. A small firm employs 9 people. The annual salaries of the employees are:

R 600 000	R 250 000	R 200 000
R 120 000	R 100 000	R 100 000
R 100 000	R 90 000	R 80 000

a) Find the mean of these salaries.

b) Find the mode.

c) Find the median.

d) Of these three figures, which would you use for negotiating salary increases if you were a trade union official? Why?

24. The stem-and-leaf diagram below indicates the pulse rate per minute of ten Grade 10 learners.

7	8
8	1 3 5 5
9	0 1 1
10	3 5

Key: $7|8 = 78$

a) Determine the mean and the range of the data.

b) Give the five-number summary and create a box-and-whiskers plot for the data.

25. The following is a list of data: $3; 8; 8; 5; 9; 1; 4; x$
In each separate case, determine the value of x if the:

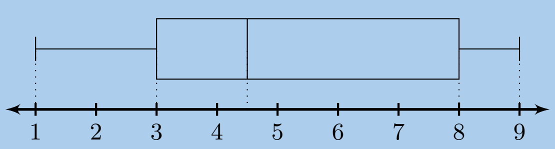
a) range = 16

b) mode = 8

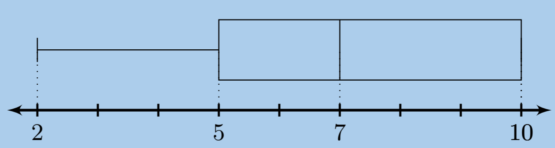
c) median = 6

d) mean = 6

e) box-and whiskers plot



26. Write down one list of numbers that satisfies the box-and-whisker plot below:



27. Given ϕ (which represents the golden ratio) to 20 decimal places: 1,61803398874989484820

a) For the first 20 decimal digits of (Φ), determine the:

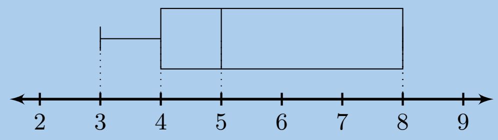
i. median

ii. mode

iii. mean

b) If the mean of the first 21 decimal digits of (Φ) is 5,38095 determine the 21^{st} decimal digit.

c) Below is a box-and-whisker plot of the 21^{st} - 27^{th} decimal digits. Write down one list of numbers that satisfies this box-and-whisker plot.



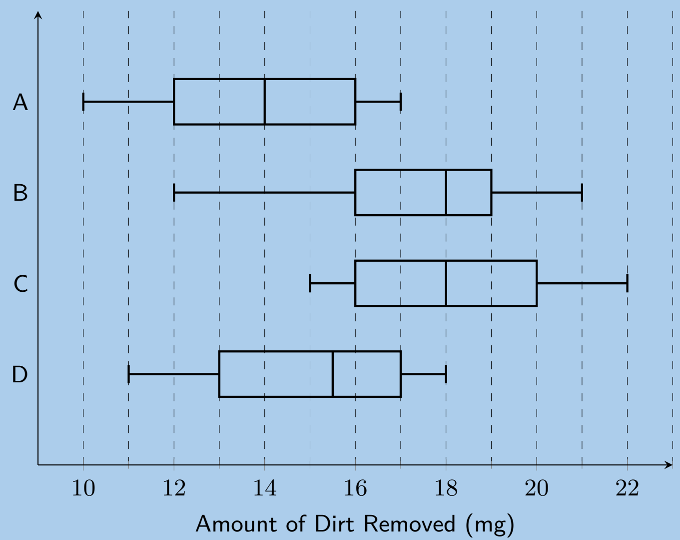
28. There are 14 men working in a factory. Their ages are : $22; 25; 33; 35; 38; 48; 53; 55; 55; 55; 55; 56; 59; 64$

a) Write down the five number summary.

b) If 3 men had to be retrenched, but the median had to stay the same, show the ages of the 3 men you would retrench.

c) Find the mean age of the men in the factory using the original data.

29. The example below shows a comparison of the amount of dirt removed by four brands of detergents (brands A to D).



a) Which brand has the biggest range, and what is this range?

b) For brand C, what does the number 18 mg represent?

c) Give the interquartile range for brand B.

d) Which brand of detergent would you buy? Explain your answer.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GNM	2. 2GNN	3. 2GNP	4. 2GNQ	5. 2GNR	6. 2GNS
7. 2GNT	8. 2GNV	9. 2GNW	10. 2GNX	11. 2GNY	12. 2GNZ
13. 2GP2	14. 2GP3	15. 2GP4	16. 2GP5	17. 2GP6	18. 2GP7
19. 2GP8	20. 2GP9	21. 2GPB	22. 2GPC	23. 2GPD	24. 2GPF
25. 2GPG	26. 2GPH	27. 2GPJ	28. 2GPK	29. 2GPM	

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